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Linear Autotopism Subgroups of Semifield Projective Planes

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Abstract. We investigate the well-known hypothesis of D. R. Hughes that the full collineation group of non-Desarguesian semifield projective plane of a finite order is solvable (the question 11.76 in Kurovka notebook was written down by N. D. Podufalov). This hypothesis is reduced to autotopism group that consists of collineations fixing a triangle. We describe the elements of order 4 and dihedral or quaternion subgroups of order 8 in the linear autotopism group when the semifield plane is of rank 2 over its kernel. The main results can be used as technical for the further studies of the subgroups of even order in an autotopism group for a finite non-Desarguesian semifield plane. The results obtained are useful to investigate the semifield planes with the autotopism subgroups from J. G. Thompson's list of minimal simple groups.

Keywords: semifield plane, autotopism, homology, Baer involution, Hughes' problem.

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Introduction

It is well-known that the geometric properties of projective plane are closely connected with the algebraic properties of its coordinatizing set. So, a finite Desarguesian projective plane is coordinatized by the field, a translation plane by the quasifield.

The study of finite semifields and semifield planes started ago with the first examples constructed by L. E. Dickson in 1906. A *semifield* is called a non-associative ring $Q = (Q, +, \cdot)$ with identity where the equations $ax = b$ and $ya = b$ are uniquely solved for any $a, b \in Q$, $a \neq 0$. The absence of an associative law in a semifield leads to a number of anomalous properties in comparison with a field or a skewfield or even a near-field.

By the mid-1950s, some classes of finite semifield planes had been found. All of them had the common property that the collineation group (automorphism group) is solvable. So D. R. Hughes conjectured in 1959 that any finite projective plane coordinatized by a non-associative semifield has the solvable collineation group. This hypothesis is presented in the monography [1]; it is proved also that the hypothesis is reduced to the solvability of an autotopism group as a group fixing a triangle. Moreover, the hypothesis is reduced to linear autotopism subgroup over the kernel. The Hughes' problem attracted the interest of a wide range of researchers who proved the collineation group solvability for an extensive list of semifield planes with certain restrictions.

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In 1990 the problem was written down by N.D. Podufalov in the Kourovka notebook ([2], the question 11.76).

We represent the approach to study Hughes' problem based on the classification of finite simple groups and theorem of J. G. Thompson on minimal simple groups. The spread set method allows us to identify the conditions when the semifield plane with certain autotopism subgroup exists. This method can be used also to construct examples, including computer calculations. The elimination of some groups from Thompson's list as autotopism subgroups allows us make progress in solving the problem.

We consider, mostly, the case when a semifield plane has a rank 2 over its kernel basing on the theory of M. Biliotti and co-authors [3]. Nevertheless, some results are generalised to N -rank case.

It is shown by the first author in [4, 5], that an autotopism of order two has the matrix representation convenient for calculations and reasoning. Here we use the spread set method to describe the geometric sense of an autotopism of order 4. The matrix representation of this autotopism allows us to prove the criterion of existence for the dihedral and quaternion subgroups of order 8 in an autotopism group. We present the examples of semifield planes of minimal order 625 with this property.

1. Main definitions and preliminary discussion

We use main definitions, according [1, 6], see also [7–9], for notations.

Consider a linear space Q , n -dimensional over the finite field $GF(p^s)$ (p is prime) and the subset of linear transformations $R \subset GL_n(p^s) \cup \{0\}$ such that:

- 1) R consists of p^{ns} square $(n \times n)$ -matrices over $GF(p^s)$;
- 2) R contains the zero matrix 0 and the identity matrix E ;
- 3) for any $A, B \in R$, $A \neq B$, the difference $A - B$ is a nonsingular matrix.

The set R is called a *spread set* [1]. Consider a bijective mapping θ from Q onto R and present the spread set as $R = \{\theta(y) \mid y \in Q\}$. Determine the multiplication on Q by the rule $x * y = x \cdot \theta(y)$ ($x, y \in Q$). Then $\langle Q, +, * \rangle$ is a right quasifield of order p^{ns} [6, 10]. Moreover, if R is closed under addition then $\langle Q, +, * \rangle$ is a semifield.

Note, that if we use a prime field $\mathbb{Z}_p$ as a basic field then the mapping θ is presented using only linear functions; it greatly simplifies reasoning and calculations (also computer).

A semifield Q coordinatizes the projective plane π of order $|\pi| = |Q|$ such that:

- 1) the affine points are the elements (x, y) of the space $Q \oplus Q$;
- 2) the affine lines are the cosets to subgroups

$$V(\infty) = \{(0, y) \mid y \in Q\}, \quad V(m) = \{(x, x\theta(m)) \mid x \in Q\} \quad (m \in Q);$$

- 3) the set of all cosets to the subgroup is the singular point;
- 4) the set of all singular points is the singular line;
- 5) the incidence is set-theoretical.

The solvability of a collineation group $Aut \pi$ for a semifield plane is reduced [1] to the solvability of an autotopism group Λ (collineations fixing a triangle). Without loss of generality, we can assume that linear autotopisms are determined by linear transformations of the space $Q \oplus Q$:

$$\lambda : (x, y) \rightarrow (x, y) \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix},$$

here the matrices A and B satisfy the condition (for instance, see [11])

$$A^{-1}\theta(m)B \in R \quad \forall \theta(m) \in R. \quad (1)$$

The collineations fixing a closed configuration have special properties. It is well-known [1], that any involutory collineation is a central collineation or a Baer collineation.

A collineation of a projective plane is called *central*, or *perspectivity*, if it fixes a line pointwise (*the axis*) and a point linewise (*the center*). If the center is incident to the axis then a collineation is called *an elation*, and a *homology* in another case. The order of any elation is a factor of the order $|\pi|$ of a projective plane, and the order of any homology is a factor of $|\pi| - 1$. All the perspectivities in an autotopism group are homologies and form the cyclic subgroups [12]:

$$H_1 = \left\{ \begin{pmatrix} M & 0 \\ 0 & E \end{pmatrix} \mid M \in R_m^* \right\}, \quad H_2 = \left\{ \begin{pmatrix} E & 0 \\ 0 & M \end{pmatrix} \mid M \in R_r^* \right\},$$

$$H_3 = \left\{ \begin{pmatrix} M & 0 \\ 0 & M \end{pmatrix} \mid M \in R_l^* \right\}.$$

The matrix subsets R_l, R_m, R_r are defined by a spread set [12]:

$$R_l = \{M \in GL_n(p^s) \cup \{0\} \mid MT = TM \quad \forall T \in R\},$$

$$R_m = \{M \in R \mid MT \in R \quad \forall T \in R\},$$

$$R_r = \{M \in R \mid TM \in R \quad \forall T \in R\},$$

they are called *left, middle and right nuclei* of the plane π respectively. These subfields in $GL_n(p^s) \cup \{0\}$ are isomorphic to correspondent nuclei of the coordinatizing semifield Q :

$$N_l = \{x \in Q \mid (x * a) * b = x * (a * b) \quad \forall a, b \in Q\},$$

$$N_m = \{x \in Q \mid (a * x) * b = a * (x * b) \quad \forall a, b \in Q\},$$

$$N_r = \{x \in Q \mid (a * b) * x = a * (b * x) \quad \forall a, b \in Q\}.$$

The plane π is Desarguesian (classic) iff Q is a field, then $R \simeq Q \simeq GF(p^{ns})$.

An autotopis group of a semifield plane of odd order contains three involutory homologies:

$$h_1 = \begin{pmatrix} -E & 0 \\ 0 & E \end{pmatrix} \in H_1, \quad h_2 = \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} \in H_2, \quad h_3 = h_1 h_2 = \begin{pmatrix} -E & 0 \\ 0 & -E \end{pmatrix} \in H_3.$$

A collineation of a projective plane π of order m is called *Baer collineation* if it fixes pointwise a subplane of order $\sqrt{|\pi|} = \sqrt{m}$ (*Baer subplane*). We use the results on the matrix representation of a Baer involution $\tau \in \Lambda$ and of a spread set obtained by M. Biliotti with co-authors [3] and by the first author in [4, 5].

2. Linear autotopisms of order 4

We consider now the case when a semifield plane π has a rank 2 over its kernel, $|\pi| = |N_l|^2$. To simplify the notation we use $K = N_l \simeq GF(q)$, $q = p^n$. The point set of the plane is

$$\pi = \{(x_1, x_2, y_1, y_2) \mid x_i, y_i \in GF(q)\},$$

the spread set R consists of (2×2) -matrices determined its second row:

$$R = \left\{ \theta(v, u) = \begin{pmatrix} f(v, u) & g(v, u) \\ v & u \end{pmatrix} \mid v, u \in GF(q) \right\}.$$

Here the functions f and g are additive:

$$\begin{aligned} f(v_1, u_1) + f(v_2, u_2) &= f(v_1 + v_2, u_1 + u_2), \\ g(v_1, u_1) + g(v_2, u_2) &= g(v_1 + v_2, u_1 + u_2), \quad v_1, v_2, u_1, u_2 \in GF(q), \end{aligned}$$

so f and g are the additive polynomials:

$$f(v, u) = \sum_{j=0}^{n-1} (f_j u^{p^j} + F_j v^{p^j}), \quad g(v, u) = \sum_{j=0}^{n-1} (g_j u^{p^j} + G_j v^{p^j}), \quad f_j, F_j, g_j, G_j \in GF(q).$$

The autotopism group Λ consists of *semi-linear* transformations of the linear space:

$$\lambda : (x, y) \rightarrow (x^\sigma, y^\sigma) \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix},$$

where σ is a basic field automorphism:

$$x^\sigma = (x_1, x_2)^\sigma = (x_1^{p^t}, x_2^{p^t}).$$

Evidently, that the subgroup Λ_0 of *linear* autotopisms ($t = 0$) is normal in Λ and the factor Λ/Λ_0 is isomorphic to a subgroup of $Aut K$. Therefore, the solvability problem is reduced to the linear autotopism subgroup Λ_0 .

G. E. Moorhouse in 1989 proved [13]:

Lemma 1. *Let π be a projective plane of order n^2 , $n \equiv 2$ or $3 \pmod{4}$, and G is a cyclic collineation group of order 4. Then the involution in G is central.*

We will expand Moorhouse' result for $|\pi| = p^{2n}$ if $p \not\equiv -1 \pmod{4}$.

Let π be a non-Desarguesian semifield plane of order q^2 with the kernel $K \simeq GF(q)$ ($q = 2^n$). If $\tau \in \Lambda_0$ is an involution then it is Baer, and we can propose that, in appropriate base, it has a Jordan normal form (see [3]):

$$\tau = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} L & 0 \\ 0 & L \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \quad (2)$$

The spread set $R \in GL_2(q) \cup \{0\}$ consists of matrices

$$\theta(v, u) = \begin{pmatrix} v + u + m(v) & f(v) + m(u) \\ v & u \end{pmatrix}, \quad v, u \in GF(2^n). \quad (3)$$

Lemma 2. *The linear autotopism group Λ_0 of a semifield projective plane π of order 2^{2n} with the kernel $K \simeq GF(2^n)$ does not contain elements of order 4.*

Proof. Let $\alpha \in \Lambda_0$ be an autotopism of order 4, $\alpha^4 = \varepsilon$. Then $\alpha^2 = \tau$ is a Baer involution (2), because $h_1, h_2, h_3 \notin \Lambda$ for $p = 2$. Let

$$\alpha = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix},$$

then $A^2 = B^2 = L$, $AL = LA$, $BL = LB$. So we have

$$A = \begin{pmatrix} a_1 & a_2 \\ 0 & a_1 \end{pmatrix}, \quad B = \begin{pmatrix} b_1 & b_2 \\ 0 & b_1 \end{pmatrix},$$

where

$$\begin{cases} a_1^2 = 1, \\ a_1 a_2 + a_2 a_1 = 1; \end{cases} \quad \begin{cases} b_1^2 = 1, \\ b_1 b_2 + b_2 b_1 = 1. \end{cases}$$

The systems have no solution in a field of the characteristic 2, the lemma is proved. $\square$

Theorem 2.1. *Let π be a semifield non-Desarguesian plane of order 2^{2n} with the kernel $K \simeq GF(2^n)$. Then the Sylow 2-subgroup of the linear autotopism group Λ_0 has an order at most 2.*

Proof. From the lemma, the Sylow 2-subgroup $S \subset \Lambda_0$ is elementary Abelian. Let $\tau, \alpha \in S$, where τ is (2). Then

$$\alpha = \begin{pmatrix} 1 & a & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & b \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}.$$

Consider the condition (1)

$$A^{-1}\theta(v, u)B \in R \quad \forall v, u \in GF(2^n)$$

for the spread set (3). For $\theta(0, 1) = E$ we have

$$A^{-1}B = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & b+a \\ 0 & 1 \end{pmatrix} \in R, \quad b = a.$$

Further, for $\theta(v, 0)$:

$$\begin{aligned} & \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v+m(v) & f(v) \\ v & 0 \end{pmatrix} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} = \\ & = \begin{pmatrix} v+m(v)+av & f(v)+av+m(v)a+a^2v \\ v & va \end{pmatrix} = \theta(v, va), \end{aligned}$$

and

$$m(va) = av + m(v)a + a^2v, \quad \forall v \in GF(2^n).$$

Consider the polynomial $m(v)$:

$$m(v) = m_0v + m_1v^2 + m_2v^4 + \dots + m_{n-1}v^{2^{n-1}},$$

$$m_0va + m_1v^2a^2 + m_2v^4a^4 + \dots = m_0va + m_1v^2a + m_2v^4a + \dots + va + va^2, \quad a + a^2 = 0.$$

If $a = 0$ then $\alpha = \varepsilon$; if $a = 1$ then $\alpha = \tau$. The theorem is proved. $\square$

Let now $p > 2$, $|\pi| = p^{2n}$, $K \simeq GF(p^n)$. We will not consider the case $p \equiv -1 \pmod{4}$: this case is more complicated for a semifield plane of arbitrary rank, see [14]. If $p \equiv 1 \pmod{4}$ then the prime field $\mathbb{Z}_p$ of K contains an element i such that $i^2 = -1$. We have $iE = E + \dots + E$, therefore $iE \in R_r \cap R_m$ and the linear autotopism group Λ_0 contains the homologies of order 4:

$$\alpha_1 = \begin{pmatrix} iE & 0 \\ 0 & E \end{pmatrix} \in H_1, \quad \alpha_2 = \begin{pmatrix} E & 0 \\ 0 & iE \end{pmatrix} \in H_2, \quad \alpha_1 \alpha_2 \in H_3.$$

As has been proven in [4], a Baer involution $\tau \in \Lambda_0$ can be written as

$$\tau = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} L & 0 \\ 0 & L \end{pmatrix}, \quad L = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (4)$$

for appropriate Jordan base. The spread set R consists of matrices

$$\theta(v, u) = \begin{pmatrix} m(u) & f(v) \\ v & u \end{pmatrix}, \quad v, u \in GF(p^n), \quad (5)$$

where the functions m and f are injective additive polynomials, $m(1) = 1$, $f(1) \neq \pm 1$.

The following theorem expands Moorhouse' lemma 1.

Theorem 2.2. *Let π be a non-Desarguesian semifield plane of order p^{2n} with the kernel $K \simeq GF(p^n)$, p is prime, $p \equiv 1 \pmod{4}$, and $\alpha \in \Lambda_0$ is a linear autotopism of order 4. Then α^2 is a homology and either $\alpha \in \langle \alpha_1, \alpha_2 \rangle$ or π admits a linear Baer involution τ and $\alpha \in \langle \alpha_1, \alpha_2, \tau \rangle$.*

Proof. Let α is not homology,

$$\alpha = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix},$$

then α^2 is the homology h_1 , h_2 , h_3 or a Baer involution τ (4).

If $\alpha^2 = \tau$ then from $A^2 = B^2 = L$, $AL = LA$, $BL = LB$ we have

$$A = \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix}, \quad B = \begin{pmatrix} b_1 & 0 \\ 0 & b_2 \end{pmatrix},$$

where $a_1, b_1 \in \{i, -i\}$, $a_2, b_2 \in \{1, -1\}$. Thus, the autotopism α can be represented as a product

$$\alpha = \begin{pmatrix} \pm i & 0 & 0 & 0 \\ 0 & \pm 1 & 0 & 0 \\ 0 & 0 & \pm i & 0 \\ 0 & 0 & 0 & \pm 1 \end{pmatrix} = \begin{pmatrix} L & 0 \\ 0 & L \end{pmatrix} \begin{pmatrix} M & 0 \\ 0 & M \end{pmatrix} \begin{pmatrix} -E & 0 \\ 0 & E \end{pmatrix} = \tau \mu h_1$$

or $\alpha = \tau \mu h_2$ or $\alpha = \tau \mu h_3$, where $\mu = \begin{pmatrix} M & 0 \\ 0 & M \end{pmatrix}$, $M = \begin{pmatrix} i & 0 \\ 0 & 1 \end{pmatrix}$. We consider the condition (1) for the collineation μ for any matrix (5): $M^{-1}\theta(v, u)M \in R$. For any $v \in K$ and $u = 0$ we have:

$$M^{-1}\theta(v, 0)M = \begin{pmatrix} -i & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & f(v) \\ v & 0 \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -if(v) \\ iv & 0 \end{pmatrix} \in R,$$

$f(iv) = -if(v)$. It contradicts with the additivity of $f(x)$, because $i \in \mathbb{Z}_p$ and so $f(iv) = if(v)$. Therefore, the case $\alpha^2 = \tau$ is impossible, and α^2 is a homology.

Let $\alpha^2 = h_1$. Then $A^2 = -E$, $B^2 = E$, and the following Jordan normal form are possible: $\pm iE$, $\pm iL$ for A and $\pm E$, $\pm L$ for B . If $A = \pm iE$ or $B = \pm E$ then $\alpha \in \langle \alpha_1, \alpha_2 \rangle$. The remaining possibility

$$\alpha = \begin{pmatrix} iL & 0 \\ 0 & L \end{pmatrix} \cdot h_1^{k_1} h_2^{k_2}$$

leads to $\alpha \cdot (\alpha_1 h_1^{k_1} h_2^{k_2})^{-1} = \begin{pmatrix} L & 0 \\ 0 & L \end{pmatrix} = \tau$ is the Baer involution, $\alpha \in \langle \alpha_1, \alpha_2, \tau \rangle$.

For $\alpha^2 = h_2$ and $\alpha^2 = h_3$ we obtain the analogous result. The theorem is proved. $\square$

Note that the homologies generate the normal subgroup in the autotopism group [1]. Therefore, if $F < \Lambda_0$ is a simple non-Abelian subgroup then it does not contain elements of order 4 for $p \not\equiv -1 \pmod{4}$. Further, any elementary Abelian subgroup of Λ_0 is of order at most 8. Maximal its order is for $\langle \tau, h_1, h_2 \rangle$; but $\langle h_1, h_2 \rangle$ is normal in Λ_0 , thus $|F|$ is either odd or $2 \cdot (2m+1)$. This contradicts to conjecture that F is simple non-Abelian group.

Corollary 1. *Let π be a non-Desarguesian semifield plane of order p^{2n} with the kernel $K \simeq GF(p^n)$, p is prime, $p \equiv 1 \pmod{4}$. Then its autotopism group contains no simple non-Abelian subgroups.*

For $p = 2$ the more significant result has been proven by M. J. Ganley in 1974 ([15], see also [16]).

Theorem 2.3. *Let Q be a finite semifield of order 2^s . If Q has dimension 2 over one of its nuclei then its autotopism group is solvable.*

Extend the results obtained by the information on coordinatizing semifield automorphisms. It has been proven in [11], that the linear transformation $x \rightarrow xA$ is the automorphism of a semifield Q iff the matrix

$$\begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix} \quad (6)$$

is the autotopism of the semifield plane π with the condition $A^{-1}\theta(m)A = \theta(mA)$ for any $m \in Q$. Therefore, we have the following result.

Corollary 2. *Let Q be a non-associative semifield of order p^{2n} with the left nucleus $K \simeq GF(p^n)$, p is prime, $p \not\equiv -1 \pmod{4}$. Then the automorphism subgroup $\text{Aut}_K Q$ of Q that fixes K has a Sylow 2-subgroup of order at most 2.*

Proof. For the even case $p = 2$ the result is a direct consequence of the theorem 2.1. Let $p \equiv 1 \pmod{4}$ and the transformation $x \rightarrow xA$ be an automorphism from $\text{Aut}_K Q$. Then we can assume that $A \in GL_2(p^n)$. According to [11], the involution (6) may be Baer only. Up to base chosen, we can suppose $A = L$. The centralizer of τ in Λ_0 contains involutions τ , $h_1\tau$, $h_2\tau$, $h_3\tau$ only. All these possibilities lead to the contradiction, because the involution $\tau \cdot (h_i\tau)$ is not Baer. Thus, the elementary Abelian 2-subgroup of $\text{Aut}_K Q$ is of order at most 2. Finally, if the autotopism α (6) is of order 4 then α^2 is an involutory homology h_3 with $A = -E$; we have the contradiction. $\square$

3. Dihedral and quaternion subgroups

The question on autotopism subgroups isomorphic to D_8 or Q_8 is explained by the fact that such subgroups are contained in the Sylow 2-subgroup of a large number of simple non-Abelian groups. For semifield plane of arbitrary rank over the kernel, the first author proved [8] that a

dihedral autotopism subgroup of order 8 must contain the homology if $p \equiv 1 \pmod{4}$. Now we describe the matrix representation of subgroup

$$H = \langle \alpha, \beta \mid \alpha^4 = \beta^2 = 1, \beta\alpha\beta = \alpha^{-1} \rangle \simeq D_8 \quad (7)$$

and the spread set matrices. The generalization of this result for N -dimensional case see in the next section.

Theorem 3.4. *Let π be a non-Desarguesian semifield plane of order p^{2n} with the kernel $K \simeq GF(p^n)$, p is prime, $p \equiv 1 \pmod{4}$, and the linear autotopism group Λ_0 contains a subgroup isomorphic to the dihedral group of order 8 (7). Then the base of 4-dimensional vector space over K can be chosen such that $H = \langle \alpha, \beta \rangle$,*

$$\alpha = \begin{pmatrix} -i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \end{pmatrix} \cdot h_1^{k_1} h_2^{k_2}, \quad \beta = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad i^2 = -1, \quad i \in \mathbb{Z}_p,$$

$k_1, k_2 \in \{0, 1\}$. The spread set of π consist of matrices (5), where m and f are injective involutory functions on $GF(p^n)$, $m(m(x)) = x$, $f(f(x)) = x$.

Proof. If $\alpha \in \langle \alpha_1, \alpha_2 \rangle$ then $\alpha \in Z(\Lambda)$ and the condition $\beta\alpha\beta = \alpha^{-1}$ is not satisfied for any autotopism β . Therefore, the plane π admits the Baer involution τ , it has the spread set (5), and $\alpha \in \langle \alpha_1, \alpha_2, \tau \rangle$, from the theorem 2.2. Then α^2 is one of involutory homologies h_1, h_2, h_3 . Consider the possible cases.

Let $\alpha^2 = h_1$. Then, up to involutory homologies, $\alpha = \begin{pmatrix} iL & 0 \\ 0 & L \end{pmatrix} = \alpha_1\tau$, and $\beta(\alpha_1\tau)\beta = (\alpha_1\tau)^3\beta$, $\beta\tau\beta = h_1\tau$. Denote $\beta = \begin{pmatrix} C & 0 \\ 0 & D \end{pmatrix}$, then $C^2 = D^2 = E$, $CLC = -L$, $DLD = L$, and either $D = \pm E$ or $D = \pm L$. If $D = \pm E$ then $\begin{pmatrix} C & 0 \\ 0 & E \end{pmatrix}$ is the involutory homology, and so $C = -E$, this contradicts to condition. Therefore, up to involutory homology, $D = L$,

$$\beta = \begin{pmatrix} C & 0 \\ 0 & L \end{pmatrix} = \begin{pmatrix} CL & 0 \\ 0 & E \end{pmatrix} \cdot \tau,$$

and $\beta\tau$ is the homology of order 4: $\beta\tau = \alpha_1$ or $\beta\tau = \alpha_1^3$, $CL = \pm iE$, $C = \pm iL$, $C^2 = -E \neq E$. Thus, the case $\alpha^2 = h_1$ is impossible.

By similar reasoning we come to a contradiction in the case $\alpha^2 = h_2$.

Let now $\alpha^2 = h_3$, and $\alpha = \begin{pmatrix} iL & 0 \\ 0 & iL \end{pmatrix} = \alpha_1\alpha_2\tau$, without the involutory homologies. Then $\beta(\alpha_1\alpha_2\tau)\beta = (\alpha_1\alpha_2\tau)^3 = \alpha_1\alpha_2h_3\tau$, $\beta\tau\beta = h_3\tau$, $CLC = DLD = -L$, and we have

$$C = \begin{pmatrix} 0 & c \\ c^{-1} & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & d \\ d^{-1} & 0 \end{pmatrix}.$$

Consider the transition matrix T :

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & d \end{pmatrix}. \quad (8)$$

Then, for the new base,

$$T\alpha T^{-1} = \alpha, \quad T\beta T^{-1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

To complete the proof, it is enough to test the involutory of the functions m and f . Indeed, β is a collineation, and, for any matrix $\theta(v, u)$ from the spread set R , the product

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} m(u) & f(v) \\ v & u \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} u & v \\ f(v) & m(u) \end{pmatrix}$$

must belong to R , see (1). Therefore, $m(m(u)) = u$ and $f(f(v)) = v$ for all $v, u \in GF(p^n)$. The theorem is proved. $\square$

Now let the linear autotopism group Λ_0 contains a quaternion subgroup of order 8:

$$F = \langle \alpha, \gamma \mid \alpha^4 = \gamma^4 = 1, \alpha^2 = \gamma^2, \alpha\gamma\alpha = \gamma \rangle \simeq Q_8. \quad (9)$$

Theorem 3.5. *Let π be a non-Desarguesian semifield plane of order p^{2n} with the kernel $K \simeq GF(p^n)$, p is prime, $p \equiv 1 \pmod{4}$, and the linear autotopism group Λ_0 contains a subgroup isomorphic to the quaternion group of order 8 (9). Then the base of 4-dimensional vector space over K can be chosen such that $F = \langle \alpha, \gamma \rangle$,*

$$\alpha = \begin{pmatrix} -i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \end{pmatrix} \cdot h_1^{k_1} h_2^{k_2}, \quad \gamma = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad i^2 = -1, \quad i \in \mathbb{Z}_p,$$

$k_1, k_2 \in \{0, 1\}$. The spread set of π consist of matrices (5), where m and f are injective involutory functions on $GF(p^n)$, $m(m(x)) = x$, $f(f(x)) = x$.

Proof. If either α or γ belongs to subgroup $\langle \alpha_1, \alpha_2 \rangle < Z(\Lambda_0)$ then $\alpha\gamma = \gamma\alpha$, it is impossible. Therefore, the plane π admits a Baer involution τ (4) and, for instance, $\alpha \in \langle \alpha_1, \alpha_2, \tau \rangle$. Evident, that we can ignore the involutory homologies factors, because $\langle \alpha, \gamma \rangle \simeq Q_8$ leads to $\langle \alpha h_1^{k_1} h_2^{k_2}, \gamma \rangle \simeq Q_8$. So, we suppose further $\alpha = \alpha_1 \tau$, $\alpha = \alpha_2 \tau$ or $\alpha = \alpha_1 \alpha_2 \tau$.

In the first case $\alpha^2 = \gamma^2 = h_1$. Notify

$$\gamma = \begin{pmatrix} C & 0 \\ 0 & D \end{pmatrix}, \quad C^2 = -E, \quad D^2 = E,$$

and consider the condition $\alpha\gamma\alpha = \gamma$:

$$(\alpha_1 \tau) \gamma (\alpha_1 \tau) = \gamma, \quad h_1 \tau \gamma \tau = \gamma, \quad -LCL = C, \quad LDL = D.$$

Then $D \in \{E, -E, L, -L\}$, but the case $D = \pm E$ is impossible: $\begin{pmatrix} C & 0 \\ 0 & E \end{pmatrix}$ is a homology of order 4, $C = \pm iE$, $\alpha\gamma = \gamma\alpha$. Therefore, up to involutory homologies, $D = L$,

$$\gamma = \begin{pmatrix} C & 0 \\ 0 & L \end{pmatrix} = \begin{pmatrix} CL & 0 \\ 0 & E \end{pmatrix} \cdot \tau,$$

then $\gamma\tau$ is the involutory homology too (from $CLCL = E$), that is $\gamma\tau = h_1$, $\gamma = h_1\tau$, and $\alpha\gamma = \gamma\alpha$. This contradiction shows that the case $\alpha^2 = h_1$ is impossible; for $\alpha^2 = h_2$, similarly.

Let now $\alpha^2 = \gamma^2 = h_3$, $\alpha = \begin{pmatrix} iL & 0 \\ 0 & iL \end{pmatrix} = \alpha_1\alpha_2\tau$. Then

$$\alpha\gamma\alpha = (\alpha_1\alpha_2\tau)\gamma(\alpha_1\alpha_2\tau) = h_1h_2\tau\gamma\tau = h_3\tau\gamma\tau = \gamma,$$

and we have the conditions $-LCL = C$, $-LDL = D$, $C^2 = D^2 = -E$, leading to

$$C = \begin{pmatrix} 0 & c \\ -c^{-1} & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & d \\ -d^{-1} & 0 \end{pmatrix}.$$

Choose the transition matrix T (8), then for the new base we obtain

$$T\alpha T^{-1} = \alpha, \quad T\gamma T^{-1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

Prove the involutory of m and f form the condition (1):

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{-1} \theta(v, u) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} u & -v \\ -f(v) & m(u) \end{pmatrix} \in R \quad \forall v, u \in GF(p^n).$$

Thus, $f(-f(v)) = -v$, $m(m(u)) = u$, and the additivity leads to $f(f(v)) = v$. The theorem is proved. $\square$

Remark 1. Note that, according to the theorem 2.2 on autotopisms of order 4, the collineation γ must be the product of homologies to a Baer involution. Indeed,

$$\gamma = \alpha_1\alpha_2\sigma, \quad \sigma = \begin{pmatrix} 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & -i & 0 \end{pmatrix},$$

where σ is the Baer involution fixing pointwise the subplane

$$\pi_\sigma = \{(x_1, ix_1, y_1, iy_1) \mid x_1, y_1 \in GF(p^n)\}.$$

Rewrite the autotopism γ of order 4 as

$$\gamma = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \beta\tau.$$

We see that $\langle \alpha, \beta \rangle \simeq D_8$ and $\langle \alpha, \beta\tau \rangle \simeq Q_8$, and the following corollary is proved.

Corollary 3. Let π be a non-Desarguesian semifield plane of order p^{2n} with the krenel $K \simeq GF(p^n)$, p is prime, $p \equiv 1 \pmod{4}$. The linear autotopism group Λ_0 contains a subgroup isomorphic to the quaternion group of order 8 iff Λ_0 contains a subgroup isomorphic to the dihedral group of order 8.

4. Examples

Construct the semifield planes of minimal order satisfying the condition of theorems 3.4 and 3.5. It is well-known [1] that a semifield of order p^2 is a field $GF(p^2)$. Therefore the minimal examples are the planes of order $5^4 = 625$. Let the field $K \simeq GF(25)$ is an algebraic extension of $\mathbb{Z}_5$, $K = \mathbb{Z}_5(a)$, where a is the root of the irreducible polynomial $x^2 + 3x + 3 \in \mathbb{Z}_5[x]$. Then the semifield Q of order 625 is the vector space $Q = \{x = (v, u) \mid v, u \in K\}$ with the multiplication law $y * x = y \cdot \theta(x)$ ($x, y \in Q$); the spread set R consists of matrices (5). The functions m and f are the polynomials from $K[x]$:

$$m(u) = m_0u + m_1u^5, \quad \varphi(v) = f_0v + f_1v^5,$$

which satisfy the conditions $m(m(u)) = u$, $f(f(v)) = v$ for all $u, v \in K$, $m(1) = 1$, $f(1) \neq \pm 1$. There exist 34 pairs of functions m, f such that $\det \theta(v, u) = 0$ only for $(v, u) = (0, 0)$. Therefore, we obtain 34 semifield planes of order 625 with the kernel of order 25 which admit the linear autotopism subgroup isomorphic to D_8 (or Q_8). At most 11 pairwise non-isomorphic planes are among them. The isomorphism is either multiplication by a suitable matrix (i.e. changing of base) or the automorphism of K :

$$(m_0, m_1, f_0, f_1) \rightarrow (m_0^5, m_1^5, f_0^5, f_1^5)$$

The table below represents the coefficients m_i, f_i together with the nuclei of the semifields.

Table 1. Information on the planes of order 625

N ^o	m_0	m_1	f_0	f_1	N_l	$N_m = N_r$
1	0	1	0	a	K	$\{(0, y) \mid y \in K\}$
2	0	1	0	$a + 1$		$ N_m = N_r = 25$
3	0	1	$2a + 1$	2	K	$\{(0, y) \mid y \in \mathbb{Z}_5\}$ $ N_m = N_r = 5$
4	0	1	$2a + 1$	3		
5	$4a + 2$	$a + 4$	$a + 3$	$a + 2$		
6	$4a + 2$	$a + 4$	$a + 3$	$3a + 4$		
7	$3a + 4$	$2a + 2$	0	$a + 1$		
8	$3a + 4$	$2a + 2$	$2a + 1$	2		
9	$3a + 4$	$2a + 2$	$2a + 1$	$2a$		
10	$3a + 4$	$2a + 2$	$2a + 1$	$2a + 2$	K	$\{(x, y) \mid$ $x \in \{0, a + 3, 2a + 1,$ $3a + 4, 4a + 2\}, y \in \mathbb{Z}_5\}$ $ N_m = N_r = 25$
11	$a + 3$	$4a + 3$	$a + 3$	$a + 2$		

5. Generalization for arbitrary dimension

Here we consider the case when a semifield plane π has the order p^N , without restriction to the order of the kernel. In this case we can represent the point set of π as a $2N$ -dimensional vector space over $\mathbb{Z}_p$, with the spread set $R \subset GL_N(p) \cup \{0\}$. Some results from the previous sections can be generalized for any N and $p \equiv 1 \pmod{4}$.

Let π be a non-Desarguesian semifield plane of order p^N ($p > 2$ be prime). According to the results of [5], if the autotopism group Λ contains the Baer involution τ then $N = 2n$ is even and we can choose the base of $4n$ -dimensional linear space over $\mathbb{Z}_p$ such that

$$\tau = \begin{pmatrix} L & 0 \\ 0 & L \end{pmatrix}, \quad L = \begin{pmatrix} -E & 0 \\ 0 & E \end{pmatrix}. \quad (10)$$

The spread set R in $GL_{2n}(p) \cup \{0\}$ consists of matrices

$$\theta(V, U) = \begin{pmatrix} m(U) & f(V) \\ V & U \end{pmatrix}, \quad (11)$$

where $V \in Q$, $U \in K$, Q, K are the spread sets in $GL_n(p) \cup \{0\}$, K is the spread set of the Baer subplane π_τ , m, f are additive injective functions from K and Q into $GL_n(p) \cup \{0\}$, $m(E) = E$. Note that throughout the section, the blocks-submatrices have the same dimension by default. Instead of linear autotopism group Λ_0 we will consider the autotopism group Λ :

$$\Lambda = \left\{ \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \mid A, B \in GL_N(p), A^{-1}\theta(m)B \in R \forall \theta(m) \in R \right\}.$$

Unfortunately, we can not now extend the result of the theorem 2.2 to the general case. The geometric sense of order 4 autotopism α was presented in [14] when $\alpha^2 = \tau$ is a Baer involution. Perhaps, one will construct the examples illustrating the matrix representation of a spread set in this case; there is no evident contradiction.

Theorem 5.6. *Let π be a non-Desarguesian semifield plane of order p^N , p is prime, $p \equiv 1 \pmod{4}$, and $\alpha \in \Lambda$ is an autotopism of order 4. If α^2 is a homology then either $\alpha \in \langle \alpha_1, \alpha_2 \rangle$ or π admits a Baer involution τ and $\alpha \in \langle \alpha_1, \alpha_2, \tau \rangle$.*

Proof. Let $\alpha^2 = h_1$ and $\alpha \notin \langle \alpha_1, \alpha_2 \rangle$. Then $\alpha = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$ where $A^2 = -E$, $B^2 = E$. Therefore $A = \text{diag}(i, -i) \neq \pm iE$, $B = \text{diag}(1, -1) \neq \pm E$. The number of 1 among the diagonal elements of B equals to the number of -1 , because else we have the autotopism that fixes more than a Baer subplane, it is impossible. So, we can assume, up to base changing and involution homologies factors, that $B = L$ and $A = iL$, $\alpha = \alpha_1\tau$. For $\alpha^2 = h_2$ and $\alpha^2 = h_3$ the consideration is similar. $\square$

We extend now the main result of [8]:

Theorem 5.7. *Any non-Desarguesian semifield plane π of order p^N , where $p > 2$ is prime and $p \equiv 1 \pmod{4}$, does not admit an autotopism subgroup isomorphic to the dihedral group of order 8 without homologies.*

Denote the following autotopisms:

$$\alpha = \begin{pmatrix} -iE & 0 & 0 & 0 \\ 0 & iE & 0 & 0 \\ 0 & 0 & -iE & 0 \\ 0 & 0 & 0 & iE \end{pmatrix} h_1^{k_1} h_2^{k_2}, \quad (12)$$

$$\beta = \begin{pmatrix} 0 & E & 0 & 0 \\ E & 0 & 0 & 0 \\ 0 & 0 & 0 & E \\ 0 & 0 & E & 0 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 0 & E & 0 & 0 \\ -E & 0 & 0 & 0 \\ 0 & 0 & 0 & E \\ 0 & 0 & -E & 0 \end{pmatrix},$$

where $i \in \mathbb{Z}_p$, $i^2 = -1$, $k_1, k_2 \in \{0, 1\}$. Using the result of [18], we prove.

Theorem 5.8. *Let π be a non-Desarguesian semifield plane of order p^N , p is prime, $p \equiv 1 \pmod{4}$. The autotopism group Λ contains a subgroup $H \simeq D_8$ (7) iff it contains a subgroup $F \simeq Q_8$ (9). Then $N = 2n \geq 4$, the subgroups H and F contains the involutory homology h_3 , the plane π admits a Baer involution τ (10). The base of linear space can be chosen such that $H = \langle \alpha, \beta \rangle$, $F = \langle \alpha, \gamma \rangle$, where the autotopisms are (12). The spread set R consists of matrices (11), where $V \in Q$, $U \in K$, the sets $Q, K \subset GL_n(p) \cup \{0\}$ are closed under addition. The additive injections $m : K \rightarrow K$ and $f : Q \rightarrow Q$ are non-trivial involutions.*

Proof. The result for Q_8 had been proven in [18]. We will repeat now the proof of the theorem 3.4 with generalization for N -dimensional case. Let $\alpha \in H \simeq D_8$, $\alpha^4 = \varepsilon$. Then α^2 is the homology by the theorem 5.7.

1. If $\alpha^2 = h_1$ then $\alpha \notin \langle \alpha_1, \alpha_2 \rangle \subset Z(\Lambda)$. Therefore we can assume, up to base changing, that $\alpha = \alpha_1 \tau$, the spread set R consists of matrices (11). Further, let the Baer involution β be the matrix $\beta = \begin{pmatrix} C & 0 \\ 0 & D \end{pmatrix}$. Then, from $\alpha\beta = \beta\alpha^{-1}$ we have

$$C = \begin{pmatrix} 0 & C_1 \\ C_2 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} D_1 & 0 \\ 0 & D_2 \end{pmatrix}, \quad C_1 C_2 = E, \quad D_1^2 = D_2^2 = E.$$

We can use the block-diagonal transition matrix T similar (8) and obtain $C_1 = C_2 = E$. Moreover, we can assume that D_1 and D_2 are either diagonal matrices $\text{diag}(1, -1)$ or $\pm E$. From the condition (1) we have

$$\begin{pmatrix} 0 & E \\ E & 0 \end{pmatrix} \begin{pmatrix} D_1 & 0 \\ 0 & D_2 \end{pmatrix} = \begin{pmatrix} 0 & D_2 \\ D_1 & 0 \end{pmatrix} = \theta(D_1, 0) \in R,$$

but either the matrix $E + \theta(D_1, 0) = \theta(D_1, E) \neq 0$ or the matrix $\theta(D_1, iE) \neq 0$ is singular, it is impossible. Thus, the conjecture $\alpha^2 = h_1$ leads to contradiction, similar for h_2 .

2. Let $\alpha^2 = h_3$. Then α is the matrix (12), up to base changing. The transition matrix

$$T = \begin{pmatrix} E & 0 & 0 & 0 \\ 0 & C_1 & 0 & 0 \\ 0 & 0 & E & 0 \\ 0 & 0 & 0 & D_1 \end{pmatrix}$$

preserves α and maps the Baer involution

$$\beta = \begin{pmatrix} 0 & C_1 & 0 & 0 \\ C_1^{-1} & 0 & 0 & 0 \\ 0 & 0 & 0 & D_1 \\ 0 & 0 & D_1^{-1} & 0 \end{pmatrix}$$

to the matrix (12). The condition (1) for $\theta(V, U)$ leads to the involutivity $m(m(U)) = U$, $f(f(V)) = V$. The theorem is proved. $\square$

Conclusion

We can see that the properties and the structure of the linear autotopism group for a two-dimensional semifield plane may be considerably generalized to the N -dimensional case. The proof technique can be used with more careful consideration.

In order to study Hughes' problem on the solvability of the full collineation group of a finite non-Desarguesian semifield plane, the authors consider it possible to use the obtained results to further investigations. The method applied will probably be useful to consider simple non-Abelian groups and to exclude an extensive list from possible autotopism subgroups.

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Подгруппы линейных автотопизмов полуполевых проективных плоскостей

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Аннотация. Изучается известная гипотеза Д. Хьюза 1959 г. о разрешимости полной группы коллинеаций недезарговой полуполевой проективной плоскости конечного порядка (также вопрос 11.76 Н. Д. Подуфалова в Коуровской тетради). Эта гипотеза редуцируется к группе автотопизмов, состоящей из коллинеаций, фиксирующих треугольник. В работе описаны элементы порядка 4 и диэдральные либо кватернионные подгруппы порядка 8 в группе линейных автотопизмов полуполевой плоскости ранга 2 над ядром. Основные доказанные результаты являются техническими и необходимы для дальнейшего изучения подгрупп четного порядка в группе автотопизмов конечной недезарговой полуполевой плоскости. Результаты могут быть использованы для изучения полуполевых плоскостей, допускающих подгруппы автотопизмов из списка Д. Г. Томпсона минимальных простых групп.

Ключевые слова: полуполевая плоскость, автотопизм, гомология, бэровская инволюция, проблема Хьюза.

EDN: BCTNDO

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A Multifaceted Approach to Forming Mathematical Digital Competency of Future Engineers in Teaching Applied Mathematics

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Abstract. The paper deals with the problem of forming mathematical digital competency of engineering students. Authors suggest a comprehensive approach to solve the problem which is implemented in a course of applied mathematics for future engineers. The concept of mathematical digital competency of an engineering specialist, the formation of which is the aim of students' training, is articulated. The paper provides rationale for the use of professionally oriented task system for mastering methods of mathematical modelling, computer simulators as well as game simulation models for teaching applied mathematics with the use of +ACI-Teacher–Student+ACI- automated working space. It facilitates quick mastering of basic methods of applied mathematics, computer science, some elements of algorithmization and programming by engineering students. The contents of the virtual laboratory complex for the course of applied mathematics powered by AnyLogic platform for simulation model implementation is described. The description of the developed system for managing student individual work is given. Tools for diagnosing the formation of mathematical digital competency are also presented.

Keywords: applied mathematics, mathematical digital competency, future engineers, mathematical modeling, digital education tools.

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Introduction

The model-based process of developing science and technology requires strong mathematical foundation. It allows one to build models, elaborating algorithms using the apparatus of computational mathematics for calculations and estimating model accuracy in analysis and optimization problems. It means that teaching mathematical modelling based on integration of mathematics and applied science combined with digital technologies is a current trend of development in modern engineering education [1].

One of the important subjects combining mathematical and computational modelling is applied mathematics. Its main objective is to develop competency to use mathematical apparatus to solve problems in professional activity of future engineers, including computer modelling. It is applied mathematics that is considered to be one of the main subjects forming mathematical competency of future engineers.

We regard *mathematical digital competency of an engineering specialist* as a competency constituting in the knowledge and understanding of mathematical language and digital tools to be used in engineering. Both mathematical and digital competencies determine readiness and ability to solve engineering problems by the means of mathematical and computer modelling [2].

Having created a special system of teaching applied mathematics that combines mathematical and applied knowledge based on information and communication technologies and digital tools, we aim to develop mathematical digital competency in future engineers.

The objective of the article is to describe an approach to the process of teaching future engineers a course of applied mathematics that combines mathematical and computer modelling aimed to develop mathematical digital competency.

1. Materials and methods

We chose the research methods based on the approaches to teach applied mathematics combining algorithmic and heuristic approaches to solve problems. The algorithmic approach suggests searching solution of a problem in accordance with the predetermined order of actions. In the heuristic approach the problem solving is based on the adopted strategy of finding solutions [3]). This approach agrees with developing mathematical competencies for future engineers.

A major problem that emerges in the course of teaching applied mathematics to engineering students is forming the mathematical mindset, the ability to discover and experience new patterns, and the interest in studying mathematical models. The listed features are mainly developed in the course of solving professionally oriented tasks (POT).

We regard *the professionally oriented task system for mastering methods of mathematical modelling by engineering students* as a combination and sequence of professional content tasks in the courses of further and applied mathematics which facilitate the development of mathematical digital competency of future engineers.

The tool that helps students to solve professionally oriented tasks which constitute the basis of mathematical and computer modelling of actual technical processes is chosen to be an organizational and technical system containing:

- computer simulators allowing for interaction with students via built-in control elements (button, check box, combo box, link label, radio button, text box, numeric up-down, and more);

- edutainment models for teaching applied mathematics embedded in "Teacher-Student" automated working space" software [4], mathematical suites simplifying routine calculations.

We regard *computer simulators* as imitation computer models which represent actual or hypothetical technological processes. They can be used to study dynamic changes in parameters of initial technical processes or to build up a hypothesis to explore them.

Such models for teaching applied mathematics are presented as:

- a simulator of interactive curve plotting set in explicit, parametric form and polar coordinates;
- a simulator of the interactive analysis of a queueing system;
- a simulator of the graphic method of solving linear programming problems (including Gomory's cutting plane method for integer programming problems);
- a simulator of the graphic method of solving game models with the dimension of $2 \times n$ etc.

The simulators allow one to simulate various operations from construction of lines to modelling complex production processes. Students use them while searching for a mathematical model, analyse them in the course of applied mathematics, and apply them in studying technical processes that allows one to form digital competencies of students.

The key component of teaching applied mathematics is "Teacher-Student" automated working space" software (AWS). The use of this software was particularly described in articles [4–6] etc.

Students are offered to apply the author-developed software for individual work. It can be used both during a lab session under the supervision of the instructor and in home conditions following this procedure: Stage 1 — model testing; Stage 2 — working with demonstration program; Stage 3 — individual model building and implementation check; Stage 4 — controlling student learning achievements.

The developed AWS can be considered as a tool to quickly master basic methods of applied mathematics, computer science, elements of algorithmization and programming. Further, the skills developed can widely be used to state and solve complex problems with the help of high-end mathematical suites. These skills also help students create game models while solving applied mathematics problems.

Imitation models can be ergonomically developed on AnyLogic platform as a virtual laboratory complex for the course of applied mathematics. The complex is focused on training road transport engineer and contains laboratory works.

The main objective of laboratory works is to meet the demands of actual technological processes, and to form students' ability to carry out scientific activity.

For instance, in the laboratory work «Queueing systems» it is suggested to combine the elements of classical form of subject presentation with developing the imitation model based on AnyLogic, namely, to get the visualization of the model with the opportunity to present actual (simulated) processes of queueing systems after providing the mathematical rationale. The main thing that students are supposed to do in the course of performing laboratory works is organizing their own project research activity, for example, to design gas filling stations capable of bringing service to a big number of automobiles.

Scientific research activities of future engineers have prime importance as they develop engineering mindset, and form professional competency [7]. This is the reason why students present

the laboratory work completion report to a teacher as a design of a gas filling station for a specific residential area. Students present the best research projects at scientific conferences.

Using the above mentioned materials and methods not only allows establishing hierarchy of studying applied mathematics but also paves the way to the use of integrative, research, and practice oriented approaches to teach students, and to form their mathematical and digital competencies.

2. Research results

A pedagogical experiment in developing mathematical digital competency in future engineers was carried out at Gorlovka Automobile and Highway Institute of Donetsk National Technical University over the period of 2017–2021. The experiment was focused on testing, adjustment, and implementation of the mathematical modeling teaching system in the course of applied mathematics in the scope of education digitalization.

In particular, in the course of organizing applied mathematics learning activity in the experimental group interactive methods of digital didactics (heuristic methods, gaming methods, problem based learning methods) were used; conventional organizational forms of teaching applied mathematics were combined with hybrid and mixed forms of training.

For example, in the course of learning applied mathematics, a classical model called «travelling salesman» with the purpose to search for the most advantageous closed walk going through a net of given points (locations) [8, p. 438].

Students are offered to study the lecture on the topic "Branch and Bound Method" on their own, the lecture is presented in a Moodle-based e-learning course. Students must go over the lecture material and review the solution to the problem implementing the travelling salesman model. Let us provide an example of methodology for solving applied problems.

Problem 1. *A bus tour route over Donetsk is organized. The tour must start at the "Donbass Palace" hotel. The main points are: "Nemo" dolphinarium, Donetsk Forged Figures Park, planetarium, botanical garden, "Donbass Arena" stadium. Plan the tour route with the shortest bus path length [9].*

In order to implement this model we use two algorithms of branch and bound method:

- a) Algorithm 1 — binary partition;
- b) Algorithm 2 — random partition.

After the problem analysis it is suggested that students answer the following comprehension questions:

- 1) define "combinatorial optimization";
- 2) define "the upper bound of the problem";
- 3) define "the lower bound of the subset";
- 4) continue the following statement: "if the lower bound for a subset is greater than the upper bound, then ...";
- 5) explain how a smaller distance matrix is built;
- 6) describe the stages of lower bound building;
- 7) characterize "branch and bound" mathematical model;
- 8) explain the algorithm of choosing the starting point;
- 9) explain the difference in using binary partition and random partition in branch and bound method;

10) describe the stages of building a route tree in branch and bound method;

11) provide the rationale for the applicability of the travelling salesman model in your specialist field.

12) analyze the stages of implementing the breach and bound method in logistics and highway transport management.

Further, students are suggested to perform the following activities:

- study the lecture material on this topic;
- answer the questions;
- make the list of problematic questions which need discussing during a practical session in class.

The class session starts with the problem statement: *"How do we choose the most advantageous closed walk going through a net of given points?"* Many students immediately suggest using Google Maps (Yandex Karty) with intuitive interface to specify the starting, the ending, and the intermediate points and also providing automated search of the optimal route. Some students suggest identifying the distance matrix as an array of data with a subsequent processing of the matrix as an array of data. Speculations and discussion that follow go to show that the future engineers are making active use of the theoretical material to create model situations.

Continuing the heuristic dialogue with the students we pose another problem: *"Can you obtain a large distance matrix (big data)?"* Many students reply that it will increase labour intensity even from the standpoint of performance, brute forcing all possible closed walks with further specifying the optimal (shortest) route. Some students who put forward this technology agree with the difficulty of big data processing.

The teacher gives various examples (for instance, with the use of the dichotomy method and the chord method) where the number of iterations can be drastically reduced (in iterative methods). Based on the discussion and the examples presented the teacher leads the students to the idea of using the branch and bound method, the essence of which is determining the upper bound, the lower bounds for subsets with the follow-up application of the statement: *if the lower bound on the subset of the initial set is greater than the upper bound, the solution lies in the opposite subset.*

Thus, a precise algorithm of operations (calculations) is formed in the student's mind. We present the implementation of the algorithm in "Travelling Salesman Model" laboratory work.

Any valid route can be taken as the upper bound. To calculate the lower bounds we suggest using Lower Bound Calculator (Fig. 1).

Students start performing the laboratory work applying the breach and bound method to combinatorial models.

As a result of the laboratory work students obtain the result as a solution tree (Fig. 2).

There are no more interrupted branches left (all of them have been crossed out), so the route obtained is the shortest.

The main objective of the "Travelling Salesman Model" laboratory work is not obtaining a final solution in the first place, but consistent work on applying the branch and bound method to combinatorial models.

To self-check the performed task, the student is given the macro of the automatic search of the optimal route.

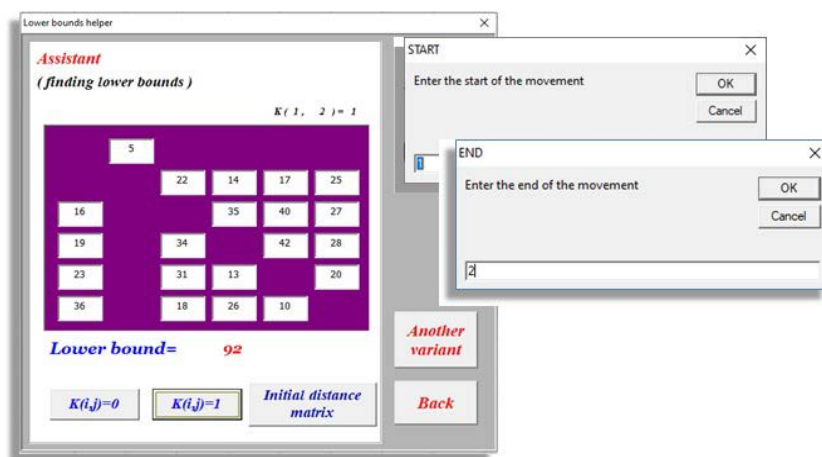


Fig. 1. Lower Bound Calculator

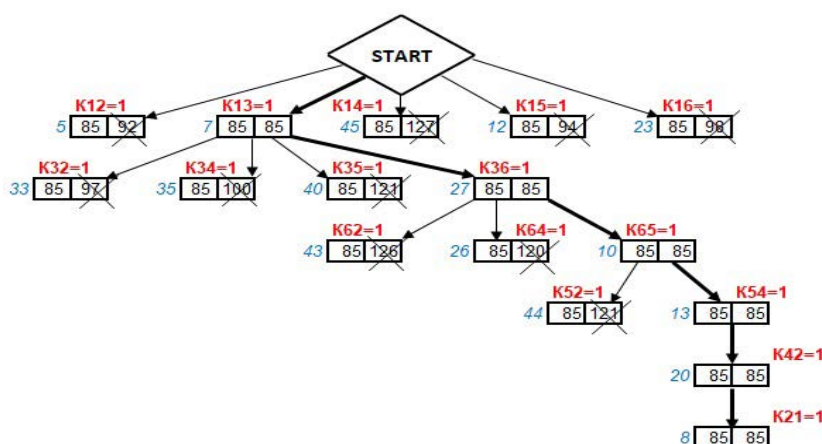


Fig. 2. Search Tree for the Most Optimal and Shortest Route

Besides, in experimental groups virtual laboratory works were constantly carried out, some of these sessions were described in the articles [6, 10, 11]. The tool to perform these works was the virtual laboratory complex.

An important factor of assuring the quality of studying the course of applied mathematics is advisably arranged student independent work (SIW). According to O. L. Prokhorova, SIW provides consistent gain of knowledge and its qualitative complication, mastering rational methods and techniques of absorbing new ways of educational activity [12].

In the course of independent work with the use of digital technologies student activity becomes more deliberate being characterized by concentration, velocity and accuracy, completeness and meaningfulness of mathematical model building process.

Each student fills Textbox in Form of the dialogue mode of "Teacher-Student" automated working space in order to receive an individual task for independent work (Fig. 3).

SIW is managed with the help of "Teacher-Student" AWS, which allows students to go

Task condition (block 0)

	B1	B2	B3	B4	ai
A1	1	2	1	2	10
A2	5	8	7	6	12
A3	6	4	2	5	8
A4	9	8	5	4	10
A5	0	0	0	0	
b _j	20	10	17	18	

Basic plan (block 3)

	B1	B2	B3	B4	ai
A1	10				10
A2					12
A3			8		8
A4					10
A5					
b _j	20	10	17	18	

Buttons: Exit, Another variant, Print all, Calculator, Examination, Print, Further

Fig. 3. Receiving SIW Form and Step-by-step Implementation of Algorithm Scheme

through each step of the model algorithm interactively.

In the end, the program reports the correctness of task performance or current mistakes, filling the Textbox background green and red respectively (Fig. 4).

Determination of potentials (block 4)

Transportation cost = 128

	B1	B2	B3	B4	ui
A1	10				0
A2	10				-4
A3			8		6
A4					-2
A5					2
v _j	1	2	2	2	

Buttons: Print theory, Examination, Printing (detailed), Print (briefly), Exit, Another variant, Print all, Calculator

Improvement of the plan (blocks 7, 8)

Cell $\lambda = (1, 3)$

Volume $\lambda = 2$

Transportation cost = 126

	B1	B2	B3	B4	ui
A1					0
A2					-4
A3					0
A4					-2
A5					2
v _j	1	2	2	2	

Buttons: Examination, Printing (detailed), Print (briefly), Further

Fig. 4. Colour Prompts of SIW Management System

We have introduced learning material acquisition tests into the management system of the student independent work. They are distributed as follows:

- basic test taken in class (occasional selective);
- self-check (self-control), performed by the student individually (with the use of AWS);
- written reproduction of the material;
- training exercises and check points.

Due to the self-check with the use of AWS, the management of the student independent work within the course of applied mathematics takes on the form of cognitive, task-oriented, and creative activity.

At the end of the course of applied mathematics we organized a test work to determine the *mathematical and digital competency* proficiency level in the control group (CG) and the experimental group (EG) with the total number of 110 students (CG – 56 ppl., EG – 54 ppl.).

The tasks for the test work were picked from the study guide for higher vocational education institution students titled "Applied Aspects of Mathematics" [9].

At this stage of the experiment, with the maximum score of 20, transitioning from the ratio scale measurement to the ordinal scale measurement, the following scale of levels of the mathematical and digital competency (MDC) was used: low (with *the score* ≤ 5); middle ($5 < \text{the score} \leq 14$); high ($14 < \text{the score} \leq 20$). The results of the test work from the ratio scale to the ordinal scale are shown in Tab. 1.

Table 1. Student Distribution by Academic Achievement Levels in EG and CG after Completing Test Work.

MDC levels	Number of students CG, ppl.	Number of students EG, ppl.
Low	8	5
Middle	31	22
High	15	29
Total:	54	56

The visualization of the test work results as a percentage of the total number of students of each group is given in Fig. 5.

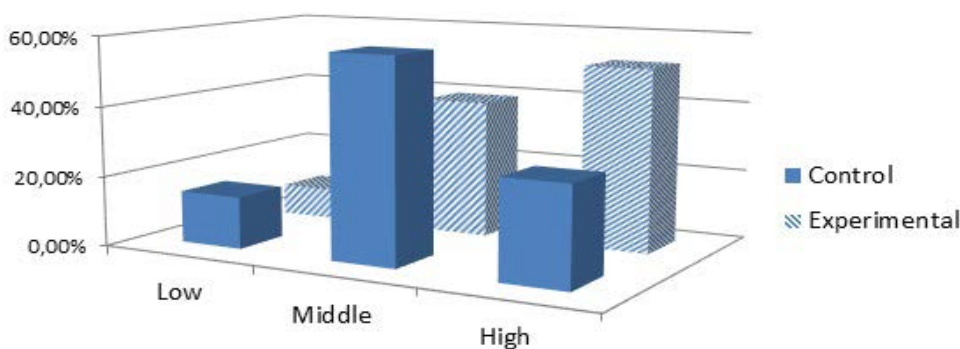


Fig. 5. Results of Applied Mathematics Final Test Work in EG and CG (level of mathematical and digital competency, %)

At the end of the experiment, the control and experimental groups were offered a questionnaire that characterizes the attitude of engineering students to the need to study mathematical modeling for future professional activities.

The experimental group students' replies were more conscious.

For instance, to the question "*Express your attitude towards the necessity to study applied mathematics for a future engineer*" 56% of the students of the experimental group replied: "I am

happy to study applied mathematics and mathematical modeling, because it is necessary for my future engineering activities".

To the question: *"In your opinion, does a future engineer need to master the methods of mathematical modeling?"* 72% of the students of the experimental group gave a positive answer, the students of the control group mostly (65%) gave the following answer: "No, because there are modern computer programs that allow modeling complex systems".

To the question *"Is the level of your knowledge and skills in mathematical modeling sufficient to create professional models?"* all respondents studying in both control and experimental groups gave similar answers: "Rather no than yes" — answered the students of the control group; "Rather yes than no" — replied the students of the experimental group. These answers make it possible to conclude that the approach to teaching mathematical modeling should be multifaceted; using the example of just one course, it is impossible to form the concepts of mathematical models effectively and learn how to deal with them.

The question *"Do you know how mathematical models of technical objects, phenomena, and processes are created?"* was answered negatively by the majority of students in both groups. This answer goes to show that realizing the importance of studying the methods of mathematical modeling it is necessary to utilize not only applied mathematics, but also various professional subjects to expand the essence of modeling, to familiarize with virtual models using the examples of engineering processes, to use mathematical and computer modeling in researching while working upon a graduation thesis.

"Can the computer be applied to solve problems with the use of mathematical modeling?" — a unanimous positive answer was received from the experimental group students, as they all built models based on computer experiments when studying applied mathematics in integrated and virtual laboratory works.

Thus, the student questionnaire to identify their attitude to the need to study mathematical modeling for its use in future professional activities showed that the effect of changes is caused precisely by the use of experimental teaching methods.

Based on the results of the experiment, a conclusion was made about the effectiveness of the author-developed methodology for teaching mathematical modeling to engineering students in the course of applied mathematics aimed at developing students' mathematical digital competency.

3. Discussion

As engineering education researchers point out, teaching future engineers must be aimed at their mastering both mathematical and digital competencies necessary for designing and solving technical systems modeling problems [13–16].

Applied mathematics is one of the subjects in which it is possible to combine the processes of studying mathematical and computer modeling. We agree with the authors of applied mathematics educational manuals who state the necessity of studying the methods of mathematical and computer modeling to develop professional competency of students [17–20]. At the same time, there have been many articles that characterize the importance of implementing high-end training aids [21], virtual laboratory works [22], computer simulators [23, 24], designed with the help of digital tools. However, the multifaceted approach to teaching applied mathematics designed for maintaining the process of teaching mathematical and computer modeling in various forms of classes and student independent work is poorly presented in research and methodological literature.

The experiment carried out at Gorlovka Automobile and Highway Institute of Donetsk National Technical University has shown the effectiveness of the represented teaching system. As a result, students' mathematical and digital competency is developed, which corresponds to the idea of the engineer of the new technological paradigm.

Conclusion

In our opinion, the course of applied mathematics must be organized as a system of developing of mathematical digital competency of future engineers which provides for the integration of mathematical and computer modeling.

The main components of the multifaceted approach to developing mathematical digital competency of future engineers in the course of applied mathematics are:

- a system of professionally oriented tasks aimed at mastering the techniques of mathematical modeling;
- laboratory works based on students' mastering mathematical modeling operations with the use of ICT;
- virtual laboratory works that facilitate mastering imitation modeling and digital tools;
- a management system of the student independent work based on "Teacher – Student" automated working space.

The implementation of such a system into the process of teaching applied mathematics in engineering educational institutions will enable researching complex processes and phenomena in real time, using application program suites to perform engineering calculations when solving problems, as well as facilitating the development of student mathematical and digital competency.

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Комплексный подход к формированию математической цифровой компетентности будущих инженеров в обучении прикладной математике

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Аннотация. В статье рассматривается проблема формирования математической цифровой компетентности студентов технических направлений подготовки. Авторами предложен комплексный подход решения проблемы, реализуемый в обучении будущих инженеров дисциплине «Прикладная математика». Раскрыто понятие математической цифровой компетентности специалиста в области инженерии, формирование которой у студентов является целью обучения. Обосновано применение системы профессионально ориентированных задач по овладению приемами математического моделирования, компьютерных симуляторов, а также игровых имитационных моделей обучения прикладной математике на основе автоматизированного рабочего места «Преподаватель – студент», являющегося средством для достаточно быстрого овладения студентами технических направлений подготовки базовыми методами прикладной математики, информатики, элементами алгоритмизации и программирования. Описано содержание авторского виртуального лабораторного комплекса по дисциплине «Прикладная математика», разработанного на платформе AnyLogic, предназначенного для реализации имитационных моделей.

Ключевые слова: прикладная математика, математическая цифровая компетентность, будущие инженеры, математическое моделирование, цифровые средства обучения.

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On the Closedness of Carpets of Additive Subgroups Associated With a Chevalley Group Over a Commutative Ring

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Dedicated to the blessed memory of my teacher Vladimir M. Levchuk

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Abstract. Let $\mathfrak{A} = \{\mathfrak{A}_r \mid r \in \Phi\}$ be a carpet of additive subgroups of type Φ over an arbitrary commutative ring K . A sufficient condition for the carpet $\mathfrak{A}$ to be closed is established. As a corollary, we obtain a positive answer to question 19.63 from the Kourovka notebook and a confirmation of one conjecture by V. M. Levchuk, provided that the type of Φ is different from C_l , $l \geq 5$ when the characteristic of the ring K is 0 or $2m$ for some natural number $m > 1$. Also, a partial answer to question 19.62 has been obtained.

Keywords: Lie algebra and ring, Chevalley group, commutative ring, carpet of additive subgroups, carpet subgroup.

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1. Introduction and preliminaries

Let Φ be a reduced indecomposable root system, $E(\Phi, K)$ be an elementary Chevalley group of type Φ over the commutative ring K with unit 1. The groups $E(\Phi, K)$ is generated by the root subgroups $x_r(K) = \{x_r(t) \mid t \in K\}$, $r \in \Phi$. The subgroups $x_r(K)$ are Abelian, and

$$x_r(t)x_r(u) = x_r(t+u), \quad (1)$$

for any $r \in \Phi$ and $t, u \in K$. We call a *carpet of type Φ over K* a collection of additive subgroups $\mathfrak{A} = \{\mathfrak{A}_r \mid r \in \Phi\}$ of the field K with the condition

$$C_{ij,rs}\mathfrak{A}_r^i\mathfrak{A}_s^j \subseteq \mathfrak{A}_{ir+js}, \quad r, s, ir+js \in \Phi, \quad i, j > 0, \quad (2)$$

where $\mathfrak{A}_r^i = \{a^i \mid a \in \mathfrak{A}_r\}$, and constants $C_{ij,rs}$ are equal to ± 1 , ± 2 or ± 3 . Inclusions (2) come from the Chevalley commutator formula

$$[x_s(u), x_r(t)] = \prod_{i,j>0} x_{ir+js}(C_{ij,rs}(-t)^i u^j), \quad r, s, ir+js \in \Phi. \quad (3)$$

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Every carpet $\mathfrak{A}$ defines a *carpet subgroup* $E(\Phi, \mathfrak{A})$ generated by the subgroups $x_r(\mathfrak{A}_r)$, $r \in \Phi$. A carpet $\mathfrak{A}$ is called *closed* if its carpet subgroup $E(\Phi, \mathfrak{A})$ has no new root elements, i.e., if

$$E(\Phi, \mathfrak{A}) \cap x_r(K) = x_r(\mathfrak{A}_r).$$

Note that the following fact follows from the condition (2). If $t \in \mathfrak{A}_r$ and $u \in \mathfrak{A}_s$, then each factor from the right-hand side of (3) lies in $E(\Phi, \mathfrak{A})$.

The above definition of a carpet was introduced by V.M. Levchuk, and it was first written down in the next question from the Kourovka notebook [1].

Question A). *What are the conditions on the carpet $\mathfrak{A} = \{\mathfrak{A}_r \mid r \in \Phi\}$ (in terms of $\mathfrak{A}_r$) over a commutative ring K necessary and sufficient for $\mathfrak{A}$ to be closed? [1, question 7.28, 1980]*

The difficulty of this question lies in the fact that the answer to it, in fact, should be the following statement: the defining relations of the carpet subgroup $E(\Phi, \mathfrak{A})$ over an arbitrary commutative ring K are exhausted by the relations (1), (3) (which give rise to the carpet conditions (2)), and relations in the subgroup $\langle x_r(\mathfrak{A}_r), x_{-r}(\mathfrak{A}_{-r}), r \in \Phi$. In the case when the carpet subgroup $E(\Phi, \mathfrak{A})$ coincides with the entire Chevalley group $E(\Phi, K)$ over the field K , this is the well-known result of R. Steinberg on defining relations in Chevalley groups over the field [2, Sec. 6, Theorem 8] In 1983, the author of question A) himself gave the answer to it for a locally finite field K [3] and reformulated it in the case when the main ring K is a field, in the following form.

Question B). *Is it true that the carpet $\mathfrak{A} = \{\mathfrak{A}_r \mid r \in \Phi\}$ of type Φ over a field K is closed if and only if the subcarpets $\{\mathfrak{A}_r, \mathfrak{A}_{-r}\}$, $r \in \Phi$, of rank 1 are closed? [1, question 15.46, 2002]*

The carpet $\{\mathfrak{A}_r, \mathfrak{A}_{-r}\}$, $r \in \Phi$, of rank 1 corresponds to an elementary matrix carpet of degree 2, and its carpet subgroup is isomorphic to the group generated by opposite elementary transvections $t_{12}(u)$, $u \in \mathfrak{A}_r$ and $t_{21}(u)$, $u \in \mathfrak{A}_{-r}$.

Lemma 14 is noted in [4] without proof, however, after more than 40 years, its proof has not appeared. Therefore, we formulate the assertion of this lemma as a conjecture.

Conjecture C). (V.M. Levchuk, 1982) *Inclusions $\mathfrak{A}_r \mathfrak{A}_{-r} \mathfrak{A}_r \subseteq \mathfrak{A}_r$, $r \in \Phi$, are sufficient for the carpet of additive subgroups $\mathfrak{A} = \{\mathfrak{A}_r \mid r \in \Phi\}$ over a commutative ring K to be closed.*

The next question of the author of this note is a strengthening of conjecture C), and in the case when the characteristic of the main coefficient ring is odd, they are equivalent.

Question D). *Are the inclusions $\mathfrak{A}_r^2 \mathfrak{A}_{-r} \subseteq \mathfrak{A}_r$, $r \in \Phi$, sufficient for the carpet of additive subgroups $\mathfrak{A} = \{\mathfrak{A}_r \mid r \in \Phi\}$ over a commutative ring K to be closed? [1, Question 19.63]*

In the articles [5,6] give examples of irreducible (if all additive subgroups are nonzero) carpets of any type Φ over rings of any even characteristic, for which the inclusions $\mathfrak{A}_r^2 \mathfrak{A}_{-r} \subseteq \mathfrak{A}_r$, $r \in \Phi$, are valid, but the inclusions from hypothesis C) are not satisfied. Question D) was first written in 2012 in [7, p. 199] and this is due to the fact that the inclusions $\mathfrak{A}_r^2 \mathfrak{A}_{-r} \subseteq \mathfrak{A}_r$, $r \in \Phi$, are necessary and sufficient conditions for the carpet subring $L(\Phi, \mathfrak{A})$ to be invariant under the carpet subgroup $E(\Phi, \mathfrak{A})$ according to the following definitions.

Let Π be the fundamental root system for Φ . The structure constants in the Chevalley basis $\{e_r, r \in \Phi; h_s, s \in \Pi\}$ of the simple complex Lie algebra $L(\Phi, \mathbb{C})$ are integers, so one can define a Lie ring (algebra) $L(\Phi, K)$ with a Chevalley basis over an arbitrary commutative ring

K (see, for example, [8, p. 62]). By definition, we assume that the subring $L(\Phi, \mathfrak{A})$ is generated (with respect to both operations) by all the sets $\mathfrak{A}_r e_r$, $r \in \Phi$. We will call $L(\Phi, \mathfrak{A})$ a *carpet* Lie subring. Note that the basis elements e_r , h_s do not have to lie in $L(\Phi, \mathfrak{A})$. A carpet $\mathfrak{A}$ is called *L-closed* if $L(\Phi, \mathfrak{A}) \cap K e_r = \mathfrak{A}_r e_r$, $r \in \Phi$. The elementary Chevalley group $E(\Phi, K)$ acts on the Lie ring $L(\Phi, K)$ as an automorphism group. We will say that the subring $R \subseteq L(\Phi, K)$ is *invariant* under $G \subseteq E(\Phi, K)$ if $gr \in R$ for any $g \in G$ and $r \in R$. Any carpet subring $L(\Phi, \mathfrak{A})$ that is invariant under the corresponding carpet subgroup $E(\Phi, \mathfrak{A})$ is *L-closed* [7, p. 199].

For any $r \in \Phi$, the automorphism $x_r(t)$ of the Lie ring $L(\Phi, K)$ acts on the Chevalley basis as follows:

$$\begin{aligned} e_r &\rightarrow e_r, \\ e_{-r} &\rightarrow e_{-r} + t h_r - t^2 e_r, \\ h_s &\rightarrow h_s - t A_{sr} e_r, \quad s \in \Pi, \\ e_s &\rightarrow \sum_{i=0}^q C_{i1,rs} t^i e_{ir+s}, \quad s \in \Phi \setminus \{\pm r\}, \end{aligned} \tag{4}$$

where $A_{rs} = \frac{2(r,s)}{(r,r)}$, $q = q(r,s)$ is the largest non-negative integer such that $s + qr \in \Phi$ and by definition $C_{01,rs} = 1$. Therefore, if $s \in \Phi \setminus \{\pm r\}$ and $t \in \mathfrak{A}_r$, $u \in \mathfrak{A}_s$, then under the automorphism $x_r(t)$, the vector ue_s , goes into a linear combination $\sum_{i=0}^q C_{i1,rs} t^i u e_{ir+s}$, each term of which, by virtue of the carpet condition, lies in the subring $L(\Phi, \mathfrak{A})$.

The main result of the article is

Theorem 1. *The inclusions $\mathfrak{A}_r^2 \mathfrak{A}_{-r} \subseteq \mathfrak{A}_r$, $r \in \Phi$, are sufficient for closedness of the carpet of additive subgroups $\mathfrak{A} = \{\mathfrak{A}_r \mid r \in \Phi\}$ over an arbitrary commutative ring K provided that the type of Φ is different from C_l , $l \geq 5$, when the characteristic of the ring K is 0 or $2m$ for some natural number $m > 1$.*

2. Proof of Theorem 1

Let $x_p(t) \in E(\Phi, \mathfrak{A})$ for fixed $p \in \Phi$ and some $t \in K$. Then, for suitable $r_i \in \Phi$ and $t_i \in \mathfrak{A}_{r_i}$ we have the equality

$$x_p(t) = x_{r_1}(t_1) x_{r_2}(t_2) \dots x_{r_n}(t_n). \tag{5}$$

Our task is to establish the inclusion $t \in \mathfrak{A}_p$. By virtue of the third equality from (4), for any root $q \in \Phi$ the following equality holds

$$x_p(t) h_q = h_q - A_{qp} t e_p. \tag{6}$$

Consider the integer $E(\Phi, \mathfrak{A})$ -submodule M of the ring $L(\Phi, K)$ generated by the element h_q and show the inclusion

$$M \leq \mathbb{Z} h_q + L(\Phi, \mathfrak{A}). \tag{7}$$

For any $r \in \Phi$ we put $\mathfrak{H}_r = \{u \in K \mid u h_r \in M\}$. By virtue of (4), the the following inclusions hold:

$$\mathfrak{H}_q \leq \mathbb{Z} + \sum \mathfrak{A}_q \mathfrak{A}_{-q}, \tag{8}$$

$$\mathfrak{H}_s \leq \sum \mathfrak{A}_s \mathfrak{A}_{-s}, \quad s \neq q, \tag{9}$$

where by definition $\sum \mathfrak{A}_r \mathfrak{A}_{-r}$ consists of sums of the form $u_1 v_1 + \dots + u_k v_k$ for $u_i \in \mathfrak{A}_r$, $v_i \in \mathfrak{A}_{-r}$, $i = 1, \dots, k$, for any natural k . Indeed, the generation of the module M begins with the equality

$$x_r(\mathfrak{A}_r)h_q = h_q - A_{qr}\mathfrak{A}_r e_r.$$

Thus, there is a base of induction. By virtue of (4), we have the equalities:

$$x_r(\mathfrak{A}_r)\mathfrak{A}_s e_s = \sum_{i=0}^q C_{i1,rs} \mathfrak{A}_r^i \mathfrak{A}_s e_{ir+s}, \quad s \neq -r, \quad (10)$$

$$x_r(\mathfrak{A}_r)\mathfrak{A}_{-r} e_{-r} = \mathfrak{A}_{-r} e_{-r} + \mathfrak{A}_r \mathfrak{A}_{-r} h_r - \mathfrak{A}_r^2 \mathfrak{A}_{-r} e_r, \quad (11)$$

$$x_r(\mathfrak{A}_r)\mathfrak{H}_s h_s = \mathfrak{H}_s h_s - A_{sr}\mathfrak{A}_r \mathfrak{H}_s e_r. \quad (12)$$

Further, by the definition of the M , these equalities are applied in any order. By Theorem 3.1 in [7] the subring $L(\Phi, \mathfrak{A})$ is invariant under the carpet subgroup $E(\Phi, \mathfrak{A})$ if and only if $\mathfrak{A}_r^2 \mathfrak{A}_{-r} \subseteq \mathfrak{A}_r$, $r \in \Phi$. Therefore $Ke_r \cap L(\Phi, \mathfrak{A}) = \mathfrak{A}_r e_r$ for all $r \in \Phi$ [7, p. 199]. This gives us the inclusions (8) and (9), since $\mathfrak{A}_r \mathfrak{A}_{-r} h_r \in L(\Phi, \mathfrak{A})$, $r \in \Phi$, due to (11). Thus, the inclusion (7) also holds.

In what follows, we need one lemma, the assertion of which for systems all of whose roots have the same length is obvious. For a system of type F_4 it follows from the fact that any of its root lies in a subsystem of type A_2 . For root systems of types B_l , C_l and G_2 it can be verified directly.

Lemma 1. *For any root $p \in \Phi$ there exists a root $q \in \Phi$ such that*

$$A_{qp} = \begin{cases} \pm 2 & \text{if } \Phi \text{ of type } C_l, \quad l \geq 1, \quad \text{and } p \text{ is a long root,} \\ \pm 1 & \text{otherwise.} \end{cases} \quad (13)$$

So now, due to (5), (6) and (7) we have $A_{qp}t \in \mathfrak{A}_p$. Hence, by Lemma 1, we obtain the desired inclusion $t \in \mathfrak{A}_p$ if Φ is different from C_l , $l \geq 1$, in the case of an even characteristic of the ring K . The constraint " Φ is different from C_l , $l \geq 1$ " can be relaxed to " Φ is different from C_l , $l \geq 5$ " for an ring K of even characteristic due to the following equalities and inclusions for root systems $A_1 = C_1 \subset C_2 = B_2 \subset C_3 \subset C_4 \subset F_4$. Indeed, we embed a carpet $\mathfrak{A}$ of type C_l for $l = 1, 2, 3, 4$ into a carpet of type F_4 such that $\mathfrak{A}_r = 0$ for all $r \in F_4 \setminus C_l$. Since Lemma 1 is true for a root system of type F_4 , the above proof also holds in this case. We only note that in this case the root q in Lemma 1 must be taken from the difference $F_4 \setminus C_l$.

It remains to consider the case when Φ is of type C_l , $l \geq 5$, and the characteristic of the ring K is equal to 2. In this case, there exists a homomorphism φ of the group $E(C_l, K)$ into the group $E(B_l, K)$ such that

$$\varphi(x_r(u)) = \begin{cases} x_{r/2}(u) & \text{if } r \text{ is a long root,} \\ x_r(u^2) & \text{if } r \text{ is a short root,} \end{cases} \quad (14)$$

for any $u \in K$ (see, for example, [2, Theorem 28] and [9, Sec. 3]). Since the carpet conditions come from the Chevalley commutator formula, φ induces a homomorphism of the carpet subgroup $E(C_l, \mathfrak{A})$ onto the carpet subgroup $E(B_l, \mathfrak{A}')$, where the carpet $\mathfrak{A}' = \{\mathfrak{A}'_r \mid r \in B_l\}$ according to (14) is defined as

$$\mathfrak{A}'_r = \begin{cases} \mathfrak{A}_{2r} & \text{if } r \text{ is a short root of root system of type } B_l, \\ \mathfrak{A}_r^2 & \text{if } r \text{ is a long root of root system of type } B_l. \end{cases} \quad (15)$$

The correspondence (15) between additive subgroups $\mathfrak{A}'_r$ and $\mathfrak{A}_s$ is given by the bijection $p_i \leftrightarrow q_i$, $i = 1, 2, \dots, l$, between fundamental root systems of type C_l and B_l , where p_1 is a long root for type C_l , and q_1 is a short root for type B_l , and the sum $p_i + p_{i+1}$ and $q_i + q_{i+1}$ are roots for all $i = 1, 2, \dots, l-1$. The inclusions $\mathfrak{A}'^2_r \mathfrak{A}'_{-r} \subseteq \mathfrak{A}'_r$, $r \in B_l$, remain valid, since from $S^2 T \subseteq S$ follows $(S^2)^2 T^2 \subseteq S^2$ for any subsets S and T of the ring K . For Φ of type C_l , only the answer to the following question was not known above. Does the inclusion $x_p(t) \in E(C_l, \mathfrak{A})$, when p is a long root and the characteristic of the ring K is even, imply the inclusion $t \in \mathfrak{A}_p$? Since $\varphi(x_p(t)) = x_{p/2}(t)$ by definition, then $\mathfrak{A}'_{p/2} = \mathfrak{A}_p$. The inclusion $t \in \mathfrak{A}'_{p/2}$ is fulfilled, since we are dealing with a carpet of type B_l . Hence $t \in \mathfrak{A}_p$. Which is what needed to be shown.

The Theorem 1 is proved. $\square$

3. Corollaries from Theorem 1

Since the inclusions $\mathfrak{A}_r \mathfrak{A}_{-r} \mathfrak{A}_r \subseteq \mathfrak{A}_r$, $r \in \Phi$, entails the inclusions $\mathfrak{A}_r^2 \mathfrak{A}_{-r} \subseteq \mathfrak{A}_r$, $r \in \Phi$, then Theorem 1 implies

Corollary 1. *Conjecture C) is true if the type of Φ is different from C_l , $l \geq 5$, when the characteristic of the ring K is 0 or $2m$ for some natural numbers $m > 1$.*

Given an elementary carpet $\mathfrak{A} = \{\mathfrak{A}_r \mid r \in \Phi\}$ of type Φ of rank $l \geq 2$, we define a collection of additive subgroups $\mathfrak{B}_p = \sum C_{ij,rs} \mathfrak{A}_r^i \mathfrak{A}_s^j$, $p \in \Phi$, where the sum is over all natural numbers i, j and roots $r, s \in \Phi$ for which $ir + js = p$. The collection $\mathfrak{B} = \{\mathfrak{B}_p \mid p \in \Phi\}$ is a carpet and is called the *derived* of $\mathfrak{A}$ [10]. It is known that for $\Phi = A_l$ the carpet $\mathfrak{B}$ is closed [11, Proposition 1]. The following question was first written in [10, p. 534, 2011].

Question E). *Is every derived carpet of type Φ over a commutative ring closed?* [1, Question 19.62, 2018]

For each root system Φ we define the number $m = m(\Phi) = \max_{r,s \in \Phi} \frac{(r,r)}{(s,s)}$. In fact,

$$m = \begin{cases} 1 & \text{if } \Phi = A_l, D_l, E_l, \\ 2 & \text{if } \Phi = B_l, C_l, F_4, \\ 3 & \text{if } \Phi = G_2. \end{cases}$$

In [10, Theorem 1] it is proved that any derived carpet $\mathfrak{B}$ of type Φ of rank $l \geq 2$ satisfies the inclusions $m! \mathfrak{B}_p \mathfrak{B}_{-p} \mathfrak{B}_p \subseteq \mathfrak{B}_p$, $p \in \Phi$. Combining this result with the assertion of Theorem 1, we obtain

Corollary 2. *A derived carpet of type Φ of rank $l \geq 2$ over a commutative ring K of characteristic p is closed if $\gcd(p, 2) = 1$ for Φ of type B_l , C_l or F_4 , and if $\gcd(p, 6) = 1$ for Φ of type G_2 .*

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О замкнутости ковров аддитивных подгрупп, ассоциированных с группой Шевалле над коммутативным кольцом

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Аннотация. Установлено достаточное условие замкнутости ковра аддитивных подгрупп $\mathfrak{A} = \{\mathfrak{A}_r \mid r \in \Phi\}$ типа Φ над произвольным коммутативным кольцом K . В качестве следствий получаем положительный ответ на вопрос 19.63 из Коуровской тетради и подтверждение одной гипотезы В. М. Левчука при условии, что тип Φ отличен от C_l , $l \geq 5$, когда характеристика кольца K есть 0 или $2m$ для некоторого натурального числа $m > 1$. Также получен частичный ответ на вопрос 19.62.

Ключевые слова: алгебра и кольцо Ли, группа Шевалле, коммутативное кольцо, ковер аддитивных подгрупп, ковровая подгруппа.

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Mellin Transforms for Rational Functions with Quasi-elliptic Denominators

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Abstract. The paper deals with residue representations of n -dimensional Mellin transforms for rational functions with quasi-elliptic denominators. These representations are given by integrals over $(n - 1)$ -dimensional relative cycles. The amount of representations (or cycles) equals to the number of facets of the Newton polytope for the denominator of the rational function.

Keywords: multidimensional Mellin transform, quasi-elliptic polynomial, Leray residue form, amoeba.

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Introduction

The fundamental property of the Mellin transform, which largely determines the scope of its applications, is the correspondence between the asymptotic behavior of the original function $g(x)$ and the singularities of the transformed function. The role of this fundamental correspondence for the Mellin transform of a function of one variable is noted in numerous papers by F. Flajolet, in particular, in [4] in relation with the calculation of harmonic sums.

We recall that the Mellin transform of the function $g(x)$ is defined by the integral

$$M[g](z) = \int_{\mathbb{R}_+^n} g(x) x^{z-I} dx, \quad (1)$$

where the differential form

$$x^{z-I} dx := x_1^{z_1} \cdot \dots \cdot x_n^{z_n} \frac{dx_1}{x_1} \wedge \dots \wedge \frac{dx_n}{x_n}$$

acts as a kernel. Inversion formulae for multidimensional Mellin transforms and classes of holomorphic functions that can be translated into each other by direct and inverse Mellin transforms are studied in [1]. In this paper we deal with the Mellin transform of rational functions with

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quasi-elliptic denominators f . Due to the specifics of the kernel, it suffices to consider the transform of the function $g(x) = 1/f(x)$. The concept of a quasi-elliptic polynomial was introduced by T. Ermolaeva and A. Tsikh in [3].

So, we consider the following polynomial in n variables

$$f(x) = f(x_1, \dots, x_n) = \sum_{\alpha \in A} c_\alpha x^\alpha = \sum_{\alpha \in A} c_{\alpha_1 \dots \alpha_n} x_1^{\alpha_1} \dots x_n^{\alpha_n} \quad (2)$$

with coefficients $c_\alpha \in \mathbb{C} \setminus \{0\}$ and the support $A \subset \mathbb{Z}_{\geq}^n$.

Definition 1. A polynomial f is called to be quasi-elliptic if for any non-zero covector $a \in \mathbb{R}^{n*}$ its truncation f_a does not vanish in the torus $(\mathbb{R} \setminus 0)^n$.

Recall that the *truncation* of the polynomial f in the direction $a \in \mathbb{R}^{n*}$ is determined to be the polynomial

$$f_a = \sum_{\alpha \in \Delta^a} c_\alpha x^\alpha,$$

where Δ^a is the face of the Newton polytope of f in the direction a . The *Newton polytope* $\mathcal{N}_f$ of a polynomial f is defined to be the convex hull in $\mathbb{R}^n$ of the support A of f .

Meromorphic continuations of the Mellin transforms for rational functions with quasi-elliptic denominators were studied by L. Nilsson and M. Passare in [9], where the following representation

$$M[1/f](z) = \Phi(z) \prod_{k=1}^N \Gamma\left(\nu^{(k)} - \langle \mu^{(k)}, z \rangle\right), \quad (3)$$

was proved. Here $\Phi(z)$ is an entire function, vectors $\mu^{(k)} \in \mathbb{Z}^n$ are primitive and define outward normal directions to facets of the Newton polytope $\mathcal{N}_f$, and $\nu^{(k)} \in \mathbb{Z}$. The Newton polytope can be given by the system of inequalities

$$\mathcal{N}_f = \bigcap_{k=1}^N \left\{ u \in \mathbb{R}^n : \langle \mu^{(k)}, u \rangle \leq \nu^{(k)} \right\},$$

so each $\nu^{(k)}$ is interpreted as the weighted power of the polynomial f with respect to the corresponding weight $\mu^{(k)}$. As it follows from the results of [3], the Mellin transform $M[1/f](z)$ is a holomorphic function in the tube domain over the interior of the Newton polytope $\mathcal{N}_f$, and formula (3) reveals that its polar set is the finite set of families of parallel hyperplanes. In each family, the hyperplanes are obtained by shifting of some facet of the Newton polytope of the polynomial f .

This approach was generalized for Euler–Mellin integrals in [2], and also found application in the theory of Feynman integrals, see [7, 8].

In this paper, we present alternative representations for the Mellin transform of rational functions of the specified class. Note that we can define the quasi-ellipticity concept on the set $\mathbb{R}_+^n$ by assuming that truncations f_a do not vanish on it, because the $\mathbb{R}_+^n$, being a connected component of the real torus $(\mathbb{R} \setminus 0)^n$, is its subgroup under the operation of coordinatewise multiplication.

Theorem 1. Let us assume that the polynomial f is quasi-elliptic on $\mathbb{R}_+^n$. Then for each normal vector $\mu^{(k)}$ of the Newton polytope $\mathcal{N}_f$ there is a representation for the Mellin transform $M[1/f](z)$ of the following form

$$M_k(z) = e^{-i\pi \langle \mu^{(k)}, z \rangle} \Gamma(-\langle \mu^{(k)}, z \rangle) \Gamma(1 + \langle \mu^{(k)}, z \rangle) \Phi_k(z), \quad (4)$$

in which

$$\Phi_k(z) = v.p. \int_{V_k} \text{Res } \omega, \quad (5)$$

where $\text{Res } \omega$ is the Leray residue form of the integrand in (1), V_k is a surface of real dimension $n - 1$, and *v.p.* denotes the principal value with respect to the set of singular points of the V_k .

The function $\Phi_k(z)$ defined by the integral (5) is holomorphic in the tube domain $U_{[k]} + i\mathbb{R}^n$, where

$$U_{[k]} = \bigcap_{j \neq k} \left\{ u \in \mathbb{R}^n : \langle \mu^{(j)}, u \rangle < \nu^{(j)} \right\}.$$

1. Quasi-ellipticity and hypoellipticity

In this section, we characterize quasi-elliptic polynomials in more detail. First, note that the polytope $\mathcal{N}_f$ has only a finite number of faces, so the condition in Definition 1 needs to be verified only for a finite number of truncations f_a .

Following [3], we can single out two classes of polynomials that are quasi-elliptic in the sense of Definition 1. The first class consists of polynomials in which all monomials have positive coefficients and even powers α_i in each variable x_i . The second class consists of elliptic polynomials that do not vanish on $\mathbb{R}^n$. Recall that a polynomial f is called *elliptic* if its homogeneous polynomial of highest degree vanishes in $\mathbb{R}^n$ only at the point $x = 0$.

Let us consider a few examples.

1. The polynomial

$$f(x) = 1 + 2x_1 + 2x_2 + (x_1 - x_2)^2$$

is not quasi-elliptic in $\mathbb{R}_+^2$, because its truncation

$$f_{(1,1)} = (x_1 - x_2)^2$$

vanishes on the diagonal $x_1 = x_2$.

2. The polynomial $f = 1 - x_1 + x_1^2 - x_2 + x_2^2 - x_1x_2$ is quasi-elliptic in $\mathbb{R}^2$.

3. The polynomial $f = 1 + x_1 + x_2$ is quasi-elliptic in $\mathbb{R}_+^2$ but not quasi-elliptic in $\mathbb{R}^2$.

The quasi-ellipticity property is related to the concept of hypoellipticity. A polynomial f is said to be *hypoelliptic* if for any multi-index $\alpha \neq 0$ the derivative $f^{(\alpha)}(x)$ satisfies the condition

$$\frac{f^{(\alpha)}(x)}{f(x)} \rightarrow 0$$

for $\|x\| \rightarrow \infty$ [6]. The following sufficient test for hypoellipticity holds.

Theorem 2 (E. Zubchenkova). *If f is a quasi-elliptic polynomial and its Newton polytope is full, then f is hypoelliptic polynomial.*

Regarding the convergence of A-hypergeometric integrals, see articles [10] and [11]. The fullness of the polytope means that its projections on all coordinate planes belong to it. This condition in Theorem 2 is essential, as the following example confirms. The polynomial $f(x_1, x_2) = x_1^8 x_2^2 + x_1^4 + 1$ is quasi-elliptic, but its Newton polytope is not full. The hypoellipticity condition is not satisfied for it, since for $\alpha = (4, 0)$

$$\left. \frac{f^{(\alpha)}(x)}{f(x)} \right|_{x_1=0} = 24 \not\rightarrow 0.$$

We note also that the hypoellipticity condition does not imply the quasi-ellipticity one. For instance, the polynomial $f(x_1, x_2) = (x_1^2 - 1)^2 + x_2^4$ is elliptic, and therefore hypoelliptic, but it is not quasi-elliptic, since the truncation $f_{(0,-1)} = (x_1^2 - 1)^2$ has zeros in the torus $(\mathbb{R} \setminus 0)^2$.

Following [3], we now formulate the condition for the convergence of the integral of a rational function over $\mathbb{R}^n$ with a quasi-elliptic denominator .

Theorem 3 (Ermolaeva–Tsikh). *If Q is a quasi-elliptic polynomial non-vanishing in $\mathbb{R}^n$, then the integral*

$$\int_{\mathbb{R}^n} \frac{P(x_1, \dots, x_n)}{Q(x_1, \dots, x_n)} dx_1 \dots dx_n$$

is absolutely convergent if and only if

$$I + \mathcal{N}_P \subset (\mathcal{N}_Q)^\circ,$$

that is, the translation of $\mathcal{N}_P$ by $I = (1, \dots, 1) \in \mathbb{R}^n$ lies in the interior $(\mathcal{N}_Q)^\circ$ of $\mathcal{N}_Q$.

Regarding the convergence of A-hypergeometric integrals, see articles [10] and [11].

2. Sets V_k

The orthant $\mathbb{R}_+^n$ is a group with respect to the operation of coordinatewise multiplication. This is a connected component of the real torus $(\mathbb{R} \setminus 0)^n$. Any torus $(\mathbb{R} \setminus 0)^n$ automorphism, as well as an automorphism of the $\mathbb{R}_+^n$, is defined by a monomial transformation

$$y \rightarrow x = y^\eta = (y^{\eta_1}, \dots, y^{\eta_n}),$$

where $\eta_1, \dots, \eta_n$ are rows of some integer unimodular matrix η ($\det \eta = \pm 1$). The automorphism allows to integrate over $\mathbb{R}_+^n$ with fibers on shifts of one-parameter subgroups in $\mathbb{R}_+^n$.

Let us define the construction of sets V_k . For each outward normal $\mu^{(k)}$ of the Newton polytope $\mathcal{N}_f$ of the polynomial (2), we define a one-parameter subgroup

$$\gamma^{(k)} = \left\{ y_1^{\mu^{(k)}} := (y_1^{\mu_1^{(k)}}, \dots, y_n^{\mu_n^{(k)}}) \in \mathbb{R}_+^n : y_1 \in \mathbb{R}_+ \right\}.$$

Next, we foliate the orthant $\mathbb{R}_+^n$ into shifts (cosets with respect to the subgroup $\gamma^{(k)}$) as follows

$$c \odot \gamma^{(k)} = (c_1 y_1^{\mu_1^{(k)}}, \dots, c_n y_n^{\mu_n^{(k)}}).$$

The set of all shifts can be given as $c = (y')^{\eta'}$, where $y' := (y_2, \dots, y_n)$, η' is an integer $(n \times (n-1))$ -matrix such that $\eta := (\mu^{(k)}, \eta')$ is a unimodular $(n \times n)$ -matrix. The existence of such a matrix is ensured by the condition the vector $\mu^{(k)}$ to be primitive [13, Prop. 4.2.13].

Consider a section of the complex hypersurface $V := \{x \in \mathbb{C}^n : f(x_1, \dots, x_n) = 0\}$ by a family of shifts of the subgroup $\gamma^{(k)}$. As a result, we get the set

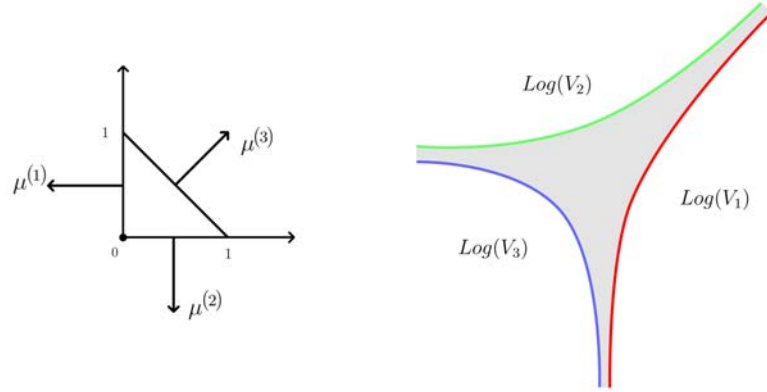
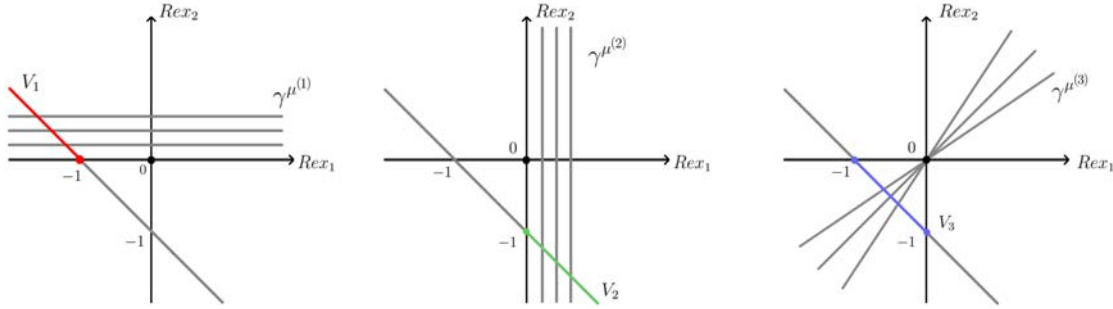
$$V_k = \bigcup_{y' \in \mathbb{R}_+^{n-1}} \left(V \cap \left\{ x = y_1^{\mu^{(k)}} \odot (y')^{\eta'^T} \right\} \right),$$

of the real dimension $n-1$. This observation allows us to apply Fubini's theorem doing integration over arbitrary one-parameter fibers.

Let us describe this construction using the example of a complex hyperplane

$$V = \left\{ x \in (\mathbb{C} \setminus 0)^2 : 1 + x_1 + x_2 = 0 \right\}.$$

The Newton polytope of the defining polynomial is a triangle, it has outward normals $\mu^{(1)} = (-1, 0)$, $\mu^{(2)} = (0, -1)$, $\mu^{(3)} = (1, 1)$ (Fig. 1 on the left). Fig. 2 shows the real part of V and its sections by shifts of the one-parameter subgroups $\gamma^{\mu^{(1)}}$ (red ray), $\gamma^{\mu^{(2)}}$ (green ray) and $\gamma^{\mu^{(3)}}$ (blue segment). These are the sets V_1 , V_2 , V_3 respectively. Their logarithmic images

Fig. 1. Newton polytope (left) and amoeba $\mathcal{A}_V$ (right) for $f = 1 + x_1 + x_2$ Fig. 2. Sets V_1, V_2, V_3

are connected components of the contour of the amoeba $\mathcal{A}_V$ of the hyperplane V (see Fig. 1 on the right). Recall that the amoeba of an algebraic hypersurface V is defined to be its image under the mapping

$$\text{Log} : (x_1, \dots, x_n) \rightarrow (\log|x_1|, \dots, \log|x_n|),$$

see, for example, [5]. The contour of the amoeba is determined as the set of critical values of the specified projection $\text{Log}|_V$, i.e. the set $\text{Log}(\gamma^{-1}(\mathbb{RP}^{n-1}))$, where $\gamma : V \rightarrow \mathbb{CP}^{n-1}$ is the logarithmic Gauss mapping [12].

3. Proof of Theorem 1

Consider a polynomial (2) that has no multiple irreducible factors, i.e. $df \neq 0$ on each irreducible component of V . According to (1), the Mellin transform of the function $1/f$ is expressed by the integral

$$M[1/f](z) = \int_{\mathbb{R}_+^n} \frac{x^{z-I}}{f(x)} dx.$$

Fix a normal vector $\mu^{(k)} = (\mu_1^{(k)}, \dots, \mu_n^{(k)})$, $k = 1, \dots, N$, of the polytope $\mathcal{N}_f$. Let us

construct an integer unimodular matrix η in which the vector $\mu^{(k)}$ is the first column:

$$\eta = \begin{pmatrix} \eta^{(1)} & \eta^{(2)} & \dots & \eta^{(n)} \\ \mu_1^{(k)} & \eta_1^{(2)} & \dots & \eta_1^{(n)} \\ \dots & \dots & \dots & \dots \\ \mu_n^{(k)} & \eta_n^{(2)} & \dots & \eta_n^{(n)} \end{pmatrix}.$$

The monomial transform $x = y^\eta$ with coordinates

$$\begin{aligned} x_1 &= y_1^{\mu_1^{(k)}} y_2^{\eta_1^{(2)}} \dots y_n^{\eta_1^{(n)}}, \\ &\dots \\ x_n &= y_1^{\mu_n^{(k)}} y_2^{\eta_n^{(2)}} \dots y_n^{\eta_n^{(n)}} \end{aligned}$$

is an automorphism of $\mathbb{R}_+^n$ due to the unimodularity of the matrix η . Let us write in variables y the expression $\frac{dx}{x}$. The result looks as follows

$$\frac{dx}{x} = J \prod_{j=1}^n (y_1^{\mu_j^{(k)}} \dots y_n^{\eta_j^{(n)}})^{-1} dy_1 \wedge \dots \wedge dy_n,$$

where

$$J = \begin{vmatrix} \mu_1^{(k)} y_1^{\mu_1^{(k)}-1} \dots y_n^{\eta_1^{(n)}-1} & \eta_1^{(2)} y_1^{\mu_1^{(k)}} y_2^{\eta_2^{(2)}-1} \dots y_n^{\eta_1^{(n)}} & \dots & \eta_1^{(n)} y_1^{\mu_1^{(k)}} \dots y_n^{\eta_1^{(n)}-1} \\ \dots & \dots & \dots & \dots \\ \mu_n^{(k)} y_1^{\mu_n^{(k)}-1} \dots y_n^{\eta_n^{(n)}-1} & \eta_n^{(2)} y_1^{\mu_n^{(k)}} y_2^{\eta_2^{(2)}-1} \dots y_n^{\eta_n^{(n)}} & \dots & \eta_n^{(n)} y_1^{\mu_n^{(k)}} \dots y_n^{\eta_n^{(n)}-1} \end{vmatrix}.$$

is the Jacobian of the monomial mapping. Multiplying the j th row of the Jacobian by $\left(y_1^{\mu_j^{(k)}} y_2^{\eta_j^{(2)}} \dots y_n^{\eta_j^{(n)}}\right)^{-1}$, we get the representation

$$\frac{dx}{x} = \begin{vmatrix} \mu_1^{(k)} y_1^{-1} & \eta_1^{(2)} y_2^{-1} & \dots & \eta_1^{(n)} y_n^{-1} \\ \dots & \dots & \dots & \dots \\ \mu_n^{(k)} y_1^{-1} & \eta_n^{(2)} y_2^{-1} & \dots & \eta_n^{(n)} y_n^{-1} \end{vmatrix} dy_1 \wedge \dots \wedge dy_n.$$

Further, from the j -th column of the determinant we take out the multiplier y_j^{-1} . As a result, we get

$$\frac{dx}{x} = \det \eta \frac{dy}{y},$$

where $\det \eta = \pm 1$, i. e. the matrix η is unimodular.

As a result of the change of variables, the Mellin transform is represented by the integral

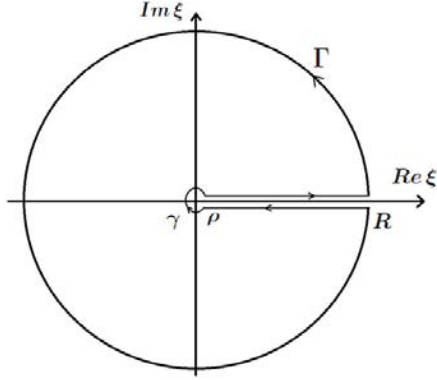
$$M[1/f(y^\eta)](z) = \det \eta \int_{\mathbb{R}_+^n} \frac{1}{f(y^\eta)} y_1^{\langle \mu^{(k)}, z \rangle} y_2^{\langle \eta^{(2)}, z \rangle} \dots y_n^{\langle \eta^{(n)}, z \rangle} \frac{dy}{y},$$

where

$$f(y^\eta) = \sum_{\alpha \in A} a_\alpha y_1^{\langle \mu^{(k)}, \alpha \rangle} y_2^{\langle \eta^{(2)}, \alpha \rangle} \dots y_n^{\langle \eta^{(n)}, \alpha \rangle}.$$

Remark that $\max_{\alpha \in A} \{\langle \mu^{(k)}, \alpha \rangle\} = \nu^{(k)}$, and this quantity is the degree of $f(y^\eta)$ over y_1 . Next, we integrate over y_1 for the fixed value $y' = (y_2, \dots, y_n) \in \mathbb{R}_+^{n-1}$:

$$M[1/f(y^\eta)](z) = \det \eta \int_{\mathbb{R}_+^{n-1}} y_2^{\langle \eta^{(2)}, z \rangle} \dots y_n^{\langle \eta^{(n)}, z \rangle} \frac{dy'}{y'} \int_0^{+\infty} \frac{y_1^{\langle \mu^{(k)}, z \rangle}}{f(y^\eta)} \frac{dy_1}{y_1}.$$

Fig. 3. The integration contour Γ

In order to calculate the inner integral over y_1 , we introduce the complex variable ξ , setting $y_1 = Re\xi$. Traditionally, we consider the integral over the contour Γ (see Fig. 3) of the following type

$$\int_{\Gamma} \frac{\xi^{\langle \mu^{(k)}, z \rangle}}{f((\xi, y')^\eta)} \frac{d\xi}{\xi} = \left(1 - e^{2\pi i(\langle \mu^{(k)}, z \rangle - 1)}\right) \int_{\rho}^R \frac{\xi^{\langle \mu^{(k)}, z \rangle - 1}}{f((\xi, y')^\eta)} d\xi + \int_{C_R} \frac{\xi^{\langle \mu^{(k)}, z \rangle - 1}}{f((\xi, y')^\eta)} d\xi + \int_{C_\rho} \frac{\xi^{\langle \mu^{(k)}, z \rangle - 1}}{f((\xi, y')^\eta)} d\xi.$$

Since the degree of $f(y^\eta)$ by y_1 equals $\nu^{(k)}$, then, by the residue total sum theorem, this integral vanishes if $\langle \mu^{(k)}, \text{Re}z \rangle < \nu^{(k)}$. Passing to the limit as $\rho \rightarrow 0$, $R \rightarrow \infty$ and applying the residue theorem, we obtain

$$\int_0^{+\infty} \frac{y_1^{\langle \mu^{(k)}, z \rangle}}{f(y^\eta)} \frac{dy_1}{y_1} = \frac{2\pi i}{1 - e^{2\pi i(\langle \mu^{(k)}, z \rangle - 1)}} \sum_j \frac{(\xi^j(y'))^{\langle \mu^{(k)}, z \rangle - 1}}{f'_{y_1}(\xi^j(y'), y')},$$

where $\xi^j(y')$ are roots of $f(y^\eta)$. Thus we get

$$\begin{aligned} M[1/f(y^\eta)] &= \frac{2\pi i}{1 - e^{2\pi i(\langle \mu^{(k)}, z \rangle - 1)}} \int_{\mathbb{R}_+^{n-1}} \sum_j \frac{(\xi^j(y'))^{\langle \mu^{(k)}, z \rangle - 1}}{f'_{y_1}(\xi^j(y'), y')} y_2^{\langle \eta^{(2)}, z \rangle} \dots y_n^{\langle \eta^{(n)}, z \rangle} \frac{dy'}{y'} = \\ &= \frac{2\pi i}{1 - e^{2\pi i(\langle \mu^{(k)}, z \rangle - 1)}} \int_{V_k} \text{Res} \left(\frac{x^{z-I}}{f(x)} dx \right). \end{aligned}$$

The factor before the integral can be rewritten as follows

$$\frac{2\pi i}{1 - e^{2\pi i(\langle \mu^{(k)}, z \rangle - 1)}} = \frac{\Gamma(-\langle \mu^{(k)}, z \rangle) \Gamma(1 + \langle \mu^{(k)}, z \rangle)}{e^{i\pi(\langle \mu^{(k)}, z \rangle)}}.$$

Thus, we obtain the first assertion of Theorem 1. The second one follows from the fact that the integral over V_k vanishes if $\langle \mu^{(k)}, \text{Re}z \rangle = \langle \mu^{(k)}, u \rangle < \nu^{(k)}$.

4. Examples

I. Consider the quasi-elliptic polynomial $f(x) = 2 - x_1 + x_1^2 - x_2 + x_2^2 - x_1x_2$ and the Mellin transform

$$M[1/f](z) = \int_{\mathbb{R}_+^2} \frac{x_1^{z_1} x_2^{z_2}}{2 - x_1 + x_1^2 - x_2 + x_2^2 - x_1 x_2} \frac{dx}{x}. \quad (6)$$

The Newton polytope of the polynomial $f(x)$ is given by three inequalities:

$$\mathcal{N}_f = \{-z_1 \leq 0\} \cap \{-z_2 \leq 0\} \cap \{z_1 + z_2 \leq 2\}.$$

It has outward normals $\mu^{(1)} = (-1, 0)$, $\mu^{(2)} = (0, -1)$, $\mu^{(3)} = (1, 1)$.

We consider the monomial change of variables

$$x_1 = t^{-1}, \quad x_2 = \tau^{-1},$$

associated with the vector $\mu^{(1)} = (-1, 0)$. As a result, the integral (6) will take the form:

$$M[1/f](z) = \int_{\mathbb{R}_+^2} \frac{t^{1-z_1} \tau^{1-z_2}}{2t^2\tau^2 - t\tau^2 + \tau^2 - t^2\tau + t^2 - t\tau} dt \wedge d\tau. \quad (7)$$

The denominator of the integrand in (7) has roots

$$t^{(1)} = \frac{\tau^2 + \tau + i\tau\sqrt{7\tau^2 - 6\tau + 3}}{2(2\tau^2 - \tau + 1)} \quad \text{and} \quad t^{(2)} = \frac{\tau^2 + \tau - i\tau\sqrt{7\tau^2 - 6\tau + 3}}{2(2\tau^2 - \tau + 1)}.$$

According to Theorem 1, the Mellin transform $M[1/f](z)$ admits the representation

$$M_1(z) = \Gamma(z_1)\Gamma(1 - z_1)e^{i\pi z_1} \int_{V_1} \text{Res } \omega,$$

where

$$V_1 = \{x_1 = 1/t^{(1)}, x_2 = 1/\tau, \tau \in [0; +\infty]\} \cup \{x_1 = 1/t^{(2)}, x_2 = 1/\tau, \tau \in [0; +\infty]\}.$$

Since the hypersurface $V = \{f(x) = 0\}$ is smooth, the principal value *v.p.* in the representation is omitted. Thus,

$$M_1(z) = \Gamma(z_1)\Gamma(1 - z_1)e^{i\pi z_1} 2^{z_1-1} \times \\ \times \int_0^\infty \frac{(\tau^2 + \tau + i\tau\sqrt{7\tau^2 - 6\tau + 3})^{1-z_1} - (\tau^2 + \tau - i\tau\sqrt{7\tau^2 - 6\tau + 3})^{1-z_1}}{\tau^{z_2} (2\tau^2 - \tau + 1)^{1-z_1} i\sqrt{7\tau^2 - 6\tau + 3}} d\tau. \quad (8)$$

Now let us define the domain of convergence of the integral in (8). At the origin, its convergence is ensured by the condition

$$u_1 + u_2 < 2,$$

where $u_1 = \text{Re} z_1$, $u_2 = \text{Re} z_2$. Next, we study the convergence in the neighborhood of the infinity using the substitution $\tau = 1/\lambda$. As a result, we obtain the integral

$$\int_0^\infty \frac{(1 + \lambda + i\sqrt{7 - 6\lambda + 3\lambda^2})^{1-z_1} - (1 + \lambda - i\sqrt{7 - 6\lambda + 3\lambda^2})^{1-z_1}}{\lambda^{1-z_2} (2 - \lambda + \lambda^2)^{1-z_1} i\sqrt{7 - 6\lambda + 3\lambda^2}} d\lambda.$$

The convergence of this integral at the origin is ensured by the condition $u_2 > 0$. Thus, the integral on the right side of (8) converges in the tube domain with the base $U_{[1]} = \{u \in \mathbb{R}^2 : u_1 + u_2 < 2, u_2 > 0\}$.

Next, consider the monomial change of coordinates

$$x_1 = \tau, \quad x_2 = t^{-1},$$

associated with the vector $\mu^{(2)} = (0, -1)$. As a result of this change of variables, the Mellin transform is expressed by the integral

$$M[1/f] = \int_{\mathbb{R}_+^2} \frac{t^{1-z_2} \tau^{z_1-1}}{2t^2 - \tau t^2 + \tau^2 t^2 - t + 1 - t\tau} dt \wedge d\tau,$$

in which the denominator of the integrand has roots:

$$t^{(1)} = \frac{\tau + 1 + i\sqrt{3\tau^2 - 6\tau + 7}}{2(\tau^2 - \tau + 2)}, \quad t^{(2)} = \frac{\tau + 1 - i\sqrt{3\tau^2 - 6\tau + 7}}{2(\tau^2 - \tau + 2)}.$$

According to Theorem 1, the Mellin transform $M[1/f](z)$ admits the representation

$$M_2(z) = \Gamma(z_2)\Gamma(1-z_2)e^{i\pi z_2} \int_{V_2} \text{Res } \omega,$$

where

$$V_2 = \{x_1 = \tau, x_2 = 1/t^{(1)}, \tau \in [0; +\infty]\} \cup \{x_1 = \tau, x_2 = 1/t^{(2)}, \tau \in [0; +\infty]\}.$$

Thus, we obtain the formula

$$M_2(z) = \Gamma(z_2)\Gamma(1-z_2)e^{i\pi z_2} 2^{z_2-1} \times \\ \times \int_0^\infty \frac{(\tau + 1 + i\sqrt{3\tau^2 - 6\tau + 7})^{1-z_2} - (\tau + 1 - i\sqrt{3\tau^2 - 6\tau + 7})^{1-z_2}}{\tau^{1-z_1} (\tau^2 - \tau + 2)^{1-z_2} i\sqrt{3\tau^2 - 6\tau + 7}} d\tau. \quad (9)$$

The integral on the right side of (9) converges in a tube domain with the base $U_{[2]} = \{u \in \mathbb{R}^2 : u_1 > 0, u_1 + u_2 < 2\}$.

Finally, consider the third normal $\mu^{(3)} = (1, 1)$ and the corresponding monomial mapping:

$$x_1 = t, \quad x_2 = \tau t.$$

The Mellin transform takes the form:

$$M[1/f](z) = \int_{\mathbb{R}_+^2} \frac{t^{z_1+z_2-1} \tau^{z_2-1}}{t^2 (\tau^2 - \tau + 1) + t(-1 - \tau) + 2} dt \wedge d\tau.$$

The denominator of the integrand in the resulting integral has two roots:

$$t^{(1)} = \frac{\tau + 1 + i\sqrt{7\tau^2 - 10\tau + 7}}{2(\tau^2 - \tau + 1)}, \quad t^{(2)} = \frac{\tau + 1 - i\sqrt{7\tau^2 - 10\tau + 7}}{2(\tau^2 - \tau + 1)}.$$

According to Theorem 1, the Mellin transform $M[1/f](z)$ admits the representation

$$M_3(z) = \Gamma(-z_1 - z_2)\Gamma(z_1 + z_2 + 1)e^{-i\pi(z_1+z_2)} \int_{V_3} \text{Res } \omega,$$

where

$$V_3 = \{x_1 = t^{(1)}, x_2 = \tau t^{(1)}, \tau \in [0; +\infty]\} \cup \{x_1 = t^{(2)}, x_2 = \tau t^{(2)}, \tau \in [0; +\infty]\}.$$

Calculating the residue, we get the representation

$$M_3(z) = \Gamma(-z_1 - z_2)\Gamma(z_1 + z_2 + 1)e^{-i\pi(z_1+z_2)} 2^{1-z_1-z_2} \times \\ \times \int_0^\infty \frac{(\tau + 1 + i\sqrt{7\tau^2 - 10\tau + 7})^{z_1+z_2-1} - (\tau + 1 - i\sqrt{7\tau^2 - 10\tau + 7})^{z_1+z_2-1}}{\tau^{1-z_2} (\tau^2 - \tau + 1)^{z_1+z_2-1} i\sqrt{7\tau^2 - 10\tau + 7}} d\tau. \quad (10)$$

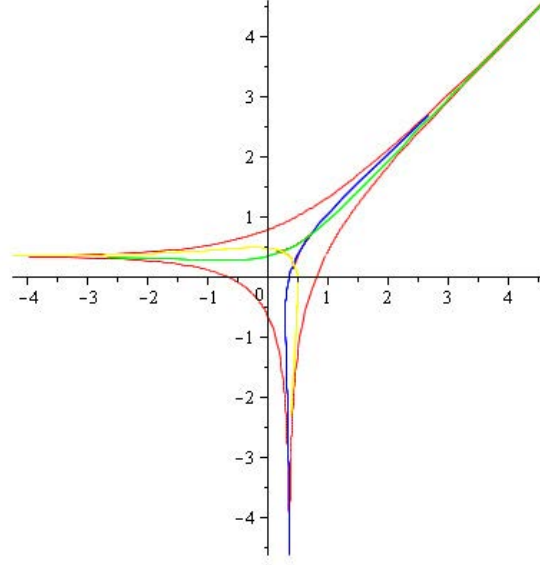


Fig. 4. Contour of the amoeba for $f(x) = 2 - x_1 + x_1^2 - x_2 + x_2^2 - x_1x_2$ and $\text{Log}(V_1), \text{Log}(V_2), \text{Log}(V_3)$

The integral in (10) converges in a tube domain with the base $U_{[3]} = \mathbb{R}_+^2$.

In Fig. 4, the logarithmic projections of the sets V_1 , V_2 , and V_3 are shown in blue, green, and yellow, respectively. The contour of the amoeba is highlighted in red.

II. Consider a quasi-elliptic polynomial $f(x) = 5 + x_1 + x_2 + x_1x_2$ and the Mellin transform:

$$M[1/f](z) = \int_{\mathbb{R}_+^2} \frac{x_1^{z_1} x_2^{z_2}}{5 + x_1 + x_2 + x_1x_2} \frac{dx}{x}. \quad (11)$$

The Newton polytope of the polynomial $f(x)$ is given by the inequalities

$$\mathcal{N}_f = \{-z_1 \leq 0\} \cap \{-z_2 \leq 0\} \cap \{z_1 \leq 1\} \cap \{z_2 \leq 1\}$$

and therefore has outward normals $\mu^{(1)} = (-1, 0)$, $\mu^{(2)} = (0, -1)$, $\mu^{(3)} = (1, 0)$, $\mu^{(4)} = (0, 1)$.

We consider the monomial change of variables

$$x_1 = t^{-1}, \quad x_2 = \tau^{-1},$$

associated with the vector $\mu^{(1)} = (-1, 0)$. The Mellin transform after the change of variables will take the form:

$$M[1/f](z) = \int_{\mathbb{R}_+^2} \frac{t^{-z_1} \tau^{-z_2}}{5t\tau + t + \tau + 1} dt \wedge d\tau.$$

According to Theorem 1, the Mellin transform $M[1/f](z)$ admits the representation

$$M_1(z) = \Gamma(z_1)\Gamma(1 - z_1)e^{i\pi z_1} \text{v.p.} \int_{V_1} \text{Res } \omega,$$

where

$$V_1 = \left\{ x_1 = -\frac{5\tau + 1}{\tau + 1}, x_2 = \tau^{-1}, \tau \in [0; +\infty] \right\}.$$

Calculating the residue, we get the following result:

$$M_1(z) = \Gamma(z_1)\Gamma(1 - z_1)e^{i\pi z_1} \int_0^\infty (-1)^{-z_1} \frac{\tau^{-z_2} (\tau + 1)^{-z_1}}{(5\tau + 1)^{1-z_1}} d\tau. \quad (12)$$

The integral in (12) converges in the domain $\{(z_1, z_2) \in \mathbb{C}^2 : 0 < \operatorname{Re} z_2 < 1\}$.

Next, consider the monomial change of coordinates

$$x_1 = \tau, \quad x_2 = t^{-1},$$

associated with the vector $\mu^{(2)} = (0, -1)$. As a result of this change of variables, the Mellin transform is expressed by the integral

$$M[1/f](z) = \int_{\mathbb{R}_+^2} \frac{\tau^{z_1-1} t^{-z_2}}{5t + \tau t + \tau + 1} dt \wedge d\tau.$$

According to Theorem 1, the Mellin transform $M[1/f](z)$ admits the representation

$$M_2(z) = \Gamma(z_2)\Gamma(1-z_2)e^{i\pi z_2} v.p. \int_{V_2} \operatorname{Res} \omega,$$

where

$$V_2 = \left\{ x_1 = \tau, x_2 = -\frac{\tau+5}{\tau+1}, \tau \in [0; +\infty] \right\}.$$

Calculating the residue, we obtain the result:

$$M_2(z) = \Gamma(z_2)\Gamma(1-z_2)e^{i\pi z_2} \int_0^\infty (-1)^{-z_2} \frac{\tau^{z_1-1}(\tau+1)^{-z_2}}{(5+\tau)^{1-z_2}} d\tau. \quad (13)$$

The integral in (13) converges in the domain $\{(z_1, z_2) \in \mathbb{C}^2 : 0 < \operatorname{Re} z_1 < 1\}$.

Further we consider the vector $\mu^{(3)} = (1, 0)$ and do the substitution

$$x_1 = t, \quad x_2 = \tau.$$

As a result, the Mellin transform takes the form

$$M[1/f](z) = \int_{\mathbb{R}_+^2} \frac{t^{z_1-1} \tau^{z_2-1}}{5+\tau+t+t\tau} dt \wedge d\tau.$$

According to Theorem 1, the Mellin transform $M[1/f](z)$ admits the representation

$$M_3(z) = \Gamma(-z_1)\Gamma(1+z_1)e^{-i\pi z_1} v.p. \int_{V_3} \operatorname{Res} \omega,$$

where

$$V_3 = \left\{ x_1 = \frac{-\tau-5}{\tau+1}, x_2 = \tau, \tau \in [0; +\infty] \right\}.$$

Calculating the residue, we get the representation

$$M_3(z) = \Gamma(-z_1)\Gamma(1+z_1)e^{-i\pi z_1} \int_0^\infty (-1)^{z_1-1} \frac{\tau^{z_2-1}(\tau+5)^{z_1-1}}{(1+\tau)^{z_1}} d\tau. \quad (14)$$

The integral in (14) converges in the domain $\{(z_1, z_2) \in \mathbb{C}^2 : 0 < \operatorname{Re} z_2 < 1\}$.

Finally, we consider the normal $\mu^{(4)} = (0, 1)$ and the corresponding monomial mapping:

$$x_1 = \tau^{-1}, \quad x_2 = t.$$

The Mellin transform after the change of variables is as follows:

$$M[1/f] = \int_{\mathbb{R}_+^2} \frac{\tau^{-z_1} t^{z_2-1}}{5\tau+1+t+t\tau} dt \wedge d\tau.$$

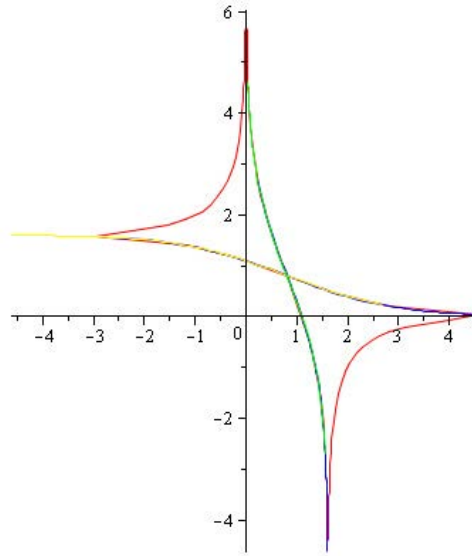


Fig. 5. Contour of the amoeba of $V = \{5 + x_1 + x_2 + x_1x_2 = 0\}$ and logarithmic projections of V_k

According to Theorem 1, the Mellin transform $M[1/f](z)$ admits the representation

$$M_4(z) = \Gamma(-z_2)\Gamma(1+z_2)e^{-i\pi z_2} \text{v.p.} \int_{V_4} \text{Res } \omega,$$

where

$$V_4 = \left\{ x_1 = \tau^{-1}, x_2 = \frac{-5\tau - 1}{\tau + 1}, \tau \in [0; +\infty] \right\}.$$

Calculating the residue, we get the representation

$$M_4(z) = \Gamma(-z_2)\Gamma(1+z_2)e^{-i\pi z_2} \int_0^\infty (-1)^{z_2-1} \frac{\tau^{-z_1}(5\tau+1)^{z_2-1}}{(1+\tau)^{z_2}} d\tau. \quad (15)$$

The integral in (15) converges in the domain $\{(z_1, z_2) \in \mathbb{C}^2 : 0 < \text{Re } z_1 < 1\}$. The contour of the amoeba of $V = \{5 + x_1 + x_2 + x_1x_2 = 0\}$ is shown in Fig. 5. The sets $\text{Log}(V_1)$ and $\text{Log}(V_3)$ coincide with the green part of the contour, and the sets $\text{Log}(V_2)$ and $\text{Log}(V_4)$ coincide with the yellow one.

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Преобразование Меллина для рациональных функций с квазиэллиптическими знаменателями

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Аннотация. В статье рассматриваются вычетные представления n -мерных преобразований Меллина для рациональных функций с квазиэллиптическим знаменателем. Эти представления задаются интегралами по $(n-1)$ -мерным относительным циклам. Количество представлений (или циклов) равно числу граней многогранника Ньютона знаменателя рациональной функции.

Ключевые слова: многомерное преобразование Меллина, квазиэллиптический полином, форма-вычет Лере, амёба.

EDN: HIOGSQ

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Manifestation of Slow Dynamics in Multilayer Nanostructures with Different Thicknesses of Magnetic Films

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Abstract. The results of a numerical Monte Carlo study of the behavior of a trilayer nanostructure with magnetization oriented in the film plane with ferromagnetic film thicknesses $N = 3$ and 21 MLs are presented. The non-equilibrium behavior of the structure has been investigated at the critical temperatures $T_c(N)$ and temperatures $T_c(N)/2$, $T_c(N)/4$, and the time dependence of the autocorrelation function during the evolution from different initial states and the dynamic susceptibility from the high-temperature initial state have been analyzed. Aging effects are revealed in the behavior of the two-temporal characteristics. The asymptotic values of the fluctuation-dissipative ratio for the structures with $N = 3$ and 21 MLs are calculated which indicate the violation of the fluctuation-dissipative theorem in the non-equilibrium behavior of these structures.

Keywords: Monte Carlo simulation, non-equilibrium behavior, anisotropic Heisenberg model, nanostructures, aging, fluctuation-dissipation ratio.

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The study of macroscopic statistical systems characterized by slow dynamics [1], as well as the study of the properties of ultrathin magnetic films [2] is currently of considerable interest. The increased interest in studying the behavior of ultrathin magnetic films is associated with the presence of unique their properties that differ significantly from those of bulk metals: sensitivity to the effects of anisotropy created by the crystal field of the substrate or nonmagnetic interlayers; the emergence of the effect of giant magnetoresistance; the manifestation of slow dynamics in a wider temperature range than in bulk materials.

It is known that multilayer magnetic structures are characterized by anomalously slow dynamics with the effects of aging and violation of the fluctuation-dissipative theorem predicted and observed in the slow evolution of systems from a non-equilibrium initial state [3, 4].

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The aim of this work is to simulate the non-equilibrium behavior of a trilayer nanostructure with different thicknesses of ferromagnetic films of $N = 3$ and 21 monolayers and magnetization oriented in the plane of the films.

1. Characteristics of non-equilibrium critical behaviour and results of simulations

The magnetic properties of ultrathin transition metal films in contact with nonmagnetic noble metal films are correctly described by the anisotropic Heisenberg model [5]:

$$H = -J_1 \sum_{\langle i,j \rangle}^{N_s^{(1,2)}} [S_i^x S_j^x + S_i^y S_j^y + (1 - \Delta(N)) S_i^z S_j^z] - J_2 \sum_{\langle i \in N_1, j \in N_2 \rangle}^{L^2} [S_i^x S_j^x + S_i^y S_j^y + (1 - \Delta(N)) S_i^z S_j^z], \quad (1)$$

where S_i^x , S_i^y , S_i^z are components of the three-dimensional unit spin vector $\vec{S}_i$ in the sites of the fcc lattice of the trilayer structure Co/Cu/Co; $N_s^{(1,2)} = N \times L \times L$ is a number of spins in film 1 and 2 with $L = 64$ as linear size of the film; $\Delta(N)$ is dimensionless anisotropy parameter depending on the thickness of the ferromagnetic film, in this work $\Delta(N = 3) = 0.665$, $\Delta(N = 21) = 0.02$; $J_1 = 1$ is integral of the exchange interaction between spins inside the film, $J_2 = -0.1J_1$ is the exchange integral of the interlayer interaction between spins on inner surface of films. The negativity of J_2 reflects the fact that the interlayer exchange interaction between the spins of ferromagnetic layers has an antiferromagnetic character with the magnetizations of neighboring layers oriented opposite to each other. The structure itself is schematically represented in Fig. 1.

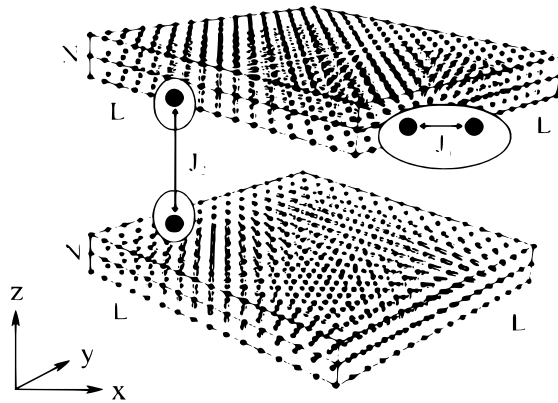


Fig. 1. Schematic representation of a multilayer nanostructure

The autocorrelation function $C(t, t_w)$ and the dynamical susceptibility $\chi(t, t_w)$ are considered as characteristics of nonequilibrium behavior:

- in this paper, the Metropolis algorithm was used to model the autocorrelation function

$$C(t, t_w) = \left\langle \frac{1}{N} \sum_{i=1}^N \vec{S}_i(t) \vec{S}_i(t_w) \right\rangle - \left\langle \frac{1}{N} \sum_{i=1}^N \vec{S}_i(t) \right\rangle \left\langle \frac{1}{N} \sum_{i=1}^N \vec{S}_i(t_w) \right\rangle; \quad (2)$$

- the Glauber dynamics algorithm was used to model the dynamic susceptibility

$$\chi(t, t_w) = \frac{1}{T_c N} \sum_{i=1}^N \langle \vec{S}_i(t) \Delta \vec{S}_i(t_w) \rangle, \quad (3)$$

$$\Delta \vec{S}_i(t_w) = \sum_{i=1}^{t_w} \vec{S}_i^{\parallel} \left(1 - \text{th} \left(\frac{1}{2T_c} \Delta H_i(t) \right) \right). \quad (4)$$

Statistical averaging, denoted in Eq. (2) and (3) by angle brackets, is implemented through averaging over runs with different initial states. Statistical averaging of the autocorrelation function $C(t, t_w)$ and dynamic susceptibility $\chi(t, t_w)$ was carried out on 3000 MC runs for every t_w .

In the course of this work, we numerically investigated the two-time dependence of the autocorrelation function $C(t, t_w)$ for different waiting times t_w during the evolution of the structure from different initial states at the critical temperature $T_c(N = 3) = 2.3918(65)$ (Fig.2). The plots show that when evolving from the low-temperature completely ordered initial state with $m_0 = 1$, the autocorrelation function is characterized by a faster decline and weakening of the influence of the waiting time t_w than when evolving from the high-temperature initial state with $m_0 \ll 1$ for all quenching temperatures $T_s = T_c, T_c/2, T_c/4$. Aging effects are revealed which are manifested not only at the critical temperature T_c , but also in the whole low-temperature phase that is expressed in the slowing down of the time decay of $C(t, t_w)$ with increasing t_w .

It is also worth noting that the autocorrelation function in the evolution from the low-temperature completely ordered initial state with $m_0 = 1$ is characterized by a slower decline with increasing quenching temperature T_s that is opposite to the behavior of $C(t, t_w)$ in the evolution of the system from the high-temperature initial state, where the decline is faster with increasing T_s .

One of the issues considered in this work was the influence of the thickness of ferromagnetic films N on the behavior of the autocorrelation function $C(t, t_w)$ and dynamic susceptibility $\chi(t, t_w)$. Analysis of the time dependencies of these characteristics (Fig. 3) shows that the autocorrelation function $C(t, t_w)$ and the dynamic susceptibility $\chi(t, t_w)$ have a stronger dimensional dependence on N at critical temperatures $T_c(N = 3) = 2.391(6)$, $T_c(N = 21) = 3.21(3)$ than in the low-temperature phase with $T_s < T_c$.

The autocorrelation function $C(t, t_w)$ and the dynamic susceptibility $\chi(t, t_w)$ at the critical temperature $T_c(N)$ are found to decrease faster with increasing film thickness N . That is, the aging effects are weakened with increasing film thickness N . This is due to the weakening of the critical correlations at the dimensional transition in the structures from films with quasi two-dimensional properties for $N = 3$ ML to films with three-dimensional bulk properties for $N = 21$ ML.

At temperatures below the critical temperature, the opposite tendency of strengthening the effects of coalescence with increasing film thickness is observed. This is due to the increase in the characteristic correlation length of transverse spin correlations with decreasing temperature, leading to an increase in the correlation time and relaxation of the structure.

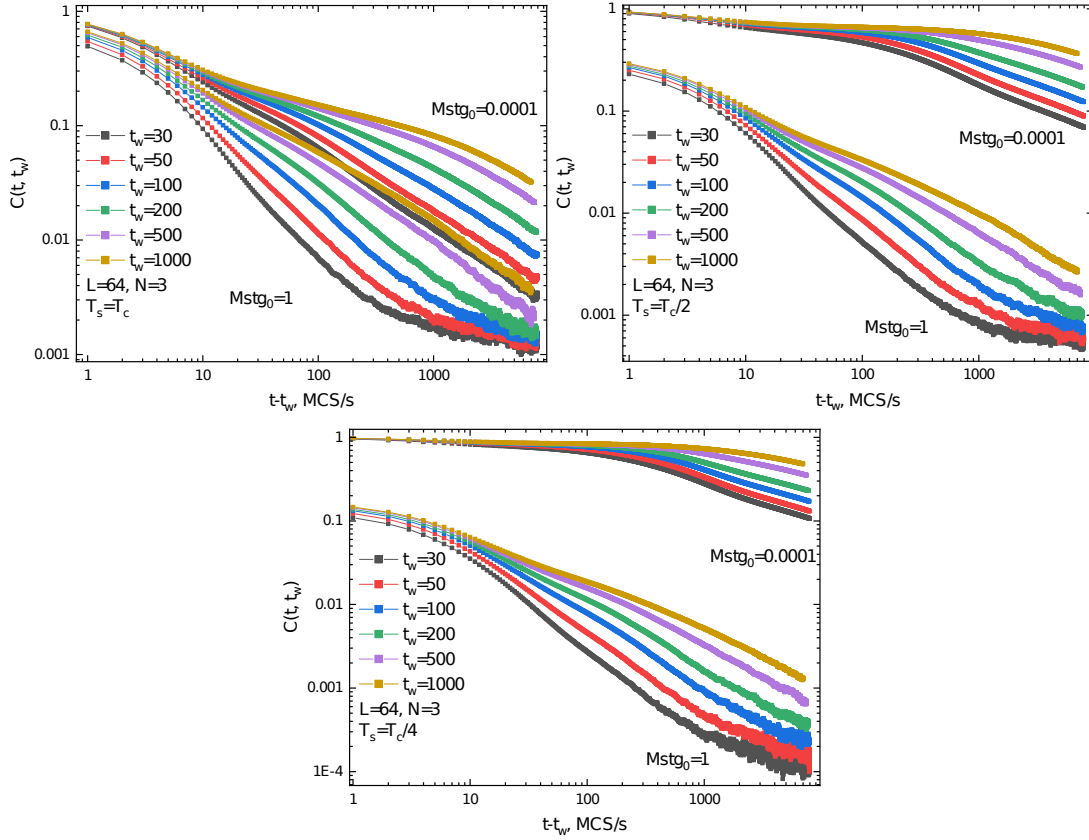


Fig. 2. Time dependence of $C(t, t_w)$ during the evolution of the structure with thickness $N = 3$ ML of ferromagnetic films from the low-temperature and high-temperature initial states for quenching temperatures $T_s = T_c, T_c/2, T_c/4$

For the first time, the calculation of the limiting fluctuation-dissipative ratio (FDR) X^∞ for multilayer nanostructures has been carried out. The FDR is an important universal characteristic of non-equilibrium processes in various systems with slow dynamics [6] and can be defined by the relation:

$$X^\infty = \lim_{t_w \rightarrow \infty} \lim_{t \rightarrow \infty} X(t, t_w) = \lim_{t_w \rightarrow \infty} \lim_{C \rightarrow \infty} T \frac{\partial \chi(t, t_w)}{\partial C(t, t_w)}. \quad (5)$$

Based on the calculated bivariate relationships for the autocorrelation function $C(t, t_w)$ and dynamic susceptibility $\chi(t, t_w)$ (Fig. 4), the asymptotic values of the FDR X^∞ were determined and presented in Tab. 1.

For structures with ferromagnetic film thicknesses $N = 3$ ML, the obtained asymptotic value of the FDR at the critical temperature T_c $X^\infty = 0.448(6)$ agrees very well with the value $X^\infty = 0.444(26)$ [7] for the 2d XY model at the Berezinskii–Kosterlitz–Towless phase transition temperature, and for structure with $N = 21$ ML $X^\infty = 0.385(4)$ agrees with the value $X^\infty = 0.383(6)$ [8] for the 3d isotropic Heisenberg model because of the small influence of anisotropy effects in these structures ($\Delta(N = 21) = 0.02$).

For temperatures $T_s < T_c$, the asymptotic values of the FDR exhibit much larger values than at the critical temperature T_c .

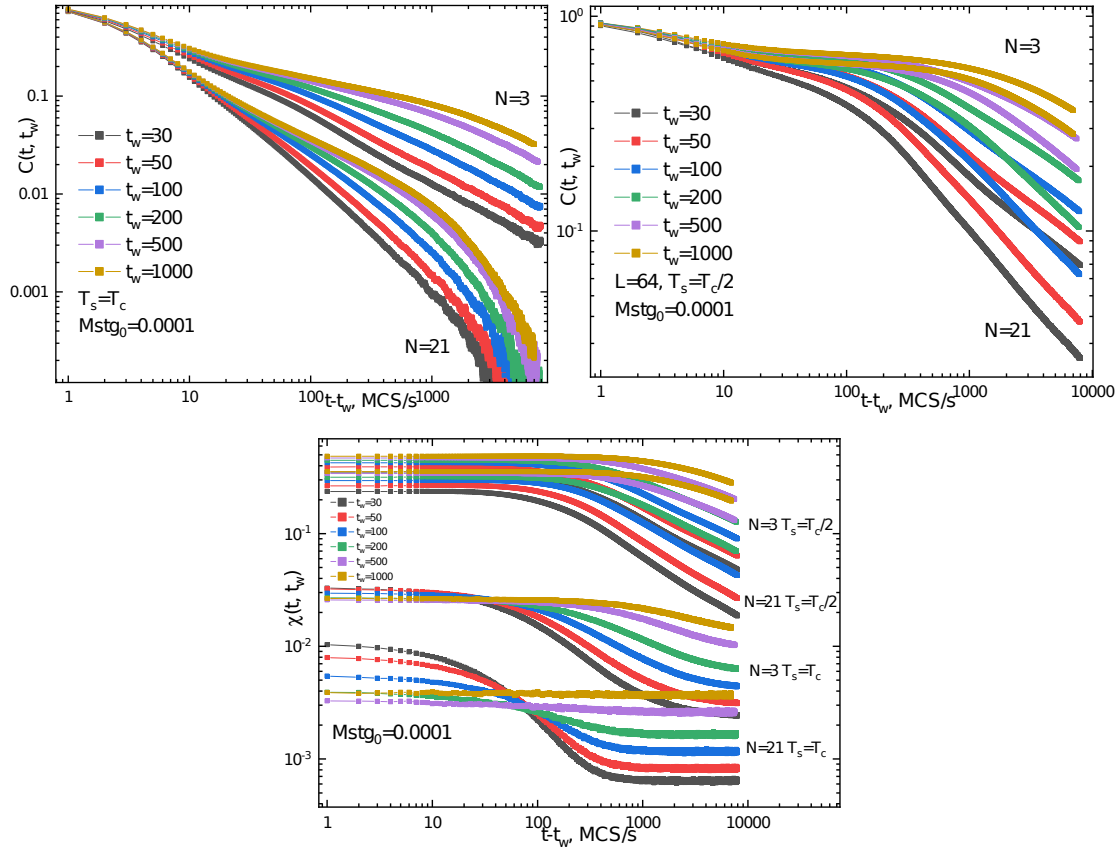


Fig. 3. Time dependence of $C(t, t_w)$ and $\chi(t, t_w)$ during the evolution of the structures from the high-temperature initial state for film thicknesses $N = 3, 21$ ML at quenching temperatures $T_s = T_c$ and $T_s = T_c/2$

Table 1. Asymptotical values of the FDR for different quenching temperatures T_s and film thicknesses $N = 3$ and 21 ML

T_s	$T_c(N)$	$T_c(N)/2$	$T_c(N)/4$
$X^\infty(N = 3)$	0.448(6)	0.909(1)	0.9672(7)
$X^\infty(N = 21)$	0.385(4)	0.850(1)	0.921(7)

Conclusions

In conclusion, we note that in the presented work we have carried out a numerical Monte Carlo study of the manifestation of slow dynamics features in the magnetic properties of a multilayer nanostructure consisting of two ferromagnetic films with thicknesses $N = 3, 21$ ML, separated by a film of nonmagnetic metal and connected by an antiferromagnetic exchange interaction. The anisotropic Heisenberg model with anisotropy of the "easy" plane type was applied to describe the properties of such a structure. The non-equilibrium behavior of the structures has been investigated not only at the critical temperature but also in the whole low-temperature phase. The aging effects have been revealed in the time dependence of the autocorrelation function and the dynamic susceptibility. The calculation of the limiting fluctuation-dissipative ratios

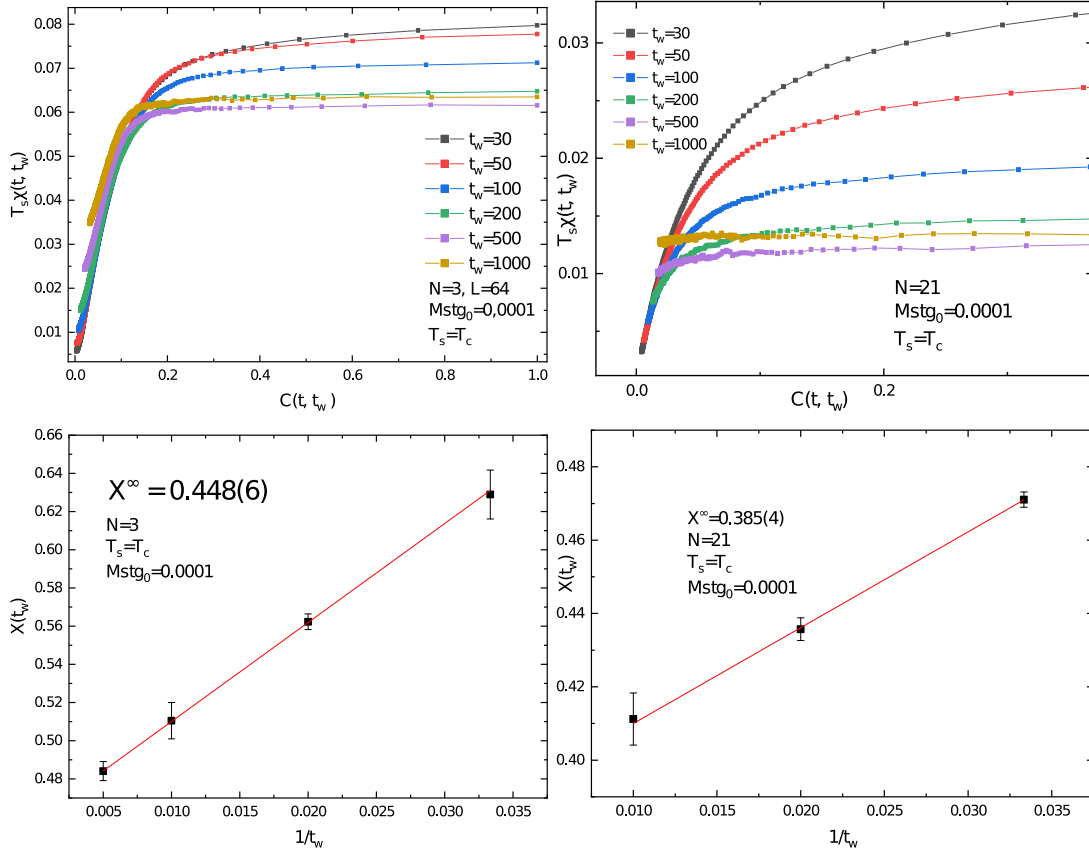


Fig. 4. Parametric dependence of $T_c \chi(t, t_w)$ on $C(t, t_w)$ and linear approximation of the FDR $X(t_w \rightarrow \infty)$ for different t_w to determine X^∞ for a film with $N = 3, 21$ at $T_s = T_c$

$X^\infty(N = 3, T_c) = 0.448(6)$ and $X^\infty(N = 21, T_c) = 0.383(6)$ for the case of structure evolution from the high-temperature initial state has been carried out. The obtained values of the FDR at the critical temperatures demonstrate the dimensional changes in non-equilibrium behavior of structures from quasi two-dimensional properties for $N = 3$ ML to three-dimensional bulk properties for $N = 21$ ML. It is revealed that the asymptotical values of the FDR lead to unity as the quenching temperature with $T_s < T_c(N)$ decreases, indicating a tendency for the non-equilibrium effects to decrease.

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Проявление медленной динамики в мультислойных наноструктурах с различными толщинами магнитных пленок

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Аннотация. Представлены результаты численного Монте-Карло исследования поведения трехслойной наноструктуры с толщиной ферромагнитной пленки $N = 3$ и 21 МС и намагниченностью, ориентированной в плоскости пленки. Исследовано неравновесное поведение структуры при критических температурах $T_c(N)$ и температурах $T_c(N)/2$, $T_c(N)/4$, а также временная зависимость автокорреляционной функции при эволюции системы из различных начальных состояний и динамической восприимчивости при эволюции из высокотемпературного начального состояния. Эффекты старения проявляются в поведении двухвременных характеристик. Рассчитаны асимптотические значения флуктуационно-диссипативного отношения для структур с $N = 3$ и 21 МС, которые указывают на нарушение флуктуационно-диссипативной теоремы при неравновесном поведении этих структур.

Ключевые слова: метод Монте-Карло, неравновесное критическое поведение, анизотропная модель Гейзенберга, наноструктуры, эффекты старения, флуктуационно-диссипативное отношение.

EDN: HRVUPS

УДК 517.55

On the Non-standard Interpolations in $\mathbb{C}^n$ and Combinatorial Coefficients for Weil Polyhedra

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Abstract. Multidimensional non-standard interpolation has been recently presented in an article by D. Alpay and A. Yger. We are talking about algebraic interpolation where discrete roots of a system of polynomial equations serve as nodes. With the help of the Grothendieck residue duality, the problem of describing the desired interpolation space of functions is reduced to solving the affine-bilinear equation. To implement this reduction, algorithms for calculating local Grothendieck residues or their sums are required. In a fairly general situation, the calculation of these residues is based on the well-known Gelfond–Khovanskii formula. This article provides examples of calculating local residues or their sums. In 2-dimensional case we generalise the Gelfond–Khovanskii formula for Newton polyhedra that are not in the unfolded position. This is done using the concept of an amoeba of an algebraic set and the notion of an homological resolvent for the boundary of Weil polyhedron.

Keywords: Grothendieck residue, interpolation, amoeba, Homological resolvent

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Introduction

By classical, or standard, interpolations we understand the Lagrange, Hermite, or Newton interpolations. Let us consider the first two of them.

Problem (Lagrange). *Given a set of distinct points $\{w_j\}_{j=1}^m \subset \mathbb{C}$ and the values $c_j \in \mathbb{C}$, find the polynomial $f(z)$ of degree $m - 1$ with the property*

$$f(w_j) = c_j, \quad j = 1, \dots, m.$$

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Note that the interpolation polynomial f is defined in terms of the polynomial $p(z) = (z - w_1) \cdot \dots \cdot (z - w_m)$ by the formula:

$$f(z) = p(z) \sum_{j=1}^m \frac{c_j}{z - w_j} \operatorname{res}_{w_j} \left(\frac{1}{p} \right).$$

Thus, specifying the interpolation nodes in the form of the null set of the polynomial p provides tools for constructing an interpolation polynomial by using residues. More general is the following

Problem (Hermite). *Let $\{w_j\}_{j=1}^m \subset \mathbb{C}$ be a set of pairwise distinct points and the following values are given*

$$c_{j,\ell} \in \mathbb{C}, \quad \text{where } j = 1, \dots, m, \quad \ell = 0, \dots, \mu_j - 1.$$

It is necessary to find a polynomial $f(z)$ of minimal degree, which at points w_j has the given values of derivatives up to orders of $\mu_j - 1$ including, that is

$$f^{(\ell)}(w_j) = c_{j,\ell}, \quad j = 1, \dots, m, \quad \ell = 0, \dots, \mu_j - 1. \quad (1)$$

In the Hermite interpolation problem, it is advisable to enumerate the set of points w_j taking into account their multiplicities, thereby considering the set $\{w_j\}_{j=1}^m$ as an algebraic set $p^{-1}(0)$, where

$$p(z) = (z - w_1)^{\mu_1} \cdot \dots \cdot (z - w_m)^{\mu_m}. \quad (2)$$

The Hermite interpolation polynomial can be represented as

$$\sum_{j=1}^m \frac{p(z)}{(z - w_j)^{\mu_j}} \sum_{\ell=0}^{\mu_j-1} \frac{c_{j,\ell}}{\ell!} \left(\sum_{s=0}^{\mu_j-\ell-1} (z - w_j)^{\ell+s} \operatorname{res}_{w_j} \left(\frac{(z - w_j)^{\mu_j-1-s}}{p} \right) \right),$$

that is, again using residues.

Recently there have been papers on so-called non-standard interpolation (see [1,2]). They pose the problem of constructing a function f on an algebraic set $p^{-1}(0)$ (considered as an analytical space), whose values lie on a given hypersurface. More precisely, the article [1] discusses the following:

Problem. *Given the complex numbers $a_{j,k}$ ($j = 1, \dots, m$; $k = 0, \dots, \mu_j - 1$) and c . It is necessary to describe the set of all functions f which are analytic in the neighborhood $\Omega \subset \mathbb{C}$ of points $w_1, \dots, w_m$ and satisfy the equation:*

$$\sum_{j=1}^m \sum_{k=0}^{\mu_j-1} a_{j,k} f^{(k)}(w_j) = c. \quad (3)$$

Note that if f is a solution of (3), then $f + ph$ is also a solution, where

$$p(z) = \prod_{j=1}^m (z - w_j)^{\mu_j}, \quad h \in \mathcal{O}(\Omega).$$

In other words, we can work in the factor ring $\mathcal{O}(\Omega)/\langle p \rangle$ by the ideal generated by the polynomial p .

In a similar form, a non-standard problem can be posed in the multidimensional case, treating the interpolation nodes $w_1, \dots, w_m \in \mathbb{C}^n$ as the zeros of an ideal $\langle \mathbf{p} \rangle = \langle p_1(\mathbf{z}), \dots, p_n(\mathbf{z}) \rangle$ in the ring of polynomials in the variable $\mathbf{z}$ from $\mathbb{C}^n$. In the main part of the article, in particular, we will demonstrate a non-trivial example of such multidimensional interpolation.

1. Multidimensional non-standard interpolation

To formulate a multidimensional non-standard interpolation problem, we need the following definition of a Noetherian operator.

Definition 1 (Ehrenpreis [15], Palamodov [14]). *Let $I \subset \mathbb{C}[s_1, \dots, s_n]$ be a primary ideal. A family of linear differential operators with polynomial coefficients $\partial_\ell(\mathbf{s}, D)$, $\ell = 1, \dots, t$ is called a Noetherian operator for I , if the conditions*

$$\partial_\ell(\mathbf{s}, D)\varphi(\mathbf{s})|_{V(I)} = 0 \quad \forall \ell = 1, \dots, t$$

are necessary and sufficient for the function $\varphi(\mathbf{s})$ to belong to ideal I .

In the one-dimensional case an arbitrary polynomial has the form:

$$p(s) = (s - w_1)^{\mu_1} \cdot \dots \cdot (s - w_k)^{\mu_k},$$

and its generated ideal is decomposed into the intersection of primary ones

$$\rho_j = \langle (s - w_j)^{\mu_j} \rangle, \quad j = 1, \dots, k.$$

A necessary and sufficient condition for a given function φ to belong to the primary component ρ_j is vanishing of φ by the following operators with constant coefficients:

$$\mathcal{L}_{j,0}, \mathcal{L}_{j,1}, \dots, \mathcal{L}_{j,\mu_j-1},$$

where $\mathcal{L}_{i,j}[\varphi(s)] = \frac{d^j \varphi}{ds^j} \Big|_{s=w_i}$.

For an arbitrary $n \geq 1$, the primary components ρ_j of a zero-dimensional polynomial ideal $\langle p_1, \dots, p_n \rangle$ are attributed to the roots w_j . The Noetherian operators ρ_j are the arrangements of differential operators with constant coefficients.

$$\mathcal{L}_{w_j,\ell}(\partial/\partial s)|_{w_j}, \quad \ell \in A_{w_j}.$$

Here A_{w_j} is a finite subset in $\mathbb{N}^n$

Now we can proceed to the formulation of the multidimensional non-standard interpolation problem.

Problem 1 ([2]). *Let $\mathbf{p}^{-1}(0) = \{w_1, \dots, w_m\}$ and U be an open subset of $\mathbb{C}^n$ containing $\mathbf{p}^{-1}(0)$. Fix $a_{j,l}$, $j = 1, \dots, m$, $l \in A_{w_j}$ and c ; all of them are complex numbers. We need to describe the space of holomorphic functions $f: U \rightarrow \mathbb{C}$ with the following property:*

$$\sum_{j=1}^m \sum_{\ell \in A_{w_j}} a_{j,\ell} \mathcal{L}_{w_j,\ell}[f](w_j) = c. \quad (4)$$

The following monomial basis

$$\mathcal{B} = \{\mathbf{s}^{\beta_k}; k = 0, \dots, N(\mathbf{p}) - 1\}$$

in the quotient space $\mathbb{C}[\mathbf{z}]/\langle \mathbf{p} \rangle$ is one of ingredients for solving the interpolation problem. In fact, this factor is the space of remainders when dividing polynomials by the ideal $\langle \mathbf{p} \rangle$.

2. Grothendieck residue and its role in interpolation theory

The Grothendieck residue is a cornerstone of complex analysis and algebraic geometry and it plays the key role in the singularity theory and foliations theory. Assume that the sequence of germs

$$f_1, \dots, f_n \in \mathbb{C}[z] = \mathbb{C}[z_1, \dots, z_n]$$

have an isolated common zero at $a \in \mathbb{C}^n$. Consider a meromorphic differential n -form

$$\omega = \frac{1}{(2\pi i)^n} \frac{h(z) dz}{f_1(z) \dots f_n(z)} \quad (\text{with } dz = dz_1 \wedge \dots \wedge dz_n).$$

Definition 2 ([4, 5]). *The Grothendieck residue, associated with $f = (f_1, \dots, f_n)$ and h , is determined as an integral*

$$\text{res}_{a, f}^{\text{Groth}}(h) = \int_{\Gamma_a} \omega$$

of the form ω over a very special cycle

$$\Gamma_a = \{z \in U_a : |f_j(z)| = \varepsilon_j, j = 1, \dots, n\},$$

where the neighborhood U_a of a and ε_j are chosen such that the closure $\overline{U}_a$ does not contain roots different from a and $\Gamma_a \subset \subset U_a$.

We call the integration set Γ_a a Grothendieck cycle. Note that this is the skeleton of the Weil polyhedron $\{z \in U_a : |f_j(z)| < \varepsilon_j, j = 1, \dots, n\}$.

In the case of a finite set of zeros, the mapping $\mathbf{p} = (p_1, \dots, p_n)$ can define the global Grothendieck residue as the sum of the local ones. The global residue is denoted by $\text{Res} \left[\begin{smallmatrix} h(\mathbf{s}) ds_1 \wedge \dots \wedge ds_n \\ p_1(\mathbf{s}), \dots, p_n(\mathbf{s}) \end{smallmatrix} \right]$, $h \in H(D)$ (D is the domain containing all zeros of the mapping $\mathbf{p}$).

Now, using the notation introduced above, we can state a theorem that gives the way to solve Problem 1.

Theorem 2.1 (Alpay, Yger [2]). *Let $\{w_1, \dots, w_m\} = \mathbf{p}^{-1}(0)$, U be an open subset in $\mathbb{C}^n$ containing $\mathbf{p}^{-1}(0)$. Let the sequence*

$$\mathbf{a} = \{a_{j,\ell}, j = 1, \dots, m, \ell \in A_{w_j}\}$$

and the complex number c be given. Let us denote the polynomials

$$h_{w_j}^{\mathbf{a}}(\mathbf{s}) = \sum_{\ell \in A_{w_j}} a_{j,\ell} (\mathbf{s} - w_j)^\ell / \ell!,$$

making up the sequence $\mathbf{h}_{\mathbf{w}}^{\mathbf{a}} = [h_{w_1}^{\mathbf{a}}, \dots, h_{w_m}^{\mathbf{a}}]$, and let

$$\alpha[\mathbf{h}_{\mathbf{w}}^{\mathbf{a}}] = (\alpha_0[\mathbf{h}_{\mathbf{w}}^{\mathbf{a}}], \dots, \alpha_{N(\mathbf{p})-1}[\mathbf{h}_{\mathbf{w}}^{\mathbf{a}}])$$

be the projection of this sequence onto the quotient space $\mathbb{C}[\mathbf{z}] / \langle p \rangle$.

- If $\alpha[\mathbf{h}_{\mathbf{w}}^{\mathbf{a}}] = 0$, then the problem has no solution in the case $c \neq 0$, and any function $f \in \mathcal{O}(U)$ is a solution in the case $c = 0$;
- If $\alpha[\mathbf{h}_{\mathbf{w}}^{\mathbf{a}}] \neq 0$, then $f \in \mathcal{O}(U)$ satisfies the condition (4) iff

$$\alpha[\mathbf{f}] \cdot \mathbf{Q}_{\mathbf{p}}[\mathcal{B}] \cdot \alpha[\mathbf{h}_{\mathbf{w}}^{\mathbf{a}}]^T = c,$$

where T is the transposition sign, and $\mathbf{Q}_{\mathbf{p}}[\mathcal{B}]$ is the Grothendieck global residues matrix:

$$\mathbf{Q}_{\mathbf{p}}[\mathcal{B}] = \text{Res} \left[\frac{\mathbf{s}^{\beta_{k_1} + \beta_{k_2}} d\mathbf{s}}{p_1(\mathbf{s}) \dots p_n(\mathbf{s})} \right]_{0 \leq k_1, k_2 \leq N(\mathbf{p})-1}$$

3. Amoeba and its complement

For further reasoning we will need to introduce the concept of the amoeba of the Laurent polynomial, as well as describe some of its properties.

Definition 3. *Given a Laurent polynomial f its amoeba A_f is the image of the hypersurface $V = f^{-1}(0)$ under the map*

$$\text{Log}: (z_1, \dots, z_n) \rightarrow (\log |z_1|, \dots, \log |z_n|).$$

For the amoeba we will also use notation A_V .

Amoeba reflects the distribution of the algebraic set V . More precisely, one can say that the amoeba depicts hollows for V .

The shape of the amoeba is closely related to the Newton polytope Δ_f of the polynomial f . Recall that Δ_f is defined as the convex hull in $\mathbb{R}^n$ of the index set A in the expression

$$f(z_1, \dots, z_n) = \sum_{\alpha \in A} a_\alpha z^\alpha.$$

The set of integer points in Δ_f admits a natural partition $\Delta_f \cap \mathbb{Z}^n = \bigcup_{\Gamma} A_\Gamma$, where Γ is a face on Δ_f and A_Γ denotes the intersection of $\mathbb{Z}^n$ with the relative interior of Γ . We shall consider the dual cone C_ν of Δ_f at ν defined as

$$C_\nu = \left\{ s \in \mathbb{R}^n : \langle s, \nu \rangle = \max_{\alpha \in \Delta_f} \langle s, \alpha \rangle \right\}.$$

Notice that $\dim C_\nu = n - \dim \Gamma$ when $\nu \in A_\Gamma$. In particular, C_ν has nonempty interior if ν is a vertex of Δ_f , and it equals $\{0\}$ whenever ν is an interior point of Δ_f .

The following theorem allows us to introduce an order on the components of the amoeba complement.

Theorem 3.1 (Forsberg, Passare, Tsikh [13]). *On the set $\{E\}$ of connected components of ${}^c A_f$ there is an injective map (the order map)*

$$\nu: \{E\} \rightarrow \Delta_f \cap \mathbb{Z}^n$$

with the property that the dual cone $C_{\nu(E)}$ is equal to the recession cone of E . That is, for any $u \in E$ one has $u + C_\nu \in E$ and no strictly larger cone is contained in E (Notice that if ν is the k -skeleton of Δ_f the C_ν has dimension $n - k$).

Thus, connected components can be numbered as E_ν with integer $\nu \in \Delta_f$. See, for examples, the figures below for the polynomial $1 + z_1^2 z_2 + z_1 z_2^2 + 5z_1 z_2$ (Fig. 1).

4. The Gelfond-Khovanskii formula

We say that the sequence of polytopes $\Delta_1, \dots, \Delta_n \in \mathbb{R}^n$ is unfolded, if for each covector $v \in (\mathbb{R}^n)^*$ there is such number i , that for vectors $x \in \Delta = \Delta_1 + \dots + \Delta_n$ (the Minkovskii sum of polytopes Δ_j) the scalar product $\langle x, v \rangle$ take its maximal value only in some vertex of Δ_i .

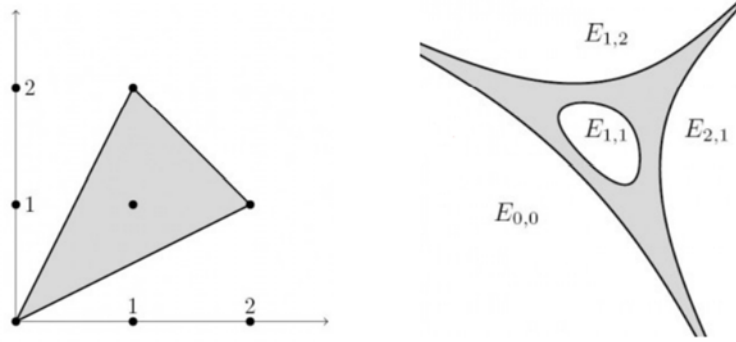


Fig. 1. An example of an amoeba.

Theorem 4.1 (Gelfond–Khovanskii [16]). *Assume that the Newton polytopes $\Delta_1, \dots, \Delta_n$ of polynomials $f_1, \dots, f_n$ are unfolded. Then the sum of all local residues in $(\mathbb{C} \setminus 0)^n$ is calculated by the formula:*

$$\sum_{\{a\}} \text{res}_f(h) = \sum_{\nu \in \text{Vert } \Delta} k_\nu \text{Res}_{E_\nu} \left(\frac{h}{f_1 \dots f_n} \right),$$

where Res_{E_ν} is the coefficient c_{-I} of the Laurent decomposition for $\frac{h}{f_1 \dots f_n}$ in the connected component E_ν .

In fact one can prove that the sum $\sum_{\{a\}} \Gamma_a$ of local Grothendieck cycles Γ_a is homologically equivalent to the sum

$$\sum_{\nu \in \text{Vert } \Delta} k_\nu \text{Log}^{-1}(u_\nu), \quad u_\nu \in E_\nu,$$

where k_ν are the combinatorial coefficients: We ascribe the combinatorial coefficient to each vertex ν of the sum Δ of unfolded polytopes. Each face $\Gamma \subset \Delta$ is a sum $\Gamma_1 + \dots + \Gamma_n$ of faces $\Gamma_i \subset \Delta_i$.

Definition 4 ([16]). *Combinatorial coefficient k_A is the local degree of the germ*

$$(\partial\Delta, A) \rightarrow (\partial\mathbb{R}_+^n, 0)$$

of the characteristic map $(h_1, \dots, h_n): \partial\Delta \rightarrow \partial\mathbb{R}_+^n$, where each component h_i is zero precisely on that face of Γ , for which the term Γ_i is a vertex of Δ_i .

5. The homological resolvent

Let $\mathcal{U} = \{U_i\}$ be a finite covering of some manifold X . Denote $S_q^{\mathcal{U}}$ the group generated by all singular simplices of dimension q which supports belong to some element U_i of the covering $\mathcal{U}$. Let $C_*(\mathcal{U}, S^*)$ be the complex (which can be called the Čech–de Rham complex in homological version) formed by the group $S_q^{\mathcal{U}}$, that is bigraduated groups

$$C_{p,q} = \bigoplus_{i_0 < i_1 < \dots < i_p} S_q(U_{i_0} \cap U_{i_1} \cap \dots \cap U_{i_p}), \quad p, q = 0, 1, \dots$$

We will need the following definition later to calculate local Grothendieck residues. The definition given below differs slightly from Gleason's definition [17].

Definition 5. *The sequence of $\mathcal{U}$ -chains $\{\xi_p\}_{p=0}^r$, $\xi_p \in C_{p,r-p}$ we will call the $\mathcal{U}$ -resolvent of the cycle $\xi \in Z_r(S_*^\mathcal{U})$ if the following two conditions are met:*

1. $\varepsilon \xi_0 = \xi$.
2. $\delta \xi_p = \partial \xi_{p-1}$, $p = 1, \dots, r$.

Here $\varepsilon : C_{0,*} \rightarrow S_*^\mathcal{U}$ is an inclusion operator $U_i \subset X$ the action of which is determined by the formula of the alternated sum:

$$\varepsilon \sigma = \sum_{i \in I} \sigma(i).$$

Boundary operator $\partial : C_{*,q} \rightarrow C_{*,q-1}$ is defined as

$$(\partial \sigma)(i_0, i_1, \dots, i_p) = \partial(\sigma(i_0, i_1, \dots, i_p)).$$

The Čech coboundary operator: Inclusions

$$U_{i_0} \cap U_{i_1} \cap \dots \cap U_{i_p} \hookrightarrow U_{i_0} \cap U_{i_1} \cap \dots \cap [i_k] \dots \cap U_{i_p}, \quad k = 0, \dots, p,$$

induce the operator $\delta : C_{p,*} \rightarrow C_{p-1,*}$ determined using the alternated sum formula:

$$(\delta \sigma)(i_0, i_1, \dots, i_{p-1}) = \sum_{i \in I} \sigma(i, i_0, \dots, i_{p-1}).$$

6. A generalisation of the Gelfond-Khovanskii formula in 2-dimensional case

Let us consider an example of the polynomial system of equations:

$$\begin{aligned} F_1 &= 3z_1^2 z_2 + z_2^4 + 2z_1 z_2^3 = 0, \\ F_2 &= z_1^3 + 4z_1 z_2^3 + 3z_1^2 z_2^2 = 0, \end{aligned} \tag{5}$$

for which the Newton polytopes are not in unfolded position: Δ_1 and Δ_2 have parallel edges.

For convenience, we introduce the notation:

$$\begin{aligned} F_1 &= z_2(3z_1^2 + z_2^3 + 2z_1 z_2^2) = z_2 f_1, \\ F_2 &= z_1(z_1^2 + 4z_2^3 + 3z_1 z_2^2) = z_1 f_2. \end{aligned}$$

Amoebas of A_{f_1} and A_{f_2} are shown in Fig. 2. In the complement of amoeba $A_{f_1 f_2}$ there are 6 connected components: E_{51}, E_{35}, E_{17} corresponds to the vertices of $\Delta = \Delta_1 + \Delta_2$, and E_{13}, E_{26}, E_{34} correspond to integer points in the relative interiors of the edges.

We will find the formula for the resolvent of the boundary ∂W of the Weil polyhedron $W = \text{Log}^{-1}(z\Delta)$, which is defined by a homothetic dilatation of the triangle Δ which contains the intersection of the amoebas Δ_1 and Δ_2 . So, W contains all roots of the system $F_1 = F_2 = 0$ in the torus $(\mathbb{C} \setminus \{0\})^2$.

Note that the sets

$$\begin{aligned} U_1 &= \{(\mathbb{C} \setminus \{0\})^2\} \setminus \{z : F_1(z) = 0\}, \\ U_2 &= \{(\mathbb{C} \setminus \{0\})^2\} \setminus \{z : F_2(z) = 0\}. \end{aligned}$$

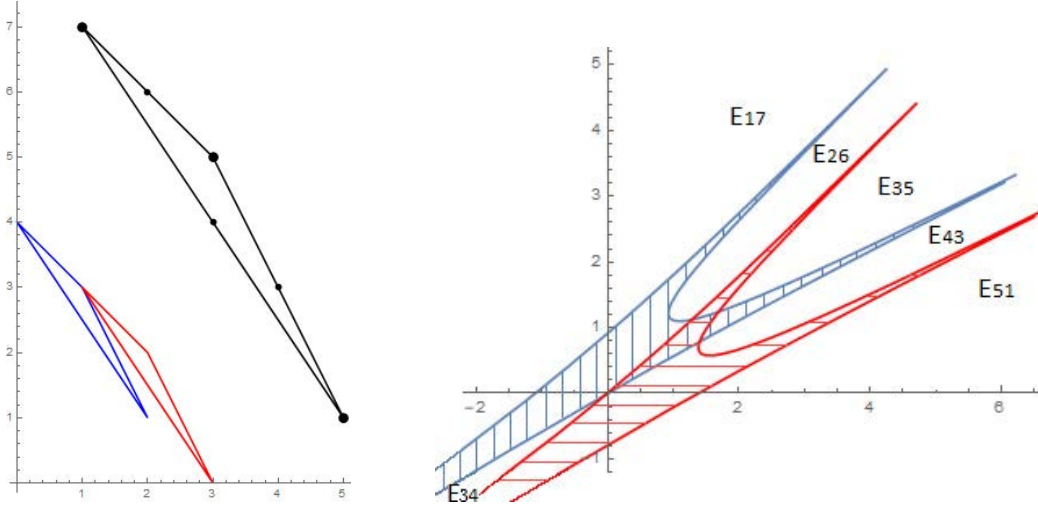


Fig. 2. Newton polytopes $\Delta_1, \Delta_2, \Delta$ and amoebas A_{f_1}, A_{f_2}

form a covering $\mathcal{U}$ of the complement $(\mathbb{C} \setminus \{0\})^2 \setminus \{z : F_1(z) = F_2(z) = 0\}$. We want to construct the resolvent for the cycle $\xi \in Z_3(S_*^{\mathcal{U}})$ which is the boundary of the polyhedron W .

1st step: Decomposition $\xi = \sigma_1 + \sigma_2$ by blue and red chains with supports $\text{supp } \sigma_j \subset U_j$ (see Fig. 3). Therefore we can take $\xi_0 \in C_{0,3}$ in Definition 5 as the following:

$$\begin{cases} \xi_0(1) = \sigma_1, \\ \xi_0(2) = \sigma_2 \end{cases}$$

(Note that each support of σ_j consists of 2 connected components).

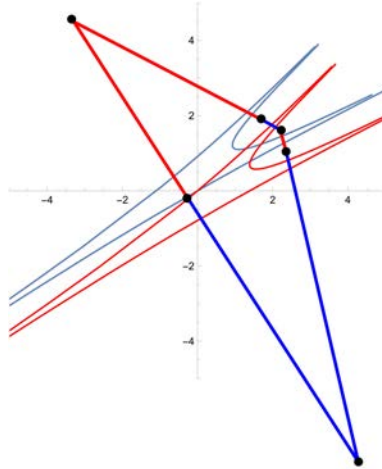


Fig. 3. The boundary of the Weil polyhedron.

2nd step: Computation of the boundary of chain ξ_0 gives

$$\begin{cases} (\partial\xi_0)(1) = \partial(\xi_0(1)) = \partial\sigma_1 = \Gamma_{26} - \Gamma_{35} + \Gamma_{43} - \Gamma_{34}, \\ (\partial\xi_0)(2) = \partial(\xi_0(2)) = \partial\sigma_2 = \Gamma_{34} - \Gamma_{26} + \Gamma_{35} - \Gamma_{43}. \end{cases}$$

Now let $\xi_1 \in C_{1,2}$ be our chain from the resolvent. Then by inclusions $U_i \cap U_j \hookrightarrow U_i$ we get the following system:

$$\begin{cases} (\delta\xi_1)(1) = \sum_{i \in I} \xi_1(i, 1) = \xi_1(1, 1) + \xi_1(2, 1) = -\xi_1(1, 2), \\ (\delta\xi_1)(2) = \sum_{i \in I} \xi_1(i, 2) = \xi_1(1, 2) + \xi_1(2, 2) = \xi_1(1, 2). \end{cases}$$

Therefore if we take into account the fact that $\delta\xi_1$ should be equal to $\partial\xi_0$ then from the system

$$\begin{cases} -\xi_1(1, 2) = (\delta\xi_1)(1) = (\partial\xi_0)(1) = \Gamma_{26} - \Gamma_{35} + \Gamma_{43} - \Gamma_{34}, \\ \xi_1(1, 2) = (\delta\xi_1)(2) = (\partial\xi_0)(2) = \Gamma_{34} - \Gamma_{26} + \Gamma_{35} - \Gamma_{43} \end{cases}$$

we have the following expression for the resolvent:

$$\xi_1(1, 2) = \Gamma_{34} - \Gamma_{26} + \Gamma_{35} - \Gamma_{43}.$$

Thus we get the following formula for the sum of the Grothendieck residues:

$$\sum_{\{a\}} \text{res}_F(h) = \text{Res}_{E_{34}} \left(\frac{h}{F_1 F_2} \right) - \text{Res}_{E_{26}} \left(\frac{h}{F_1 F_2} \right) + \text{Res}_{E_{35}} \left(\frac{h}{F_1 F_2} \right) - \text{Res}_{E_{43}} \left(\frac{h}{F_1 F_2} \right).$$

Obviously, the following statement in dimension 2 is obtained by similar reasoning.

Theorem 6.1. *Assume that the system $F_1 = F_2 = 0$ has in $(\mathbb{C} \setminus \{0\})^2$ a finite number of roots. Then the sum of Grothendieck residues in the torus $(\mathbb{C} \setminus \{0\})^2$ is calculated by the formula*

$$\sum_{\{a\}} \text{res}_F(h) = \sum_{\nu \in \mathbb{Z}^2 \cap \partial\Delta} k_\nu \text{Res}_{E_\nu} \left(\frac{h}{F_1 F_2} \right), \text{ where } k_\nu \in \{0, 1, -1\}.$$

7. Example in dimension 3

Let us consider an example of non-standard interpolation when the single point $a = 0$ is defined as an isolated zero of the polynomial system

$$\begin{aligned} P_1 &= z_1^3 - z_2 z_3 = 0, \\ P_2 &= z_2^3 - z_1 z_3 = 0, \\ P_3 &= z_3^3 - z_1 z_2 = 0. \end{aligned}$$

We have an open covering $\mathcal{U}$ for the punctured neighborhood U of the origin:

$$U_i = U \setminus \{z : P_i(z) = 0\}, \quad i = 1, 2, 3.$$

Now we want to construct the resolvent for a multicoloured cycle $\xi \in Z_5(S_*^\mathcal{U})$ (in the picture 4 below) which is homeomorphic to the sphere S^5 .

1st step: Decomposition $\xi = \sigma_1 + \sigma_2 + \sigma_3$ by blue, red and green chains (see Fig. 4) with supports $\text{supp } \sigma_j \subset U_j$. Therefore we can take $\xi_0 \in C_{0,5}$ as the following chain:

$$\{\xi_0(1) = \sigma_1, \xi_0(2) = \sigma_2, \xi_0(3) = \sigma_3\}$$

(Note that each support of σ_j consists of 2 connected components).

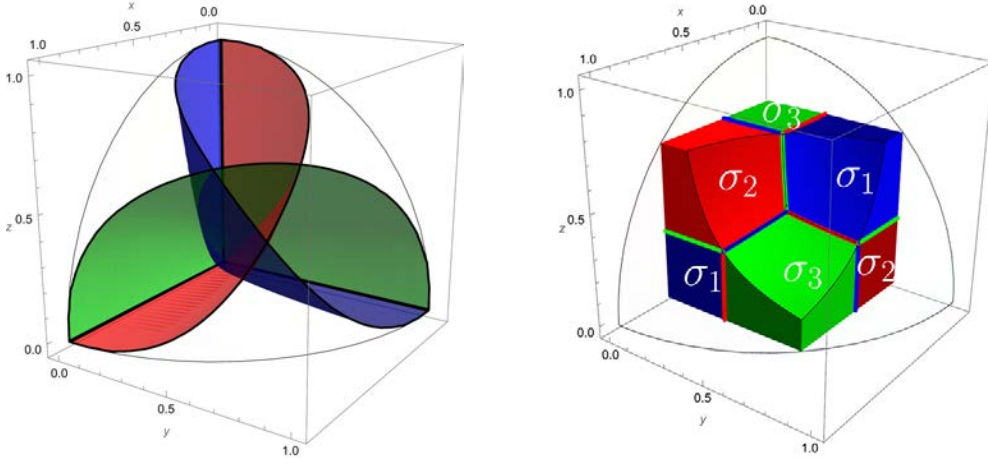


Fig. 4. Hypersurfaces (left) and toric polyhedron (right) on the Reinhardt diagram

2nd step: Computation of the boundary of chain ξ_0 (see Fig. 5):

$$\begin{cases} (\partial\xi_0)(1) = \partial(\xi_0(1)) = \partial\sigma_1 = c + b - f - g - d + h, \\ (\partial\xi_0)(2) = \partial(\xi_0(2)) = \partial\sigma_2 = -a - c + e + d - i + g, \\ (\partial\xi_0)(3) = \partial(\xi_0(3)) = \partial\sigma_3 = -h - e + f + i - b + a. \end{cases}$$

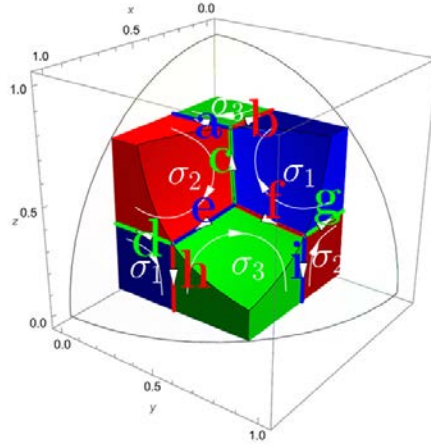


Fig. 5. Orientation and indexing of chains

Now let $\xi_1 \in C_{1,4}$ be our chain from the resolvent. Then by inclusions $U_i \cap U_j \hookrightarrow U_i$ we get the following system:

$$\begin{cases} (\delta\xi_1)(1) = \sum_{i \in I} \xi_1(i, 1) = \xi_1(1, 1) + \xi_1(2, 1) + \xi_1(3, 1) = -\xi_1(1, 2) - \xi_1(1, 3), \\ (\delta\xi_1)(2) = \sum_{i \in I} \xi_1(i, 2) = \xi_1(1, 2) + \xi_1(2, 2) + \xi_1(3, 2) = \xi_1(1, 2) - \xi_1(2, 3), \\ (\delta\xi_1)(3) = \sum_{i \in I} \xi_1(i, 3) = \xi_1(1, 3) + \xi_1(2, 3) + \xi_1(3, 3) = \xi_1(1, 3) + \xi_1(2, 3). \end{cases}$$

Note that chain $\xi_1(1, 2)$ can contain only green segments (d, c, g) [only these segments belongs to both sets U_1 and U_2], chain $\xi_1(1, 3)$ can contain only red segments (b, f, h) chain $\xi_1(2, 3)$ can contain only blue segments (a, e, i) . Therefore if we take this fact into account together with the fact that $\delta\xi_1$ should be equal to $\partial\xi_0$ then from system:

$$\begin{cases} -\xi_1(1, 2) - \xi_1(1, 3) = (\delta\xi_1)(1) = (\partial\xi_0)(1) = c + b - f - g - d + h, \\ \xi_1(1, 2) - \xi_1(2, 3) = (\delta\xi_1)(2) = (\partial\xi_0)(2) = -a - c + e + d - i + g, \\ \xi_1(1, 3) + \xi_1(2, 3) = (\delta\xi_1)(3) = (\partial\xi_0)(3) = -h - e + f + i - b + a. \end{cases}$$

we have the following solution:

$$\begin{cases} \xi_1(1, 2) = d - c + g, \\ \xi_1(1, 3) = -b + f - h, \\ \xi_1(2, 3) = a - e + i. \end{cases}$$

3rd step: Compute the boundary of the chain ξ_1 (see Fig. 6):

$$\begin{cases} (\partial\xi_1)(1, 2) = \partial(\xi_1(1, 2)) = \partial(d - c + g) = \partial d - \partial c + \partial g = -\Gamma_2 + \Gamma_0 - \Gamma_1 - \Gamma_3, \\ (\partial\xi_1)(1, 3) = \partial(\xi_1(1, 3)) = \partial(-b + f - h) = -\partial b + \partial f - \partial h = \Gamma_1 + \Gamma_3 - \Gamma_0 + \Gamma_2, \\ (\partial\xi_1)(2, 3) = \partial(\xi_1(2, 3)) = \partial(a - e + i) = \partial a - \partial e + \partial i = -\Gamma_1 + \Gamma_0 - \Gamma_2 - \Gamma_3. \end{cases}$$

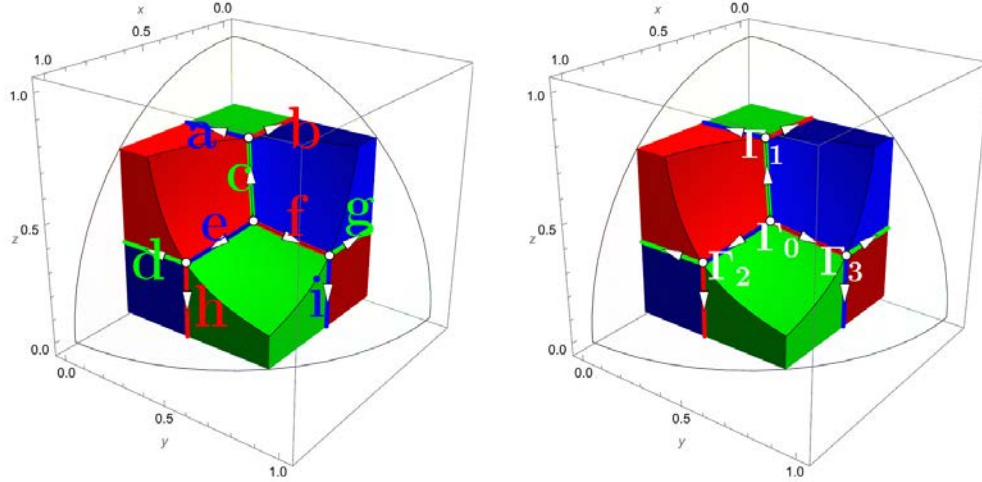


Fig. 6. 4-dimensional chains $a, b, \dots$ (left) and toric cycles $\Gamma_0, \Gamma_1, \dots$ (right)

Now let $\xi_2 \in C_{2,3}$ be our chain from the resolvent. Then by inclusions $U_1 \cap U_2 \cap U_3 \hookrightarrow U_i \cap U_j$ we get the following system:

$$\begin{aligned} (\delta\xi_2)(1, 2) &= \sum_{i \in I} \xi_2(i, 1, 2) = \xi_2(1, 1, 2) + \xi_2(2, 1, 2) + \xi_2(3, 1, 2) = \xi_2(1, 2, 3), \\ (\delta\xi_2)(1, 3) &= \sum_{i \in I} \xi_2(i, 1, 3) = \xi_2(1, 1, 3) + \xi_2(2, 1, 3) + \xi_2(3, 1, 3) = -\xi_2(1, 2, 3), \\ (\delta\xi_2)(2, 3) &= \sum_{i \in I} \xi_2(i, 2, 3) = \xi_2(1, 2, 3) + \xi_2(2, 2, 3) + \xi_2(3, 2, 3) = \xi_2(1, 2, 3). \end{aligned}$$

Therefore by resolvent condition we have:

$$\begin{cases} \xi_2(1, 2, 3) = (\delta\xi_2)(1, 2) = (\partial\xi_1)(1, 2) = -\Gamma_2 + \Gamma_0 - \Gamma_1 - \Gamma_3, \\ -\xi_2(1, 2, 3) = (\delta\xi_2)(1, 3) = (\partial\xi_1)(1, 3) = \Gamma_1 + \Gamma_3 - \Gamma_0 + \Gamma_2, \\ \xi_2(1, 2, 3) = (\delta\xi_2)(2, 3) = (\partial\xi_1)(2, 3) = -\Gamma_1 + \Gamma_0 - \Gamma_2 - \Gamma_3. \end{cases}$$

That is why $\xi_2(1, 2, 3) = \Gamma_0 - \Gamma_1 - \Gamma_2 - \Gamma_3$, and therefore the Grothendieck cycle admits a representation

$$\Gamma = \Gamma_{222} - \Gamma_{511} - \Gamma_{151} - \Gamma_{115}$$

by toric cycles. This fact helps us to construct the matrix $\mathbf{Q}_p[\mathcal{B}]$ from Theorem 2.1. For example, let us compute the following integral.

$$\begin{aligned} & \frac{1}{(2\pi i)^3} \int_{\Gamma_{222}} \frac{cz_1^\alpha z_2^\beta z_3^\gamma dz}{P_1 P_2 P_3} = \frac{c}{(2\pi i)^3} \int_{\Gamma_{222}} \frac{z_1^\alpha z_2^\beta z_3^\gamma dz_1 dz_2 dz_3}{(z_1^3 - z_2 z_3)(z_2^3 - z_1 z_3)(z_3^3 - z_1 z_2)} = \\ & = \frac{c}{(2\pi i)^3} \int_{\Gamma_{222}} \frac{z_1^\alpha z_2^\beta z_3^\gamma dz_1 dz_2 dz_3}{(-z_2 z_3)(-z_1 z_3)(-z_1 z_2)(1 - \frac{z_1^3}{z_2 z_3})(1 - \frac{z_2^3}{z_1 z_3})(1 - \frac{z_3^3}{z_1 z_2})} = \\ & = \frac{-c}{(2\pi i)^3} \int_{\Gamma_{222}} \frac{z_1^{\alpha-2} z_2^{\beta-2} z_3^{\gamma-2} dz_1 dz_2 dz_3}{(1 - \frac{z_1^3}{z_2 z_3})(1 - \frac{z_2^3}{z_1 z_3})(1 - \frac{z_3^3}{z_1 z_2})} = \\ & = \frac{-c}{(2\pi i)^3} \int_{\Gamma_{222}} \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \sum_{l=0}^{\infty} \left(\frac{z_1^3}{z_2 z_3} \right)^m \left(\frac{z_2^3}{z_1 z_3} \right)^n \left(\frac{z_3^3}{z_1 z_2} \right)^l z_1^{\alpha-2} z_2^{\beta-2} z_3^{\gamma-2} dz = \\ & = -c \sum_{m,n,l \geq 0} \frac{1}{(2\pi i)^3} \int_{\Gamma_{222}} z_1^{3m-n-l+\alpha-2} z_2^{3n-m-l+\beta-2} z_3^{3l-m-n+\gamma-2} dz = -ck. \end{aligned}$$

Here k is the number of sets of non-negative integers $\{m, n, l\}$ for which the following system of equalities holds:

$$\begin{cases} 3m - n - l + \alpha - 2 = -1, \\ -m + 3n - l + \beta - 2 = -1, \\ -m - n + 3l + \gamma - 2 = -1. \end{cases}$$

Let us consider an example of non-standard interpolation when the single point $a = 0$ is defined as an isolated zero of the polynomial system

$$\begin{aligned} P_1 &= z_1^3 - z_2 z_3 = 0, \\ P_2 &= z_2^3 - z_1 z_3 = 0, \\ P_3 &= z_3^3 - z_1 z_2 = 0. \end{aligned}$$

The multiplicity at 0 equals 11.

The Grothendieck cycle admits a representation

$$\Gamma_0 = \Gamma_{222} - \Gamma_{511} - \Gamma_{151} - \Gamma_{115}.$$

Proposition 1. *The list of Noetherian operators for the ideal $I_0\langle P \rangle$ is:*

$$\begin{aligned} \{\mathcal{L}_{\mathbf{0},\ell}\} = & \left\{ \mathcal{L}_{\mathbf{0},000} = \left(-\partial^0 - \frac{\partial^3}{\partial z_1 \partial z_2 \partial z_3} - \frac{1}{4!} \frac{\partial^4}{\partial z_1^4} - \frac{1}{4!} \frac{\partial^4}{\partial z_2^4} - \frac{1}{4!} \frac{\partial^4}{\partial z_3^4} \right); \right. \\ \mathcal{L}_{\mathbf{0},100} = & \left(-\frac{1}{3!} \frac{\partial^3}{\partial z_1^3} - \frac{\partial^2}{\partial z_2 \partial z_3} \right); \mathcal{L}_{\mathbf{0},010} = \left(-\frac{1}{3!} \frac{\partial^3}{\partial z_2^3} - \frac{\partial^2}{\partial z_1 \partial z_3} \right); \\ \mathcal{L}_{\mathbf{0},001} = & \left(-\frac{1}{3!} \frac{\partial^3}{\partial z_3^3} - \frac{\partial^2}{\partial z_1 \partial z_2} \right); \mathcal{L}_{\mathbf{0},110} = \left(-\frac{\partial}{\partial z_3} \right); \mathcal{L}_{\mathbf{0},101} = \left(-\frac{\partial}{\partial z_2} \right); \\ \mathcal{L}_{\mathbf{0},011} = & \left(-\frac{\partial}{\partial z_1} \right); \mathcal{L}_{\mathbf{0},200} = \left(-\frac{1}{4} \frac{\partial^2}{\partial z_1^2} \right); \mathcal{L}_{\mathbf{0},020} = \left(-\frac{1}{4} \frac{\partial^2}{\partial z_2^2} \right); \\ \mathcal{L}_{\mathbf{0},002} = & \left(-\frac{1}{4} \frac{\partial^2}{\partial z_3^2} \right); \mathcal{L}_{\mathbf{0},111} = (-\partial^0); \mathcal{L}_{\mathbf{0},300} = \left(-\frac{1}{3!} \frac{\partial}{\partial z_1} \right); \\ \mathcal{L}_{\mathbf{0},030} = & \left(-\frac{1}{3!} \frac{\partial}{\partial z_2} \right); \mathcal{L}_{\mathbf{0},003} = \left(-\frac{1}{3!} \frac{\partial}{\partial z_3} \right); \mathcal{L}_{\mathbf{0},400} = \left(-\frac{1}{4!} \partial^0 \right); \\ \mathcal{L}_{\mathbf{0},040} = & \left(-\frac{1}{4!} \partial^0 \right); \mathcal{L}_{\mathbf{0},004} = \left(-\frac{1}{4!} \partial^0 \right) \left. \right\}. \end{aligned}$$

Proposition 2. *The monomial basis for the factor-space $\mathcal{O}_0/I_0\langle P \rangle$ is*

$$\{1, z_1, z_2, z_3, z_1^2, z_2^2, z_3^2, z_1 z_2, z_1 z_3, z_2 z_3, z_1 z_2 z_3\}.$$

Now we can formulate the local non-standard interpolation problem and its solution.

Problem 2. *Let the complex numbers $\{a_\ell\}_{\ell \in A_0}$ and c be given. Let U_0 is an open subset of $\mathbb{C}^3$, containing the point $\mathbf{0} = (0, 0, 0)$ which is an isolated zero of mapping $P = (z_1^3 - z_2 z_3, z_2^3 - z_1 z_3, z_3^3 - z_1 z_2)$. It is necessary to describe the space of holomorphic functions $f : U_0 \rightarrow \mathbb{C}$, with the property:*

$$\sum_{\ell \in A_0} a_\ell \mathcal{L}_{\mathbf{0},\ell}[f](\mathbf{0}) = c.$$

Theorem 7.1. *If $\alpha[\mathbf{h}_w^a] \neq \mathbf{0}$, then the holomorphic function $f(\mathbf{s})$ satisfies the Alpay–Yger problem for single point ($m = 1$) iff the coordinatization of f satisfies the following condition:*

$$\begin{aligned} & \left(a_{000} + a_{111} - \frac{a_{400} + a_{040} + a_{004}}{24} \right) \alpha_1[f] + \left(a_{011} + \frac{a_{300}}{6} \right) \alpha_2[f] + \\ & + \left(a_{101} + \frac{a_{030}}{6} \right) \alpha_3[f] + \left(a_{110} + \frac{a_{003}}{6} \right) \alpha_4[f] + \frac{a_{200}}{2} \alpha_5[f] + \\ & + \frac{a_{020}}{2} \alpha_6[f] + \frac{a_{002}}{2} \alpha_7[f] + a_{001} \alpha_8[f] + a_{010} \alpha_9[f] + a_{001} \alpha_{10}[f] + a_{000} \alpha_{11}[f] = -c. \end{aligned}$$

This means that the coordinate vector of f in the local algebra lies in the prescribed affine hyperplane $\Pi_a \subset \mathbb{C}^{11}$.

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О нестандартных интерполяциях в $\mathbb{C}^n$ и комбинаторных коэффициентах для многогранников Вейля

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Аннотация. Многомерная нестандартная интерполяция была недавно представлена в статье Д. Алпая и А. Ижера. Речь идет об алгебраической интерполяции, в которой узлами служат дискретные корни системы полиномиальных уравнений. С помощью двойственности вычета Гротендика задача описания искомого интерполяционного пространства функций редуцируется к решению аффинно-билинейного уравнения. Для реализации этой редукции требуются алгоритмы вычисления локальных вычетов Гротендика или их сумм. В достаточно общей ситуации вычисление указанных вычетов основано на известной формуле Гельфонд–Хованского. В данной статье приведены примеры вычисления локальных вычетов или их сумм. В двумерном случае мы обобщаем формулу Гельфонд–Хованского для многогранников Ньютона, которые не находятся в развернутом положении. Это делается с использованием понятия амёбы алгебраического множества и понятия гомологической резольвенты для границы многогранника Вейля.

Ключевые слова: вычет Гротендика, интерполяция, амёба, гомологическая резольвента.

EDN: IQRZEA

УДК 538.955

First-principles Calculations of the Exchange Coupling Constants for an Iron Film

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Abstract. First-principles calculations of the energy and magnetic characteristics of Fe films on Cr substrates are carried out with the use of the VASP software package in PAW PBE approximation. The exchange interaction parameters for Fe/Cr(001) and Fe/Cr(001)/Fe structures as a function of ferromagnetic films thickness were calculated within a classical Heisenberg model. The results of our calculation of exchange coupling constants can be used to calculate the temperature dependence of magnetoresistance and compare them with experimental data.

Keywords: ultrathin films, ab-initio calculations, exchange interaction parameters, magnetoresistance.

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Introduction

Today, ultrathin magnetic films are the object of intensive research, both in Russia and abroad. The results obtained by researchers in this area are discussed at Russian and international conferences, numerous reviews, articles and monographs are written. Such great interest is due to the fact that the study of the properties of ferromagnetic films contributes to the solution of fundamental problems in the physics of magnetic phenomena and the development of the theory of ferromagnetism. The study of films provides new information about the magnetic properties of ferromagnets, deepening our knowledge in the field of magnetism. Thus, the study of thin films has significantly expanded the understanding of the physical nature of the anisotropy of ferromagnets, made it possible to identify and study magnetization reversal processes and discover new physical phenomena. One of these phenomena is giant magnetoresistance (GMR), which occurs in structures consisting of ferromagnetic layers separated by non-magnetic layers. Such magnetic structures are widely used as read heads of hard disks, memories, and spintronics devices. In this work, the atomic structures and magnetic properties of iron films on a Cr substrate are studied to describe exchange-correlation interactions.

A very important parameter in magnetism, in particular in micromagnetism, is the exchange interaction constant. This parameter reflects the strength of the connection between neighboring spins, which arises due to exchange interaction. The exchange interaction constant is a phenomenological parameter reflecting the magnetic symmetry of the system. It can be related to the microscopic parameters of the system, for example, to the exchange integral in the Heisenberg model. Today's interest in thin magnetic films and nanostructures makes it important to

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understand how the exchange interaction constant behaves as the size of the system changes and in the presence of interfaces.

It is very difficult to obtain the exchange interaction constant experimentally. There are very few works in which the constant for thin magnetic films [1] is estimated. Theoretically, as a rule, the exchange interaction constant is estimated by indirect methods, which include assumptions about specific models of magnetic order. There are various software packages for obtaining energy values and finding exchange interactions. For example, in the work [2] a study of exchange interaction was carried out using the Corring–Kohn–Rostoker (KKR) method. In this paper, the authors calculated the Heisenberg exchange parameter for bcc-Fe, fcc-Co, and fcc-Ni. In the work [3], using density functional theory (DFT) with the full-potential linearized augmented-plane-wave (FLAPW) method, the exchange interaction was calculated for bulk bcc-Fe and Fe/Au. It should be noted that DFT gives slightly overestimated results in relation to the mean field method in comparison with the KKR method, which describes volumetric structures well. However, for calculating thin-film structures, the KKR method works worse, so in this work we performed calculations by the VASP (Vienna Ab-Initio Simulation Package) software package [4] using the Projector AugmentedWave (PAW) method with the PBE version generalized gradient approximation (GGA).

1. Model

In this work, within the framework of a first-principles approach, using the spin density functional method, the formation of ultrathin ferromagnetic films on the surface of a non-magnetic substrate was theoretically described. The occurrence of exchange interaction in bulk material, as well as in films, has been studied.

In this work, the following structures were studied: a bulk 16-atomic bcc cell of iron, as well as a film of iron, both pure and adsorbed on a chromium surface with a (100) face orientation. Fig. 1 shows the crystal structure of the supercells of the Fe/Cr and $Fe/Cr/Fe$ systems.

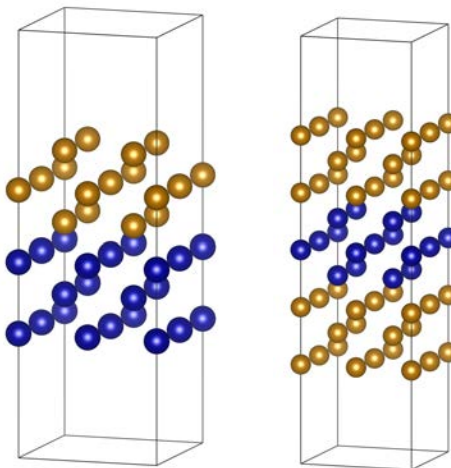


Fig. 1. The crystal structure of a supercell

Calculations were carried out for a chromium substrate with a thickness of 3 monolayers, while the thickness of the iron film varied from one to three monolayers. There is also a 5 Å vacuum layer on both sides. Also on both sides there is a vacuum layer of 5 Å. We obtained the value of the lattice constant $a=5.74$ Å.

Integration in the first Brillouin zone was carried out using a k-grid constructed using the Monkhorst–Pack method. The k-grid size was $13 \times 13 \times 1$. The plane-wave basis cutoff energy was chosen to be 500 eV.

The contribution to ab initio total energies, corresponding to exchange interaction, can be described by the following classical Heisenberg model:

$$H = - \sum_{i,j} J_{ij} (\mathbf{S}_i \mathbf{S}_j). \quad (1)$$

We got a system of equations for determination of exchange interaction integrals by calculating the difference in energies of the ferromagnetic (FM) and antiferromagnetic (AFM) configurations of spins in the film:

$$J_j = \Delta N_{ji}^{-1} \Delta E_i, \quad (2)$$

where

$$\Delta N = N_{AFM} - N_{FM}, \quad (3)$$

$$\Delta E = E_{AFM} - E_{FM}, \quad (4)$$

N is calculated as

$$N = - \sum_{ij} S_i S_j. \quad (5)$$

For the case of taking into account the interaction of the nearest neighbors ($j = 1$) and the next nearest neighbors ($j = 2$) we obtain the following expression:

$$\begin{pmatrix} J_1 \\ J_2 \end{pmatrix} = \begin{pmatrix} \Delta N_1 & \Delta N N_1 \\ \Delta N_2 & \Delta N N_2 \end{pmatrix}^{-1} \begin{pmatrix} \Delta E_1 \\ \Delta E_2 \end{pmatrix}. \quad (6)$$

The antiferromagnetic spin configuration for which the calculation was carried out is shown in Fig. 2. We will describe in more detail the mechanism of calculating the number of nearest

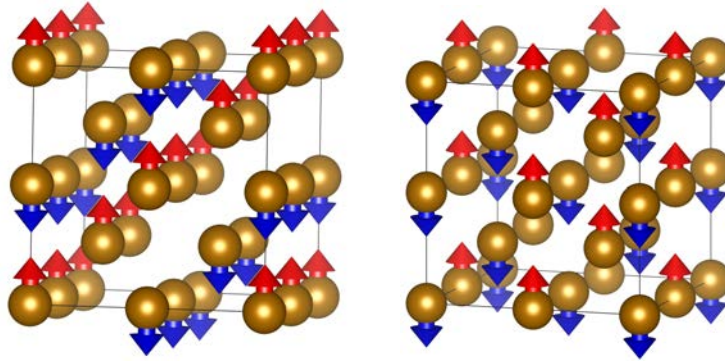


Fig. 2. Orientations of magnetic moments of atoms

neighbors. We gradually increased the radius of the coordination sphere until the array of neighbors increased, then we checked which new atoms were included in the array of neighbors. If the new atoms corresponded to the selection, then the iterations were stopped, if the increase was due to atoms that should not be taken into account in the calculations, then they were removed from the array and the search continued. For example, consider the 4Fe12Cr system. We need to get Fe-Fe neighbors. Iron atoms have indices 0, 1, 2 and 3. Gradually increasing the

radius of the coordination sphere, we get a list of neighbors for each atom. The mechanism is clearly described in (7).

$$\begin{bmatrix} 1 & 2 & 4 & 8 & \dots \\ 3 & 4 & 2 & 9 & \dots \\ 3 & 4 & 0 & 10 & \dots \\ 4 & 5 & 6 & 7 & \dots \\ 5 & 2 & 8 & 9 & \dots \\ 7 & 9 & 11 & 13 & \dots \\ 7 & 10 & 11 & 14 & \dots \\ 11 & 15 & 1 & 5 & \dots \\ 9 & 10 & 12 & 0 & \dots \\ 11 & 12 & 13 & 8 & \dots \\ 11 & 12 & 13 & 8 & \dots \\ 12 & 13 & 14 & 15 & \dots \\ 13 & 14 & 4 & 8 & \dots \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & - & 0 \\ 3 & - & 2 & - \\ 3 & - & 0 & - \\ - & - & - & 1 \\ - & 2 & - & - \\ - & - & - & - \\ - & - & 1 & - \\ - & - & - & 0 \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & - & 0 \\ 3 & - & 2 & - \\ 3 & - & 0 & - \\ - & - & - & 1 \end{bmatrix} \quad (7)$$

Each i -th row of the array is a set of indices of atoms that are neighbors of one i -th atom. After receiving the array, we remove all chromium atoms that are neighbors, that is, in each row we remove elements that correspond to the indices of chromium atoms. This leaves us with an array of Fe-Fe and Cr-Fe neighbors. After that, we delete the chromium atoms themselves, that is, the rows whose indices correspond to the indices of chromium atoms. As a result, we have Fe-Fe neighbors.

Next, we sort through the remaining pairs of atoms, if the spin direction is co-directed, then we take -1 , if it is oppositely directed, then 1 .

The values of these sums for the systems under study are given in Tab. 1.

Table 1. The number of pairs of the Fe-Fe nearest (NN) and next- nearest neighbors (NNN) with oppositely directed spins

		1 ml		2 ml		3 ml	
		NN	NNN	NN	NNN	NN	NNN
Fe	<i>FM</i>	-16	-16	-32	-64	-64	-136
	<i>AFM</i> ₁	16	-16	0	0	0	24
	<i>AFM</i> ₂	0	16	-16	48	-32	88
FeCr	<i>FM</i>	-16	-16	-32	-64	-64	-136
	<i>AFM</i> ₁	16	-16	0	0	0	24
	<i>AFM</i> ₂	0	16	-16	48	-16	72
FeCrFe	<i>FM</i>	-32	-32	-64	-128	-128	-272
	<i>AFM</i> ₁	32	-32	0	0	0	48
	<i>AFM</i> ₂	0	32	-32	96	-32	144

Thus, subtracting the values N for *AFM* and *FM* configuration, we get ΔN .

2. Result

Using the methodology described above, we calculated and presented in Tab. 2 the values of the exchange integral J_1 for interaction of the nearest neighbours and J_2 for interaction of the next-nearest neighbours for bulk iron.

Table 2. Values of the exchange integral for the nearest neighbors J_1 and for the next nearest neighbors J_2 for Fe in the bulk

	J1			J2		
	meV	erg*10 ⁻¹⁴	mRy	meV	erg*10 ⁻¹⁴	mRy
Fe_{bulk}	25.117	4.023	1.846(VASP) 0.862(MF) [1]; 1.23(KKR) 1.43(KKR) [2] 4.209(FLAPW) [3];	-3.163	-0.506	-0.232

Comparison of the results of our calculations J_1 for bulk iron using VASP and KKR methods with the results of calculations by other authors showed that the values differ depending on the method used. The results obtained by the KKR method are closer to the results obtained by the mean field approximation (MF). The results obtained using VASP, in turn, slightly overestimate the values. But this overestimation is much less than the values also obtained within the framework of the DFT FLAPW method [3].

Then we obtain the values of the exchange integral J_1 and J_2 for the iron film, both pure and adsorbed on a chromium surface with a (100) face orientation for the Fe/Cr and Fe/Cr/Fe systems Tab. 3.

Table 3. Values of the exchange integral for the nearest neighbors J_1 and for the next nearest neighbors J_2 for Fe films depending on the thickness

		J1			J2		
		meV	erg*10 ⁻¹⁴	mRy	meV	erg*10 ⁻¹⁴	mRy
Fe	1 ml	31.343	5.021	2.303	-0.928	-0.148	-0.0682
	2 ml	43.185	6.918	3.174	-1.904	-0.3051	-1.399
	4 ml	42.410	6.794	3.117	-9.981	-1.599	-0.733
FeCr	1 ml	-9.221	-1.477	-0.677	-2.319	-0.371	-0.170
	2 ml	13.626	2.182	1.001	5.739	0.919	0.421
	3 ml	30.789	4.932	2.263	6.541	1.048	0.480
FeCrFe	1 ml	-10.399	-1.665	-0.764	-0.614	-0.098	-0.045
	2 ml	20.390	3.266	0.149	5.540	0.887	0.407
	3 ml	17.981	2.881	0.132	5.184	0.830	0.381

Tab. 4 shows the energy difference values per atom for the Fe/Cr and $Fe/Cr/Fe$ systems. Comparing the obtained results J_1 and ΔE we see that the value of ΔE directly affects the sign of the obtained values J_1 . For example, in a monolayer film, for which ΔE turned out to be negative, that is, the AFM state turned out to be more energetically favorable, the value of J_1 also turned out to be negative.

Consider the values of the magnetic moments of the systems under study, given in the Tab. 5. It can be seen that the values of the magnetic moments in the outer layer of the film are almost the same as the values of the magnetic moments in the pure film, and the iron layers located closer to the substrate have a lower magnetic moment due to the magnetization of the substrate atoms.

Table 4. The difference between the reduced energies of the AFM and FM configurations for the Fe/Cr/Fe and Fe/Cr systems depending on the thickness of the Fe film. The obtained values of the radius of the coordination sphere r for the nearest and next-to-nearest neighbors

	FeCrFe. $\Delta E/\text{At}$, eV	FeCr. $\Delta E/\text{At}$, eV	r_1 , Å	r_2 , Å
1 ml	-0.133	-0.073	1.2	2.4
	-0.074	-0.055		
2 ml	0.287	0.160	0.9	1.8
	0.270	0.172		
3 ml	0.440	0.502	0.9	1.8
	0.431	0.473		

Table 5. Values of magnetic moments of atoms for different configurations depending on the thickness of the iron film

		Fe			$2Fe$				$3Fe$				
		μ_{Fe}	μ_{Cr1}	μ_{Cr2}	μ_{1Fe}	μ_{2Fe}	μ_{Cr1}	μ_{Cr2}	μ_{1Fe}	μ_{2Fe}	μ_{3Fe}	μ_{Cr1}	μ_{Cr2}
Fe	FM	3.177			2.804	2.804			2.951	2.306	2.951		
	AFM_1	3.191			2.522	2.522			2.744	1.498	2.744		
	AFM_2	3.178			2.486	2.486			2.889	1.755	2.889		
FeCr	FM	1.815	-0.646	0.150	2.903	1.781	-0.423	0.411	2.917	2.359	2.204	-1.119	1.381
	AFM_1	2.294	0.000	-0.055	2.866	1.254	-0.707	0.089	2.788	1.470	1.703	-0.040	0.226
	AFM_2	2.056	0.000	-0.009	2.903	1.543	-0.116	0.006	2.735	1.872	1.781	-0.173	0.033
FeCrFe	FM	2.258	-0.869	0.754	2.877	1.948	-0.553	0.327	2.923	2.370	2.182	-0.992	0.817
	AFM_1	2.498	0.000	0.000	2.874	1.276	-0.056	0.108	2.796	1.507	1.955	-0.053	0.217
	AFM_2	2.538	0.000	0.000	2.828	1.624	-0.205	0.086	2.744	1.862	1.866	-0.014	0.000

Due to the fact that chromium is a layer-by-layer antiferromagnet, there should be an alternation of the signs of the magnetic moments in different layers, which we see in the Tab. 5. The outer and middle layers of the chromium substrate have opposite signs.

When studying a monolayer film, we found that in antiferromagnetic cases, chromium layers have a zero magnetic moment and no magnetization occurs, which is probably why the AFM state is energetically more favorable for a monolayer film.

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Первопринципные расчеты констант обменного взаимодействия для пленки железа

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Аннотация. Используя программный пакет VASP в приближении PAW PBE, были проведены первопринципные расчеты энергетических и магнитных характеристик пленок Fe на подложках Cr. В рамках классической модели Гейзенберга рассчитаны параметры обменного взаимодействия для структур Fe/Cr(001) и Fe/Cr(001)/Fe в зависимости от толщины ферромагнитных пленок. Результаты расчета констант обменного взаимодействия могут быть использованы для расчета температурной зависимости магнитосопротивления и сравнения их с экспериментальными данными.

Ключевые слова: ультратонкие пленки, ab-initio расчеты, параметры обменного взаимодействия, магнитосопротивление.

EDN: IWKRAA

УДК 538.915

Lang-Firsov Transformation in the Generalized Tight-binding Method

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Abstract. The effects of strong electron-phonon interaction in a realistic model of a system with strong Coulomb correlations are analyzed using the Lang and Firsov transformation. It is shown that polaronic and bipolaronic transformations, widely discussed in the literature and associated with a smooth or sharp transitions in the properties of charge carriers when the strength of the electron-phonon coupling changes, determine the switching of the system between the regimes of correlated carriers, polarons or bipolarons. These transitions are controlled by the local electron-phonon interaction of the Holstein type. At the same time, the non local electron-lattice contribution associated with the modulation of the hopping integral plays a major role in the crossover of the polaron and bipolaron regimes in the limit of strong electron correlations.

Keywords: Lang–Firsov transformation, polaron and bipolaron crossover, pd-model.

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1. Lang-Firsov transformation

Let us consider the classical canonical transformation for systems with the strong electron-phonon coupling of the Holstein type. It was proposed by Lang and Firsov [1] for systematic consideration of the perturbation theory of the mobility of small-radius polarons. They managed to record a significant part of the short-ranged electron-phonon interaction in such a way that, during the transformation of the Hamiltonian, they could collect infinite series for the transformed operators. The authors [1] simplified the Hamiltonian of the system by limiting the contribution of the electron-phonon interaction to the local electron variables:

$$H = \sum_m \varepsilon a_m^\dagger a_m + \sum_{m,\mathbf{q}} T_g a_{m+g}^\dagger a_m + \sum_{\mathbf{q}} \hbar \omega_{\mathbf{q}} \left(b_{\mathbf{q}}^\dagger b_{\mathbf{q}} + \frac{1}{2} \right) + \sum_{m,\mathbf{q}} \hbar \omega_{\mathbf{q}} a_m^\dagger a_m (U_{m,\mathbf{q}} b_{\mathbf{q}}^\dagger + U_{m,\mathbf{q}}^* b_{\mathbf{q}}), \quad (1)$$

where $a_m^\dagger$ (a_m) are operators of creation (annihilation) of an electron at a site m , $b_{\mathbf{q}}^\dagger$ ($b_{\mathbf{q}}$) are operators of creation (annihilation) of a phonon with quasi momentum $\mathbf{q}$; ε is the ground state

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electron energy; the parameter T_g characterizes the hopping energy from site g to m and vice versa; $U_{m,\mathbf{q}}$ is the electron-phonon interaction (EPI) parameter and $\omega_{\mathbf{q}}$ is the phonon mode frequency.

According to the original paper, the canonical transformation $\tilde{H} = e^S H e^{-S}$ of the Hamiltonian (1) with the unitary operator S in the form:

$$S = \sum_{m,\mathbf{q}} a_m^\dagger a_m (b_{\mathbf{q}}^\dagger U_{m,\mathbf{q}} - b_{\mathbf{q}} U_{m,\mathbf{q}}^*),$$

leads to a new Hamiltonian:

$$\tilde{H} = \sum_m a_m^\dagger a_m (\varepsilon - \Delta) + \sum_{\mathbf{q}} \hbar \omega_{\mathbf{q}} \left(b_{\mathbf{q}}^\dagger b_{\mathbf{q}} + \frac{1}{2} \right) + \sum_{m,g} T_g \hat{\Phi}_{m,g} a_{m+g}^\dagger a_m,$$

which describes the energy of polarons, lattice oscillators, and their residual interaction. Here $\Delta = \sum_{\mathbf{q}} \hbar \omega_{\mathbf{q}} |U_{m,\mathbf{q}}|^2$ is the polaron shift of the atomic energy of the ground states of charge carriers and $\hat{\Phi}_{m,g} = \exp \left[\sum_{\mathbf{q}} (b_{\mathbf{q}}^\dagger \Delta_{m,m+g,\mathbf{q}} - b_{\mathbf{q}} \Delta_{m,m+g,\mathbf{q}}^*) \right]$ is the renormalization of the hopping integral, it depends on the strength of the electron-phonon coupling: $\Delta_{m,m+g,\mathbf{q}} = U_{m,\mathbf{q}} - U_{m+g,\mathbf{q}}$. In the transformed Hamiltonian, the contribution characterizing the interaction of electrons at different sites through the exchange of virtual phonons is discarded, since it is less than the Coulomb correlations of electrons not taken into account in the original model.

The modified Lang and Firsov transformation, which includes the nonlocal electron-phonon interaction, is also found in the literature. In the general case, analysis of the obtained renormalizations is complicated, however, there are some results that generalize the Holstein theory of polarons. For example, within the framework of the Holstein–Peierls model, the authors of [2, 3] obtain an expression describing the temperature narrowing of the polaron bandwidth and demonstrate that nonlocal electron-phonon coupling plays an important role in organic molecular crystals.

2. Extended pd-model

Realistic models of systems with strong electron-phonon interaction contain more numbers of contributions compared to the Hamiltonian (1). In addition to the nonlocal electron-lattice interaction, these can be multiband effects, Coulomb correlations, and much more. Let us consider an extended pd-model combining contributions essential for describing the nature of high-temperature superconductors based on copper oxides. The Hamiltonian of the model has the form:

$$H = H_{pd} + H_{e-ph} + H_{ph}, \quad (2)$$

here H_{pd} is a well-known pd model describing the hole carriers in the orbitals of oxygen p and copper d ; it takes into account both the Coulomb interaction of holes in the same and different orbitals and the strong overlap of the orbitals [4].

The term H_{e-ph} describes the interaction of electrons with lattice vibrations. We consider only one optical fully symmetric mode, the dispersion of which can be neglected. Deviations of atoms from their equilibrium positions when bonds are stretching lead to modulations of the single-site energy of charge carriers in copper orbitals (local Holstein-type EPI) and of the hopping integral between the copper and oxygen orbitals (non-local Peierls-type EPI). The H_{ph} part corresponds to the energy of free phonons.

Let us show the complete form for each part of the Hamiltonian (2):

$$H_{pd} = \sum_{\mathbf{g}, l, \sigma, \alpha} \left(\varepsilon_\alpha n_{\mathbf{g}_\alpha, \sigma}^\alpha + \frac{1}{2} U_\alpha n_{\mathbf{g}_\alpha, \sigma}^\alpha n_{\mathbf{g}_\alpha, \bar{\sigma}}^\alpha \right) + \sum_{\mathbf{g} \mathbf{g}', ll', \sigma} P_{pp} t_{pp} \left(p_{\mathbf{g}_l, \sigma}^\dagger p_{\mathbf{g}', l', \sigma} + \text{H.c.} \right) + \sum_{\mathbf{g}, l, \sigma} P_{pd} t_{pd} \left(d_{\mathbf{g}, \sigma}^\dagger p_{\mathbf{g}_l, \sigma} + \text{H.c.} \right) + \sum_{\mathbf{g}, l, \sigma \sigma'} V_{pd} n_{\mathbf{g}_l, \sigma}^p n_{\mathbf{g}, \sigma'}^d. \quad (3)$$

Here $d_{\mathbf{g}, \sigma}^\dagger$ ($d_{\mathbf{g}, \sigma}$) and $p_{\mathbf{g}_l, \sigma}^\dagger$ ($p_{\mathbf{g}_l, \sigma}$) are operators of creation (annihilation) of a hole with spin σ on copper and oxygen orbitals in positions specified by vectors $\mathbf{g}$ and $\mathbf{g}_l$ respectively. The index l enumerates oxygen atoms in the unit cell at the site $\mathbf{g}$; $n_{\mathbf{g}_\alpha, \sigma}^\alpha$ is the hole number operator, where $\mathbf{g}_\alpha = \mathbf{g}$ for $\alpha = d$, and $\mathbf{g}_\alpha = \mathbf{g}_l$ for $\alpha = p$; ε_α is the local energy of a hole in the p or d orbital; t_{pp} and t_{pd} are hopping integrals, U_α , V_{pd} are Coulomb repulsion parameters. The factors P_{pp} and P_{pd} are equal to 1 (-1) if in the region of orbital overlap their phase signs coincide (differ). In the general case, the Hamiltonian (3) describes compounds of transition metal with an incompletely filled d orbitals.

The energy of free phonons in the Hamiltonian (2) has the form:

$$H_{ph} = \sum_{\mathbf{q}} \hbar \omega_0 \left(f_{\mathbf{q}}^\dagger f_{\mathbf{q}} + \frac{1}{2} \right) = \sum_{\mathbf{g}} \hbar \omega_0 \left(f_{\mathbf{g}}^\dagger f_{\mathbf{g}} + \frac{1}{2} \right), \quad (4)$$

where $f_{\mathbf{q}}^\dagger$ is the phonon creates operator with a quasi momentum $\mathbf{q}$ and $f_{\mathbf{g}}^\dagger$ is its Fourier image, ω_0 is the frequency of lattice oscillations.

A significant short-ranged part of the electron-phonon interaction can be written as [6]:

$$H_{e-ph} = \sum_{\mathbf{g}, \sigma} M_d (f_{\mathbf{g}}^\dagger + f_{\mathbf{g}}) d_{\mathbf{g}, \sigma}^\dagger d_{\mathbf{g}, \sigma} + \sum_{\mathbf{g}, l, \sigma} M_{pd} P_{pd} (f_{\mathbf{g}}^\dagger + f_{\mathbf{g}}) (d_{\mathbf{g}, \sigma}^\dagger p_{\mathbf{g}_l, \sigma} + \text{H.c.}), \quad (5)$$

where M_d and M_{pd} are the electron-phonon interaction parameters for Holstein and Peierls type contributions, respectively.

3. Lang-Firsov transformation for intercluster part

Assuming that the strong coupling approximation is valid for the system (2) in the limit of strong electron correlations, we analyze the effects of electron-phonon interaction using the Lang and Firsov transformation. Following the ideology of the generalized tight-binding method, we divide the system into clusters and represent the Hamiltonian (2) as the sum of the intracluster contribution H_c and the intercluster interaction H_{cc} :

$$H = H_c + H_{cc}. \quad (6)$$

Such splitting is possible if we go from the operators $p_{\mathbf{g}_l, \sigma}^\dagger$ ($p_{\mathbf{g}_l, \sigma}$) to the new ones, $b_{\mathbf{g}, \sigma}^\dagger$ ($b_{\mathbf{g}, \sigma}$), centered on copper orbitals and orthogonal in the nearest cells. A similar transformation for the displacement operators [5] translates the operators $f_{\mathbf{g}}^\dagger$ ($f_{\mathbf{g}}$) into the new $A_{\mathbf{g}}^\dagger$ ($A_{\mathbf{g}}$) [6]. As a result, the intracluster part of the Hamiltonian (2) is given by:

$$H_c = \sum_{\mathbf{g}, \sigma, \beta} \left(\varepsilon_\beta n_{\mathbf{g}, \sigma}^\beta + \frac{1}{2} U_\beta n_{\mathbf{g}, \sigma}^\beta n_{\mathbf{g}, \bar{\sigma}}^\beta \right) + \sum_{\mathbf{g}, \sigma} \left(t_{00}^{bd} (d_{\mathbf{g}, \sigma}^\dagger b_{\mathbf{g}, \sigma} + \text{H.c.}) + \sum_{\sigma'} V_{bd} n_{\mathbf{g}, \sigma}^d n_{\mathbf{g}, \sigma'}^b \right) + \sum_{\mathbf{g}, \sigma} (M_{00}^d \varphi_{\mathbf{g}}^A n_{\mathbf{g}, \sigma}^d + M_{000}^{bd} \varphi_g^A [d_{\mathbf{g}, \sigma}^\dagger b_{\mathbf{g}, \sigma} + \text{H.c.}]) + \sum_{\mathbf{g}} \hbar \omega_0 \left(A_{\mathbf{g}}^\dagger A_{\mathbf{g}} + \frac{1}{2} \right). \quad (7)$$

where $\beta = d, b$ is the orbital index in the new representation. Changes in the parameters of the original Hamiltonian (2) are result from the transition to the basis of Wannier functions.

Omitting the insignificant details of the polaronic version of the generalized tight-binding method [6], we only indicate that $M_{00}^d = M_d \mu_{00}$, $M_{000}^{bd} = -2M_{pd} \mu_{00}^2$ and the coefficient $\mu_{00} \approx 0.96$.

Let us carry out the canonical transformation of the Hamiltonian H_c , limiting ourselves to local contributions of the Holstein type EPI, that is, we put $M_{pd} = 0$. Then unitary operator is:

$$S = \sum_{\mathbf{g}, \sigma} \frac{M_{00}^d}{\hbar \omega_0} (A_{\mathbf{g}}^\dagger - A_{\mathbf{g}}) n_{\mathbf{g}, \sigma}^d,$$

and the Hamiltonian (7) takes the form:

$$\begin{aligned} H_c = \sum_{\mathbf{g}, \sigma} [(\varepsilon_d - \Delta) n_{\mathbf{g}, \sigma}^d + \varepsilon_b n_{\mathbf{g}, \sigma}^b] + \sum_{\mathbf{g}, \frac{1}{2}\sigma} \left[\left(U_d - 2 \frac{(M_{00}^d)^2}{\hbar \omega_0} \right) n_{\mathbf{g}, \sigma}^d n_{\mathbf{g}, \bar{\sigma}}^d + U_b n_{\mathbf{g}, \sigma}^b n_{\mathbf{g}, \bar{\sigma}}^b \right] + \\ + \sum_{\mathbf{g}, \sigma, \sigma'} V_{bd} n_{\mathbf{g}, \sigma}^d n_{\mathbf{g}, \sigma'}^b + \sum_{\mathbf{g}, \sigma} t_{00}^{bd} (\tilde{d}_{\mathbf{g}, \sigma}^\dagger b_{\mathbf{g}, \sigma} + \text{H.c.}) + \sum_{\mathbf{g}} \hbar \omega_0 \left(A_{\mathbf{g}}^\dagger A_{\mathbf{g}} + \frac{1}{2} \right). \quad (8) \end{aligned}$$

The residual polaron-lattice interaction results from the renormalization of the orbital overlap integral, associated with the transformation $\tilde{d}_{\mathbf{g}, \sigma}^\dagger = d_{\mathbf{g}, \sigma}^\dagger \exp \left[\frac{M_{00}^d}{\hbar \omega_0} (A_{\mathbf{g}}^\dagger - A_{\mathbf{g}}) \right]$. In the polaron

shift of the atomic level $\Delta = \frac{(M_{00}^d)^2}{\hbar \omega_0} \langle n_{\mathbf{g}, \sigma}^d \rangle$ we have explicitly preserved the mean of the particle number operator. We assume that $\langle n_{\mathbf{g}, \sigma}^d \rangle$ is not necessarily close to 1 even at small doping of the system due to the redistribution of charges carries between the orbitals b and d .

The equation (8) correlates with the well-known paper [7] of A. Aleksandrov and co-authors. The differences are associated with the formulation of the Hamiltonian (2), which takes into account, in addition to Coulomb correlations, multiband effects, and the structure of the unit cell. In addition, for the estimates below we use the realistic parameters of the Hamiltonian (2), determined from *ab initio* calculations, and compare the results with the conclusions obtained [8, 9] by the exact diagonalization of the cluster H_c .

Carrying out a similar procedure, we consider the renormalizations of the Hamiltonian H_c due to the contribution of the Peierls-type EPI; so, we set $M_d = 0$. Then we define the unitary operator in a form which is similar to the nonlocal contribution of the EPI, and, using the Baker-Campbell-Hausdorff formula, we find a new representation of the Hamiltonian operators. Avoiding cumbersome expressions, we note that the transformation of a pair of hole creation (annihilation) operators in the orbitals of copper and oxygen, $d_{\mathbf{g}, \sigma}^\dagger$ ($d_{\mathbf{g}, \sigma}$) and $b_{\mathbf{g}, \sigma}^\dagger$ ($b_{\mathbf{g}, \sigma}$), formally is a rotation in two-dimensional Minkowski space with angle $\theta = \frac{M_{000}^{bd}}{\hbar \omega_0} (A_{\mathbf{g}}^\dagger + A_{\mathbf{g}})$. Changes to phonon operators have the form:

$$\tilde{A}_{\mathbf{g}} = A_{\mathbf{g}} - \sum_{\sigma'} \frac{M_{000}^{bd}}{\hbar \omega_0} \left(d_{\mathbf{g}, \sigma'}^\dagger b_{\mathbf{g}, \sigma'} + \text{H.c.} \right). \quad (9)$$

In the transformed Hamiltonian, equation (9) gives renormalizations proportional to the value $\frac{M_{000}^{bd}{}^2}{\hbar \omega_0}$ and describing (i) the change in the interorbital Coulomb interaction, (ii) spin fluctuations in the b and d orbitals without charge transfer, as well as (iii) effective interorbital electron-phonon interaction associated with charge transfer without spin-flip and (iv) effective interorbital spin-phonon interaction, corresponding to charge transfer with a change of spin. These processes are caused by the exchange of virtual phonons between carriers in different orbitals. Note also that process (i) contributes both ε_d and ε_b energies to the polaron shift.

4. Analysis of the effects of local and non-local EPI

We introduce dimensionless electron-phonon coupling constants $\lambda_d = \frac{(M_d)^2}{\hbar\omega_0 W}$ and $\lambda_{bd} = \frac{(M_{bd})^2}{\hbar\omega_0 W}$, characterizing contributions of Holstein (local) and Peierls (non-local) types. Here, W is the width of the bare bond, which is not renormalized by EPI and Coulomb effects. For ease of comparison, all electronic structures parameters of the pd-model (3) have been chosen in accordance with the paper [9], in which the effects of EPI were investigated for the cell H_c and the total 2 Hamiltonians in the framework of polaronic version of the generalized tight binding method: $\varepsilon_d = 0$, $\varepsilon_p = 1.5$, $t_{pp} = 0.86$, $t_{pd} = 1.36$, $U_d = 9$, $U_p = 4$, $V_{pd} = 1.5$, $W = 2.15$, all in eV.

The first case is $\lambda_{bd} = 0$. Let us consider the single-particle states, the atomic energy ε_d of which decreases under strong local electron-phonon interaction by the value $\Delta = \frac{(M_{00}^d)^2}{\hbar\omega_0} \langle n_{\mathbf{g},\sigma}^d \rangle$. If the polaron shift Δ is larger than the width of the charge carrier band W , then the formation of an energetically more favorable band of polaron states begins near the Fermi level of the system. The critical EPI strength is $\lambda_d^{c1} = \frac{1}{\mu_{00}^2 \langle n_{\mathbf{g},\sigma}^d \rangle}$. For given parameters, this occurs for the electron-phonon coupling strength $\lambda_d^{c1} \geq 1.1$ if $\langle n_{\mathbf{g},\sigma}^d \rangle \approx 1$; at lower values of $\langle n_{\mathbf{g},\sigma}^d \rangle$, due to the strong hybridization of the orbitals p and d , the value of λ_d^{c1} increases. In the electronic structure of the extended pd-model (2), a narrow polaron band appears [8] in certain directions of the Brillouin zone.

Now let us consider two-particle states and estimate the value of the second critical parameter of the local EPI, at which the Coulomb interaction at the copper sites effectively vanishes or becomes negative. The equation (8) gives the value of $\lambda_d^{c2} = \frac{U_d}{2\mu_{00}^2 W}$. So, for $U_d = 9$ we obtain $\lambda_d^{c2} \approx 2.2$. The screening of the Coulomb potential by electron-phonon interaction forms a tendency to the formation of bipolarons (here, local bipolarons). The process is accompanied by a significant transfer of charge carries from the p to the d orbital at $\lambda_d \geq \lambda_d^{c2}$ [8].

In the limit of strong electron correlation we get that $\lambda_d^{c1} < \lambda_d^{c2}$. Obviously, a decrease in the Coulomb interaction U_d leads to a decrease in λ_d^{c2} . In the Holstein model, ($\lambda_{bd} = 0$) the crossover of the critical values marking the polaronic λ_d^{c1} and bipolaronic λ_d^{c2} regimes take place at $U_d < \frac{1}{\langle n_{\mathbf{g},\sigma}^d \rangle} 2W$. It corresponds to the case of weak or intermediate Coulomb correlations. However, in a more realistic approach that takes into account the contribution of nonlocal EPI, such a crossover occurs in the limit of strong electron correlations for ($\lambda_{bd} \neq 0$) [9]. The replacement of the polaron and bipolaron regimes is accompanied by changes in the electronic structure associated with the redistribution of the density of charge carriers in favor of the p orbital and with the ‘relocation’ of the chemical potential into the band of the two-particle states for hole doped system. This proves the essential role of the nonlocal EPI in the formation of bipolarons in a correlated system and emphasizes the need to take its effects into account in such problems as, for example, the search for light bipolarons.

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Преобразование Ланга-Фирсова в обобщенном методе сильной связи

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Аннотация. В настоящей работе с помощью преобразования Ланга и Фирсова анализируются эффекты сильного электрон-фононного взаимодействия в реалистичной модели системы с сильными кулоновскими корреляциями. Показано, что широко обсуждаемые в литературе поляронные и биполяронные трансформации, связанные с плавным или резким изменением свойств носителей заряда при изменении силы электрон-фононной связи, определяют переключение системы между режимами коррелированных носителей, поляронов или биполяронов. При этом существенную роль в кроссовере поляронного и биполяронного режимов в пределе сильных электронных корреляций играет нелокальный электрон-фононный вклад, связанный с модуляцией интеграла перескока между различными орбиталями.

Ключевые слова: преобразование Ланга и Фирсова, кроссовер поляронов и биполяронов, pd-модель.

EDN: GYRLPK

УДК 538.945.7

Study of Trapped Magnetic Field Relaxation in a Cylindrical Micron-Sized HTS

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Abstract. Relaxation curves of a cylindrical $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8-\delta}$ superconductor several microns in diameter have been calculated by means of the Monte Carlo method within the model of a layered high-temperature superconductor in magnetic fields parallel to the cylinder axis and at different temperatures. Relaxation rates have been obtained for various sample diameters and temperatures. Power-law nature of the relaxation rate vs temperature dependences has been demonstrated for all the considered sample diameters

Keywords: magnetic relaxation, Abrikosov vortices, HTS, Monte Carlo method.

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Introduction

Nowadays, high-temperature superconductors (HTSs) of micron and submicron dimensions are increasingly becoming the objects of experimental and theoretical research. At the same time, it is often necessary to study the dynamics of trapped magnetic flux in a superconductor on time, in particular, the relaxation processes occurring in the vortex system are of great interest.

Relaxation properties of HTSs are considered in a number of papers dedicated to the construction of magnets based on stacks of HTS tapes: in [1], an electric motor with superconducting winding cooled by liquid helium in an alternating field was studied. The main result was that the exit rate of the trapped magnetic flux decreases with time.

The aim of work [2] was to obtain data on the parameters of a stack of HTS tapes in a tilted field using numerical modeling in Comsol Multiphysics and experimental methods. It was found that, for each combination of strength and direction of the magnetizing field, there is an optimal number of tapes at which the magnetization on the tape is maximum.

The authors of [3] studied the properties of superconducting stacks in a rotating mechanism using physical modeling methods in comparison with experimental data. Such devices can also

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be used in synchrotron studies. Deviations are associated with incomplete knowledge about the properties of the studied materials.

When considering relaxation in HTSs, it is impossible to ignore the work [4]. The experimental data for $\text{YBa}_2\text{Cu}_3\text{O}_{7+x}$ presented there allows us to compare the general nature of the behavior of the main HTS characteristics, including the relaxation rate.

In not too high fields (of the order of hundreds or thousands of Oe. Such fields are generally used in experiments with micron-sized superconductors), the sample is penetrated by about several thousand vortices. The vortex lattice in a superconductor with pinning centers and Meissner and transport currents is a complex system that generally does not allow an analytical description. Therefore, numerical modeling is of particular value. Samples smaller than the London penetration depth of the magnetic field can be effectively modeled by solving the Ginzburg-Landau equations. For macroscopic samples, modeling at the level of individual vortices is not possible. In the intermediate case of samples tens or hundreds of λ in size, Monte Carlo and molecular dynamics modeling is possible. In our work, we have investigated the processes of magnetic flux exit from a superconductor in the shape of a cylinder from 2 to 5 microns in diameter and in parallel field. The choice of this geometry is due to the fact that superconducting nanowires [5] are often used for the construction of micro-bridges. Also, superconductors of similar dimensions can be used in the manufacture of superconducting sensors.

1. Description of the model

The calculations were performed using the Monte Carlo method for a vortex system within the framework of the two-dimensional model of a layered HTS [6]. In this model, a system of flat layered pancake vortices can be represented as an ensemble of classical particles with a long-range potential. The energy of the vortex system can be represented in the following form:

$$G = \sum_{ij} \frac{\Phi_0^2 h}{8\pi^2 \lambda^2} K_0(r_{ij}/\lambda) + \sum_i \frac{\Phi_0^2 h}{16\pi^2 \lambda^2} (0.52 + \log(\lambda/\xi)) + E_{\text{boundary}} + \sum_{id} \alpha \frac{\exp(-r_{id}/2\xi)}{1 + r_{id}/\xi} + \sum_{ib} H \frac{\Phi_0 h}{4\pi} \frac{I_0(r_{ib}/\lambda)}{I_0(d/2\lambda)} + E_{\text{current}}. \quad (1)$$

The Gibbs energy of the system takes into account the following terms: the repulsion and attraction of vortex pairs (respectively, of identical or opposite signs), the intrinsic energy of vortices, their interaction with sample boundaries, the pinning energy on defects or impurities; the influence of external magnetic field, and the contribution of electric current. Here, d and h are the diameter and thickness of the HTS sample respectively, r_{ij} is the distance between interacting vortices, r_{id} is the distance between a vortex and a defect, r_{ib} is the distance from a vortex to the boundary, K_0 is the modified Bessel function of the second kind, I_0 is the modified Bessel function of the first kind, $\Phi_0 = hc/2e$ is the magnetic flux quantum.

The superconductor parameters are calculated as follows:

$$\lambda = \lambda_0 \left(1 - \left(\frac{T}{T_c} \right)^{3.3} \right)^{-1/2}, \quad (2a)$$

$$\xi = \xi_0 \left(1 - \left(\frac{T}{T_c} \right)^{3.3} \right)^{-1/2}. \quad (2b)$$

The magnetization was calculated from the number of trapped vortices:

$$|4\pi M| = \left| H - \frac{\Phi_0 N}{S} \right|,$$

where N is the number of vortices, S — the cylinder circular face area, H — the external magnetic field strength.

Schematically, the mutual arrangement of the vortex filaments and the external magnetic field is shown in Fig. 1 (here, the magnetic field is directed perpendicular to the HTS layers). In this geometry, the Meissner currents flow in circles inside the cylinder, so the Lorentz force acting on the vortices is directed towards the center of the cylinder.

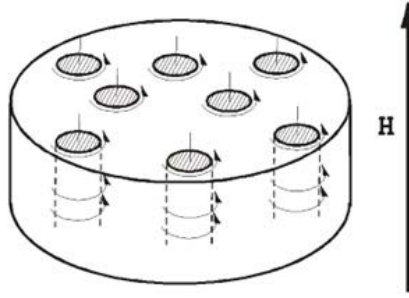


Fig. 1. Cylindrical geometry for a two-dimensional model of layered HTS. The magnetic field is oriented perpendicular to the superconducting CuO_2 layer. Vortices line up along the field

The defects are randomly scattered over the sample with a constant density of 10^{11} cm^{-2} . The effective depth α of potential wells of defects took values from 0.01 to 0.06 eV (their distribution obeyed the Gauss law). The selected potential well depth limits the pinning to one vortex per defect.

In our calculations, we neglect any changes in the temperature of superconductor arising from the motion of vortices.

An analogue of the relaxation time is the Monte Carlo (MC) step. Based on the results of earlier works [7] and [8], one may argue that one MC step corresponds to 10^{-8} – 10^{-9} s. In this paper, we have considered relaxation processes lasting up to several milliseconds.

Calculations were performed for the typical parameters of bismuthic HTSs (for certainty, we took the parameters of $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8-\delta}$).

2. Relaxation of the trapped magnetic flux

In our work, we have calculated the dependence of the magnitude of the trapped magnetic flux on the effective time (given in Monte Carlo (MC) steps). The relaxation calculations were performed for different diameters of the cylindrical sample and for temperatures from 10 to 40 K. The sample diameter varied from 2 to 5 microns. At the same time, the minimum diameter of $2 \mu\text{m}$ is about 10λ for the considered material, which suggests a noticeable influence of the boundaries on the exit of vortices from the sample.

The modeling of magnetic relaxation was carried out in two stages:

1. At a field of 3000 Gauss, vortices entered the sample forming the initial state (the first curve in the left-hand image of Fig. 2). This process lasted for 12500 MC steps.

2. The magnetic field was instantly switched off, after which the residual magnetization was collected every 50 physical MC steps.

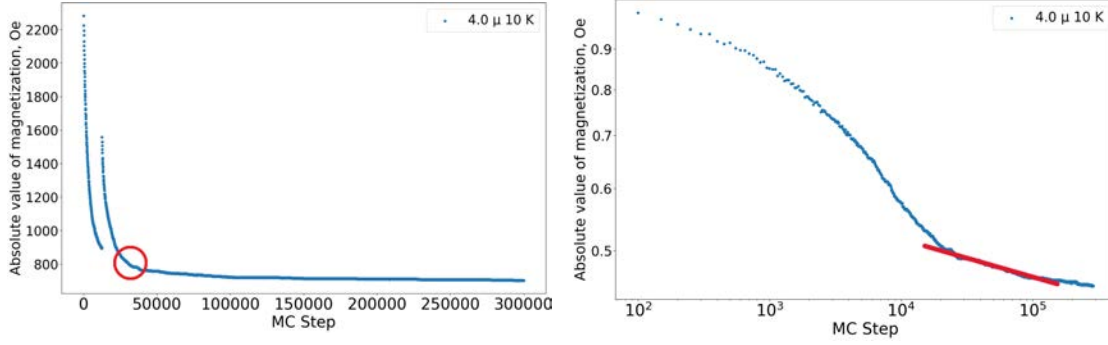


Fig. 2. Time-dependences of the absolute value of magnetization. The image on the left is presented on a linear scale, the one on the right — on a double logarithmic scale

The non-physical discontinuity is caused by the instant turning off of the field. The width of this discontinuity is less than 3000 Gs due to a delay of 50 MC steps between adjacent points of the graph, during which part of the trapped magnetic flux manages to exit the sample.

Relaxation curves were obtained for samples from 2.0 to 5.0 microns in diameter with an increment of 0.5 microns and at temperatures from 10 to 40 K with an increment of 5 K.

In Fig. 3, an example of parts of calculations is given for the entire temperature range at fixed sample diameters. The curves are normalized to the initial value and given on a double logarithmic scale. As expected, after the temperature increases, the decay rate of the residual magnetization rises — the curves are arranged sequentially in accordance with the temperature during the simulations.

Observing similar dependences at constant temperatures Fig. 4, one can also see sequentially arranged curves, albeit in accordance with the sample dimensions. The presence of steps at small diameters and high temperatures indicates the exit of single vortices from the sample. At this stage, there is a deviation from the collective flux creep theory — there is no decrease in the relaxation rate over time, which is especially noticeable on large time spans Fig. 5.

Based on the previously obtained results, we can confirm the absence of deviations from the described patterns in the behavior of residual magnetization for all temperatures and sample diameters 2 microns and greater in cylindrical geometry.

Now let us draw our attention to the relaxation rate (3) which is defined as the slope of the tangent to residual magnetization on the double logarithmic scale (Fig. 2 on the right).

$$v = \frac{\partial \log(M)}{\partial \log(t)}. \quad (3)$$

For certainty, the relaxation rate was calculated for the part of the curve marked in Fig. 2 with a circle (on the left) and a straight line (on the right). On a double logarithmic scale, the plot can be approximated by a straight line. It should be noted that the relaxation rate for a micron-diameter superconductor does not remain constant with time. This is especially noticeable for the curve corresponding to the sample diameter of 2.5 microns: its slope is maximal at times of the order of $5 \cdot 10^3$ MC steps, then the slope gradually decreases (Fig. 6).

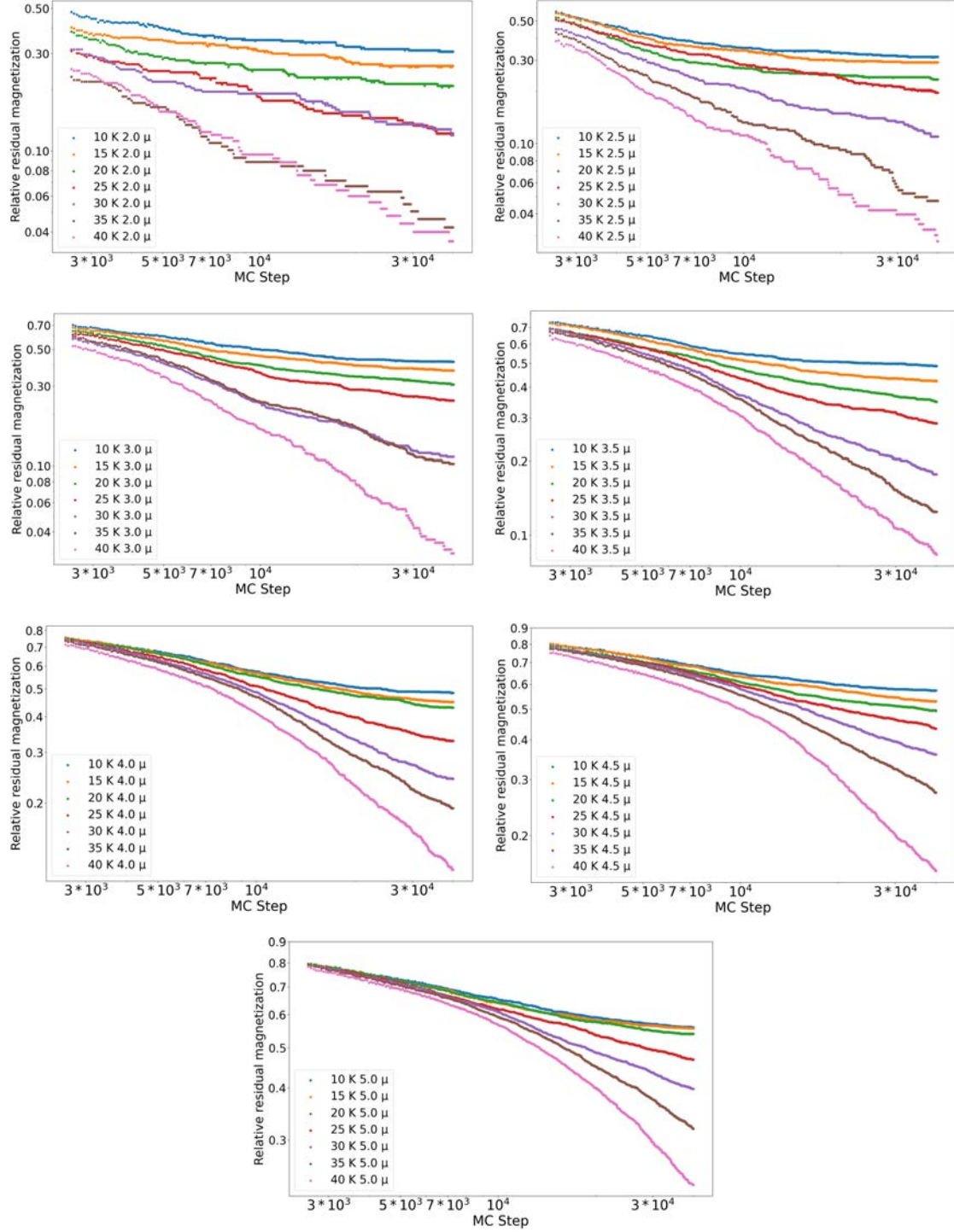


Fig. 3. Dependence of relative residual magnetization on time (in MC steps) at fixed sample sizes

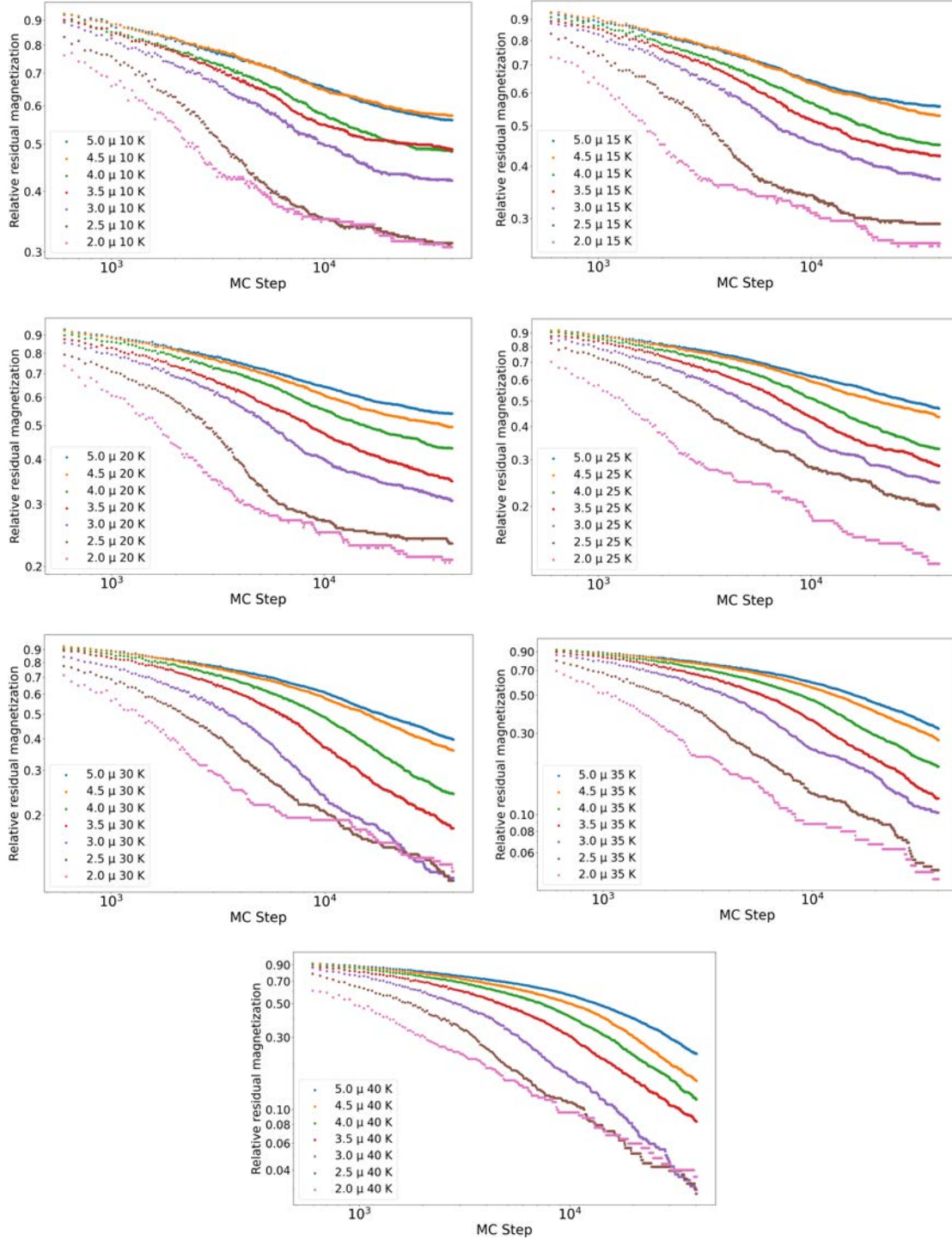


Fig. 4. Dependence of relative residual magnetization on time (in MC steps) at constant temperatures

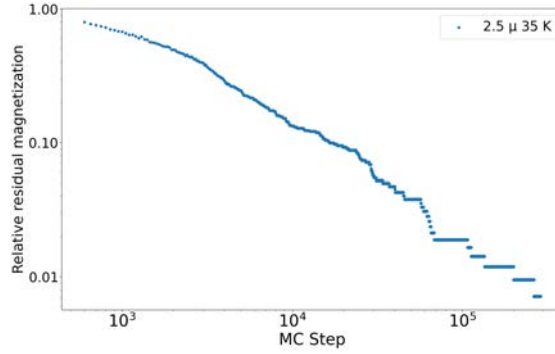


Fig. 5. A full relaxation loop after switching off of the field for a sample diameter of 2.5 microns and at a temperature of 35 K

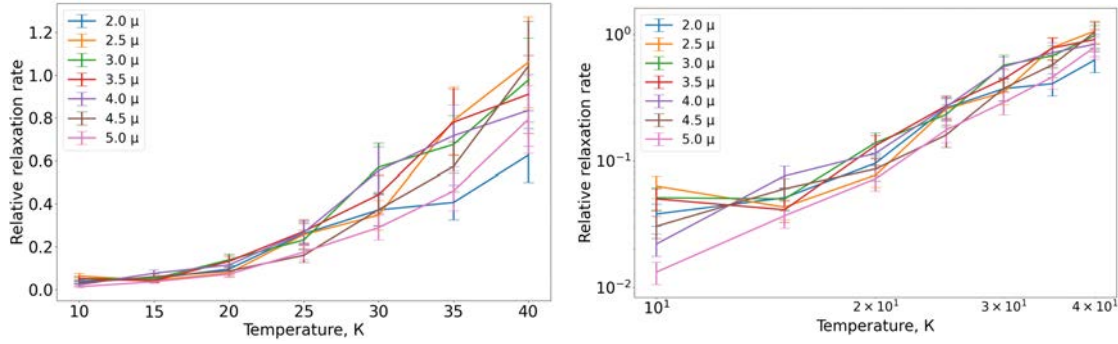


Fig. 6. Temperature dependence of the relaxation rate for samples of different diameters. The image on the left is given on a linear scale, the one on the right — on a double logarithmic scale

Fig. 6 shows the relaxation rate values for all temperatures and sample diameters studied — each line corresponds to a separate sample diameter.

When constructed on a double logarithmic scale, the relaxation rate shows a straight line, which means that there is an apparent power-law dependence of relaxation rate on temperature. It will be interesting to continue the calculations on a larger parameter span, e. g. consider different defect concentrations and pinning types.

Conclusion

The process of relaxation of the trapped magnetic flux in a cylindrical micron-diameter superconductor (analogous to a superconducting nanowire) has been studied by the Monte Carlo method within the framework of the two-dimensional model of a layered HTS. Temperature dependences of the relaxation rate have been obtained from the calculated time-dependences of the trapped magnetic field on a double logarithmic scale. It has been shown that, as the temperature increases, the decay rate of the residual magnetization also increases. The influence of the diameter of the superconducting cylinder on the relaxation processes has also been studied.

The obtained results suggest that the stability of trapped magnetic flux increases with the increase in the sample geometric dimensions. It has been shown that the temperature dependence

of the relaxation rate has a power-law shape regardless of the sample size within the considered parameter ranges. The presence of steps on the dependence of trapped magnetic field on time has also been shown, which was caused by the exit of single vortices from the sample (at the same time, no more than several dozen vortices are trapped in the sample). This behavior of magnetization at the considered times was observed at temperatures higher than 30 K and in samples with a diameter less than 3 microns. The results obtained in the present paper can be used in planning experiments related to micro-bridges and magnetic field sensors based on superconducting nanowires.

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Исследование релаксации захваченного магнитного поля в цилиндрических микронных ВТСП

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Аннотация. Методом Монте-Карло в рамках модели слоистого высокотемпературного сверхпроводника были рассчитаны релаксационные кривые цилиндрического сверхпроводника $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8-\delta}$ диаметром в несколько микрон в магнитных полях, параллельных оси цилиндра, и при разных температурах. Получены величины скорости релаксации для разных диаметров образца и температур. Продемонстрирован степенной характер зависимости скорости релаксации от температуры для всех рассмотренных диаметров образца.

Ключевые слова: магнитная релаксация, вихри Абрикосова, ВТСП, метод Монте-Карло.

EDN: RSIPYX

УДК 538.945

Changes of the Fermi Surface Topology in the Three-orbital Model for Iron Pnictides with the Spin-orbit Coupling

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Abstract. Effect of the spin-orbit coupling on the band structure and the Fermi surface of the three-orbital model is studied. The inter-orbital part of the model is reformulated to fully conform with the iron lattice symmetry. Because there are two iron ions in the unit cell, we introduce the intra- and inter-ion spin-orbit coupling constants to separate the effect of the related couplings on the band structure and the Fermi surface. Both the intra- and inter-ion parts lift the degeneracy of some bands at the $\Gamma = (0, 0)$ point and the splitting of bands along the $(0, \pi) - (\pi, \pi)$ direction of the Brillouin zone. We show that the inter-ion part of the spin-orbit coupling leads to the topological change of the Fermi surface — the splitting of two Fermi surface sheets around the $M = (\pi, \pi)$ point.

Keywords: Fe-based superconductors, spin-orbit coupling, band structure, Fermi surface.

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Iron pnictides and chalcogenides represent systems with the intriguing mixture of different states including superconductivity, magnetism, nematic state, and some coexisting states [1–3]. The variety of exotic phases is naturally stems from the multiband nature of the systems [4]. From the density-functional theory (DFT) we know that the Fermi surface (FS) in iron-based systems is formed by several bands with the dominating contribution from the three t_{2g} iron d -orbitals. Hybridization of the xz , yz , and xy orbitals results in two electron-like Fermi surface sheets around $M = (\pi, \pi)$ point and two to three hole-like sheets around $\Gamma = (0, 0)$ point of the Brillouin zone (BZ) corresponding to the crystallographic unit cell that includes two iron ions. Such a Brillouin zone often called ‘2-Fe BZ’ or ‘folded BZ’. For the single-layer iron-pnictides the folding wave vector is two-dimensional and equals to $\mathbf{Q} = (\pi, \pi)$. DFT band structures and most of the experimental results are reported in the folded BZ since crystallographically it is

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the correct one. Nesting between electron and hole sheets may produce the spin-density wave antiferromagnetic order in the undoped systems [5]. Destruction of the perfect nesting with the doping destroys long-range antiferromagnetic order, however, the scattering on the wave vector connecting hole and electron sheets remains strong. It is the most probable candidate for the interaction that drives superconducting pairing. Within the spin-fluctuation approach it results in the sign-changing extended s -wave gap ($s_{\pm}$ state) as the leading instability [4, 6–10].

There are several experimental findings pointing towards the non-vanishing role of the spin-orbit (SO) coupling. One of them is the anisotropy of the spin resonance peak violating the spin-rotational invariance, $\langle S_+ S_- \rangle = 2 \langle S_z S_z \rangle$, that have to be obeyed in the disordered system. It was observed in the inelastic neutron scattering [11]. Spin-rotational invariance violation within the three-orbital model was found to be due to the effect of the SO interaction [24] similar to the case of Sr_2RuO_4 [13]. Another intriguing fact comes from the superconducting gap studies via the multiple Andreev reflections [14, 15]. The expected number of distinct gaps in LiFeAs is two – one at the inner FS around the Γ -point, another one at the outer FS around the Γ -point and the same at the FS around the M -point. However, the analysis of the Andreev spectra reveals three different gaps [16, 17]. The observation may be explained by the FS topology change due to the SO interaction: the two crossing FS sheets around the M -point may become split into the inner and outer parts with the different gaps at each new part.

Several low-energy models with the minimal number of orbitals have been suggested for iron-based superconductors [18–23]. Here we focus on the three-orbital model from Ref. [23]. It is quite simple yet describes the most important bands coming from the t_{2g} manifold: the d_{xz} and d_{yz} orbitals are hybridized to form two electron Fermi surface sheets and one hole sheet around Γ point while the second one comes from the d_{xy} orbital. Within the model, the role of the SO coupling has been studied before [24], however, the model has a fundamental drawback – C_4 symmetry is broken for the large values of the SO coupling constant λ . Here we modify the model to avoid the violation of the symmetry and study the effect of the SO coupling on the topology of the Fermi surface and the band structure in the 2-Fe BZ.

1. Three-orbital model for the one-iron unit cell

Hamiltonian of the three-orbital model for the set of d -orbital of a *single* iron has the form [23]:

$$H_0^{(1)} = \sum_{\mathbf{k}, \sigma, l, m} \varepsilon_{\mathbf{k}}^{lm} c_{\mathbf{k}l\sigma}^\dagger c_{\mathbf{k}m\sigma}, \quad (1)$$

where $c_{\mathbf{k}l\sigma}^\dagger$ ($c_{\mathbf{k}m\sigma}$) is the creation (annihilation) operator of a particle with momentum $\mathbf{k}$, spin σ , and orbital index l (m). We choose the following numbering of orbitals, $1 \rightarrow d_{xy}$, $2 \rightarrow d_{yz}$, $3 \rightarrow d_{zx}$. We introduce vector operators in the orbital space, $\hat{\Psi}_{\mathbf{k}\sigma}^\dagger = (c_{\mathbf{k}1\sigma}^\dagger, c_{\mathbf{k}2\sigma}^\dagger, c_{\mathbf{k}3\sigma}^\dagger)$, and rewrite the Hamiltonian as

$$H_0^{(1)} = \sum_{\mathbf{k}, \sigma} \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \hat{\varepsilon}_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}\sigma}, \quad (2)$$

where the matrix of one-electron energies and hoppings has the form:

$$\hat{\varepsilon}_{\mathbf{k}} = \begin{pmatrix} \varepsilon_{1\mathbf{k}} & 0 & 0 \\ 0 & \varepsilon_{2\mathbf{k}} & \varepsilon_{4\mathbf{k}} \\ 0 & \varepsilon_{4\mathbf{k}} & \varepsilon_{3\mathbf{k}} \end{pmatrix}. \quad (3)$$

Here

$$\varepsilon_{1\mathbf{k}} = \varepsilon_{xy} - \mu + 2t_{xy}(\cos k_x + \cos k_y) + 4t'_{xy} \cos k_x \cos k_y, \quad (4)$$

$$\varepsilon_{2\mathbf{k}} = \varepsilon_{yz} - \mu + 2t_x \cos k_x + 2t_y \cos k_y + 4t' \cos k_x \cos k_y + 2t''(\cos 2k_x + \cos 2k_y), \quad (5)$$

$$\varepsilon_{3\mathbf{k}} = \varepsilon_{xz} - \mu + 2t_y \cos k_x + 2t_x \cos k_y + 4t' \cos k_x \cos k_y + 2t''(\cos 2k_x + \cos 2k_y), \quad (6)$$

with μ being the chemical potential.

Previously, $\varepsilon_{4\mathbf{k}}$ was a function of $\sin k_x/2 \sin k_y/2$ [23,24]. Such a combination does not belong to the basis functions of the tetragonal group D_{4h} . It leads to the violation of the C_4 symmetry for the large SO coupling, i.e., $(0,0) - (\pi,0)$ and $(0,0) - (0,\pi)$ directions become inequivalent. The problem comes from the $k_{x,y}/2$ argument. To avoid it, we use the new expression for $\varepsilon_{4\mathbf{k}}$:

$$\varepsilon_{4\mathbf{k}} = 4t_{xzyz} \sin k_x \sin k_y + 3t''' \sin 2k_x \sin 2k_y. \quad (7)$$

We use the following set of parameters (in eV) that allows to reproduce the topology of the FS in iron-based systems: $\varepsilon_{xy} = -0.70$, $\varepsilon_{yz} = -0.34$, $\varepsilon_{xz} = -0.34$, $t_{xy} = 0.18$, $t'_{xy} = 0.06$, $t_x = 0.26$, $t_y = -0.22$, $t' = 0.2$, $t'' = -0.07$, $t''' = 0.025$, $t_{xzyz} = 0.055$.

For the *single* iron, the spin-orbit coupling terms can be written as [25]

$$H_{SO}^{(1)} = \lambda \sum_f \mathbf{L}_f \cdot \mathbf{S}_f = i \frac{\lambda}{2} \sum_{l,m,n} \epsilon_{lmn} \sum_{\mathbf{k},\sigma,\sigma'} c_{\mathbf{k}l\sigma}^\dagger c_{\mathbf{k}m\sigma'} \hat{\sigma}_{\sigma\sigma'}^n, \quad (8)$$

where λ is the SO coupling constant, $\mathbf{L}_f$ and $\mathbf{S}_f$ are the angular momentum and spin operators at the lattice site f , respectively, ϵ_{lmn} is the completely antisymmetric tensor, indices l, m, n take values $\{d_{yz}, d_{zx}, d_{xy}\}$, and $\hat{\sigma}_{\sigma\sigma'}^n$ are the Pauli spin matrices.

Explicit form of the Hamiltonian in terms of the vector operators is

$$H_{SO}^{(1)} = \sum_{\mathbf{k},\sigma,\sigma'} \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \hat{\varepsilon}_{SO}^{\sigma\sigma'} \hat{\Psi}_{\mathbf{k}\sigma'}. \quad (9)$$

Here

$$\hat{\varepsilon}_{SO}^{\sigma\sigma'} = i \frac{\lambda}{2} \begin{pmatrix} 0 & -i\delta_{\sigma',\bar{\sigma}} \text{sgn}(\sigma) & \delta_{\sigma',\bar{\sigma}} \\ i\delta_{\sigma',\bar{\sigma}} \text{sgn}(\sigma) & 0 & \delta_{\sigma',\sigma} \text{sgn}(\sigma) \\ -\delta_{\sigma',\bar{\sigma}} & -\delta_{\sigma',\sigma} \text{sgn}(\sigma) & 0 \end{pmatrix} = \quad (10)$$

$$= i \frac{\lambda}{2} [\hat{\varepsilon}^z \delta_{\sigma',\sigma} \text{sgn}(\sigma) + \delta_{\sigma',\bar{\sigma}} (\hat{\varepsilon}^x - i\hat{\varepsilon}^y \text{sgn}(\sigma))], \quad (11)$$

where we have introduced the following matrices,

$$\hat{\varepsilon}^z = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = iJ_x, \quad \hat{\varepsilon}^x = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} = -iJ_y, \quad \hat{\varepsilon}^y = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = iJ_z. \quad (12)$$

Here, J_i are the generators of the rotation group $O(3)$.

The total Hamiltonian for the single iron, $H^{(1)} = H_0^{(1)} + H_{SO}^{(1)}$, can be written as

$$\begin{aligned} H^{(1)} &= \sum_{\mathbf{k},\sigma} \left[\hat{\Psi}_{\mathbf{k}\sigma}^\dagger \left(\hat{\varepsilon}_{\mathbf{k}} + i \frac{\lambda}{2} \hat{\varepsilon}^z \text{sgn}(\sigma) \right) \hat{\Psi}_{\mathbf{k}\sigma} + i \frac{\lambda}{2} \hat{\Psi}_{\mathbf{k}\sigma}^\dagger (\hat{\varepsilon}^x - i\hat{\varepsilon}^y \text{sgn}(\sigma)) \hat{\Psi}_{\mathbf{k}\bar{\sigma}} \right] = \\ &= \sum_{\mathbf{k}} \begin{pmatrix} \hat{\Psi}_{\mathbf{k}\uparrow}^\dagger & \hat{\Psi}_{\mathbf{k}\downarrow}^\dagger \end{pmatrix} \cdot \hat{H}^{(1)} \cdot \begin{pmatrix} \hat{\Psi}_{\mathbf{k}\uparrow} \\ \hat{\Psi}_{\mathbf{k}\downarrow} \end{pmatrix}, \end{aligned} \quad (13)$$

where

$$\hat{H}^{(1)} = \begin{pmatrix} \hat{\varepsilon}_{\mathbf{k}} + i\frac{\lambda}{2}\hat{\varepsilon}^z & i\frac{\lambda}{2}\hat{\varepsilon}^x + \frac{\lambda}{2}\hat{\varepsilon}^y \\ i\frac{\lambda}{2}\hat{\varepsilon}^x - \frac{\lambda}{2}\hat{\varepsilon}^y & \hat{\varepsilon}_{\mathbf{k}} - i\frac{\lambda}{2}\hat{\varepsilon}^z \end{pmatrix}. \quad (14)$$

Note that ε^x - and ε^y -components of the SO coupling mix spin-up and spin-down matrix elements thus effectively increasing the dimensionality of the problem by a factor of two.

2. Three-orbital model for the two-iron unit cell

To work in the crystallographic unit cell containing two irons, we make the folding procedure: momenta in Hamiltonian (1) are transformed as $(k_x + k_y)/2 \rightarrow k'_x$, $(k_x - k_y)/2 \rightarrow k'_y$ and the Hamiltonian matrix is doubled by adding the shifted $\varepsilon_{\mathbf{k}'+\mathbf{Q}}^{lm}$, where $\mathbf{Q} = (\pi, \pi)$ is the folding wave vector. For convenience, we skip prime assuming that all momenta are within the 2-Fe BZ and end up with the following Hamiltonian [24]

$$H_0 = \sum_{\mathbf{k}, \sigma, l, m} \varepsilon_{\mathbf{k}}^{lm} c_{\mathbf{k}l\sigma}^\dagger c_{\mathbf{k}m\sigma} + \sum_{\mathbf{k}, \sigma, l, m} \varepsilon_{\mathbf{k}+\mathbf{Q}}^{lm} c_{\mathbf{k}+\mathbf{Q}l\sigma}^\dagger c_{\mathbf{k}+\mathbf{Q}m\sigma}. \quad (15)$$

It can be written in the matrix form analogous to (13),

$$H = \sum_{\mathbf{k}} \begin{pmatrix} \hat{\Psi}_{\mathbf{k}\uparrow}^\dagger & \hat{\Psi}_{\mathbf{k}\downarrow}^\dagger & \hat{\Phi}_{\mathbf{k}\uparrow}^\dagger & \hat{\Phi}_{\mathbf{k}\downarrow}^\dagger \end{pmatrix} \cdot \hat{H} \cdot \begin{pmatrix} \hat{\Psi}_{\mathbf{k}\uparrow} \\ \hat{\Psi}_{\mathbf{k}\downarrow} \\ \hat{\Phi}_{\mathbf{k}\uparrow} \\ \hat{\Phi}_{\mathbf{k}\downarrow} \end{pmatrix}, \quad (16)$$

where $\hat{\Psi}_{\mathbf{k}\sigma}$ ($\hat{\Phi}_{\mathbf{k}\sigma}$) corresponds to the first (second) iron in the unit cell.

As for the SO coupling, we make the following conjecture [24]: the structure of the SO coupling between two orbitals on neighboring irons is the same as the structure between two orbitals on a single iron ion. As in Ref. [24], we assign different coupling constant to the intra-ion SO interaction (λ) and the inter-ion one (λ'). Note that the wording ‘inter-ion SO coupling’ does not necessary mean a long-range SO coupling like that in the Rashba effect. The SO interaction may be local, however, since orbitals of two neighboring irons hybridize directly and through As ion, the wave functions of electrons on those orbitals can overlap. It opens up a possibility for the effective inter-ion SO coupling.

For the intra-ion SO coupling, $\hat{H}$ has the form that immediately follows from (14):

$$\hat{H}_{SO}^{\text{intra}} = \left(\begin{array}{cc|cc} \hat{\varepsilon}_{\mathbf{k}} + i\frac{\lambda}{2}\hat{\varepsilon}^z & i\frac{\lambda}{2}\hat{\varepsilon}^x + \frac{\lambda}{2}\hat{\varepsilon}^y & 0 & 0 \\ i\frac{\lambda}{2}\hat{\varepsilon}^x - \frac{\lambda}{2}\hat{\varepsilon}^y & \hat{\varepsilon}_{\mathbf{k}} - i\frac{\lambda}{2}\hat{\varepsilon}^z & 0 & 0 \\ \hline 0 & 0 & \hat{\varepsilon}_{\mathbf{k}+\mathbf{Q}} + i\frac{\lambda}{2}\hat{\varepsilon}^z & i\frac{\lambda}{2}\hat{\varepsilon}^x + \frac{\lambda}{2}\hat{\varepsilon}^y \\ 0 & 0 & i\frac{\lambda}{2}\hat{\varepsilon}^x - \frac{\lambda}{2}\hat{\varepsilon}^y & \hat{\varepsilon}_{\mathbf{k}+\mathbf{Q}} - i\frac{\lambda}{2}\hat{\varepsilon}^z \end{array} \right). \quad (17)$$

For the inter-ion SO coupling we have

$$\hat{H}_{SO}^{\text{inter}} = \left(\begin{array}{cc|cc} 0 & 0 & i\frac{\lambda'}{2}\hat{\varepsilon}^z & i\frac{\lambda'}{2}\hat{\varepsilon}^x + \frac{\lambda'}{2}\hat{\varepsilon}^y \\ 0 & 0 & i\frac{\lambda'}{2}\hat{\varepsilon}^x - \frac{\lambda'}{2}\hat{\varepsilon}^y & -i\frac{\lambda'}{2}\hat{\varepsilon}^z \\ \hline i\frac{\lambda'}{2}\hat{\varepsilon}^z & i\frac{\lambda'}{2}\hat{\varepsilon}^x + \frac{\lambda'}{2}\hat{\varepsilon}^y & 0 & 0 \\ i\frac{\lambda'}{2}\hat{\varepsilon}^x - \frac{\lambda'}{2}\hat{\varepsilon}^y & -i\frac{\lambda'}{2}\hat{\varepsilon}^z & 0 & 0 \end{array} \right). \quad (18)$$

The total Hamiltonian matrix is the sum of matrices (17) and (18),

$$\hat{H} = \left(\begin{array}{cc|cc} \hat{\epsilon}_{\mathbf{k}} + i\frac{\lambda}{2}\hat{\epsilon}^z & i\frac{\lambda}{2}\hat{\epsilon}^x + \frac{\lambda}{2}\hat{\epsilon}^y & i\frac{\lambda'}{2}\hat{\epsilon}^z & i\frac{\lambda'}{2}\hat{\epsilon}^x + \frac{\lambda'}{2}\hat{\epsilon}^y \\ i\frac{\lambda}{2}\hat{\epsilon}^x - \frac{\lambda}{2}\hat{\epsilon}^y & \hat{\epsilon}_{\mathbf{k}} - i\frac{\lambda}{2}\hat{\epsilon}^z & i\frac{\lambda'}{2}\hat{\epsilon}^x - \frac{\lambda'}{2}\hat{\epsilon}^y & -i\frac{\lambda'}{2}\hat{\epsilon}^z \\ \hline i\frac{\lambda'}{2}\hat{\epsilon}^z & i\frac{\lambda'}{2}\hat{\epsilon}^x + \frac{\lambda'}{2}\hat{\epsilon}^y & \hat{\epsilon}_{\mathbf{k}+\mathbf{Q}} + i\frac{\lambda}{2}\hat{\epsilon}^z & i\frac{\lambda}{2}\hat{\epsilon}^x + \frac{\lambda}{2}\hat{\epsilon}^y \\ i\frac{\lambda'}{2}\hat{\epsilon}^x - \frac{\lambda'}{2}\hat{\epsilon}^y & -i\frac{\lambda'}{2}\hat{\epsilon}^z & i\frac{\lambda}{2}\hat{\epsilon}^x - \frac{\lambda}{2}\hat{\epsilon}^y & \hat{\epsilon}_{\mathbf{k}+\mathbf{Q}} - i\frac{\lambda}{2}\hat{\epsilon}^z \end{array} \right). \quad (19)$$

Number of electrons on a filled d -orbital of a single iron is 6 and we assume 2 orbitals to be completely filled, so for doping concentration x we have for the number of electrons on two irons $n = 2(6 - 2 - x) = 2(4 - x)$. Hereafter, the chemical potential μ is calculated to fulfill this criterion with $x = 0$.

3. Results for the Fermi surface and the band structure

Without the SO coupling ($\lambda = 0$) both the band structure and the Fermi surface are shown in Fig. 1. Topologically, the FS is different from that of Ref. [24] by the additional central FS sheet. Qualitatively, the FS is similar to those found in DFT calculations and five-orbital models [7,8].

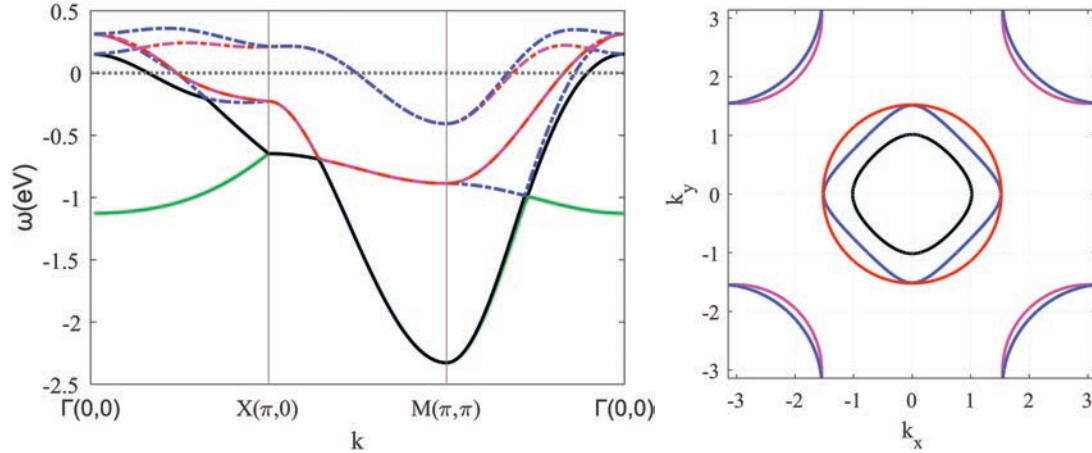


Fig. 1. Band structure and the Fermi surface in the three-orbital model without the SO coupling ($\lambda = \lambda' = 0$) in the 2-Fe BZ; $\mu = -0.0529$ eV. Bands are plotted relative to the chemical potential, so zero corresponds to μ

Effect of the intra-ion SO coupling is shown in Fig. 2, where $\lambda = 100$ meV and $\lambda' = 0$. The degeneracy of some bands is lifted at the Γ -point and the volume of the central FS sheet slightly changes. There is no topological difference to the FS results for the vanishing SO interaction.

If we switch on only the inter-ion SO interaction by setting $\lambda = 0$ and $\lambda' = 100$ meV, we end up with the splitting of the previously degenerate bands along the $X - M$ direction, see Fig. 3. It results in the topological change of the corresponding FS sheets around the M and symmetrical points — instead of the coinciding FS sheets along the $X - M$ direction, we have two separate ones.

Finite intra-ion and inter-ion SO couplings ($\lambda = \lambda' = 100$ meV) results in the band structure and FS shown in Fig. 4. Both the lifting of the degeneracy of some bands at the Γ -point and the splitting of the previously degenerate bands along the $X - M$ direction are present.

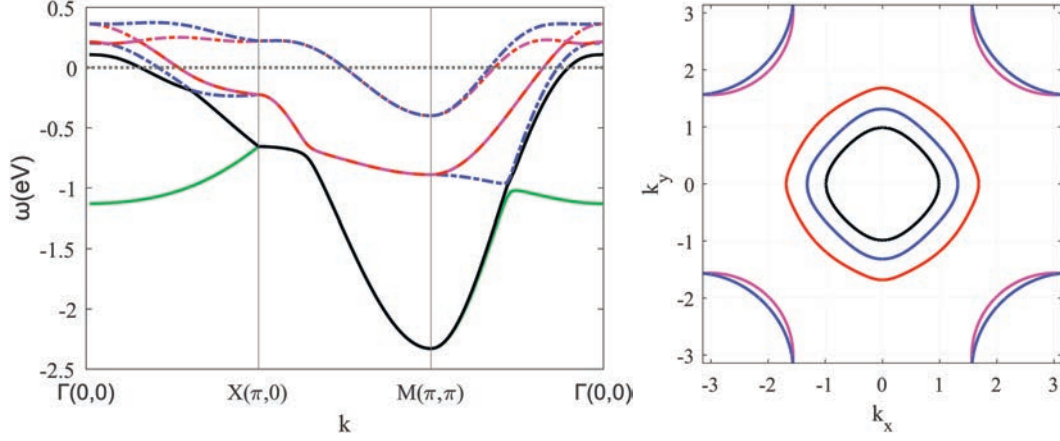


Fig. 2. The same as in Fig. 1 but for the finite intra-ion SO coupling $\lambda = 100$ meV and zero inter-ion SO coupling $\lambda' = 0$; $\mu = -0.0542$ eV

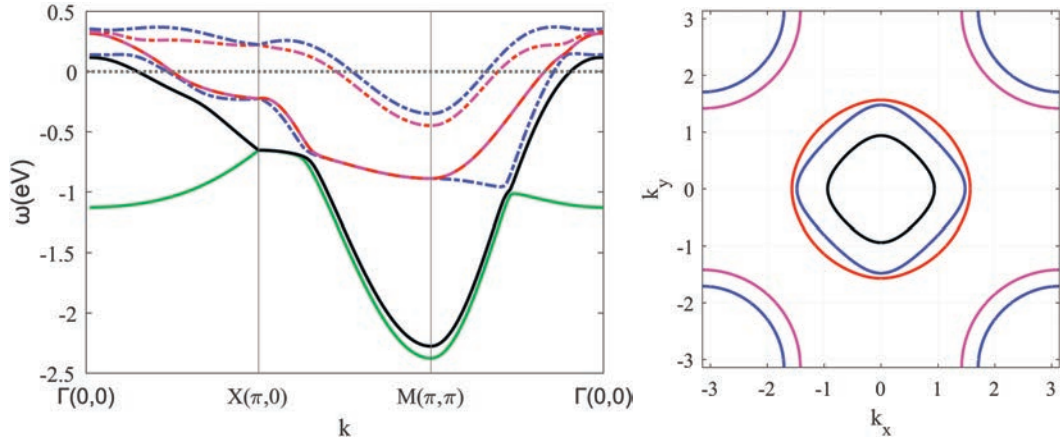


Fig. 3. The same as in Fig. 1 but for the finite inter-ion SO coupling $\lambda' = 100$ meV and zero intra-ion SO coupling $\lambda = 0$; $\mu = -0.0556$ eV

FS topologically different from the FS without the SO coupling, similar to the case of finite inter-ion SO interaction.

Conclusions

Here we modified the interorbital hopping term of the three-orbital model to avoid the unphysical violation of the C_4 symmetry. Introducing the intra- and inter-ion SO coupling constants λ and λ' allows us to separate the effect of the related couplings on the band structure and the Fermi surface. We show that the SO interaction results in the topological changes of the Fermi surface. In particular, the inter-ion part of the SO coupling leads to the splitting of two FS sheets around the M -point. Without SO, they may be represented as two ellipses rotated by the $\pi/2$ relative to each other. With the finite SO coupling, they resemble two concentric circles with the center at the M -point. Apart from that, the lifting of the degeneracy of some bands at the Γ -point and the splitting of the previously degenerate bands along the $X - M$ direction appear due to the SO interaction.

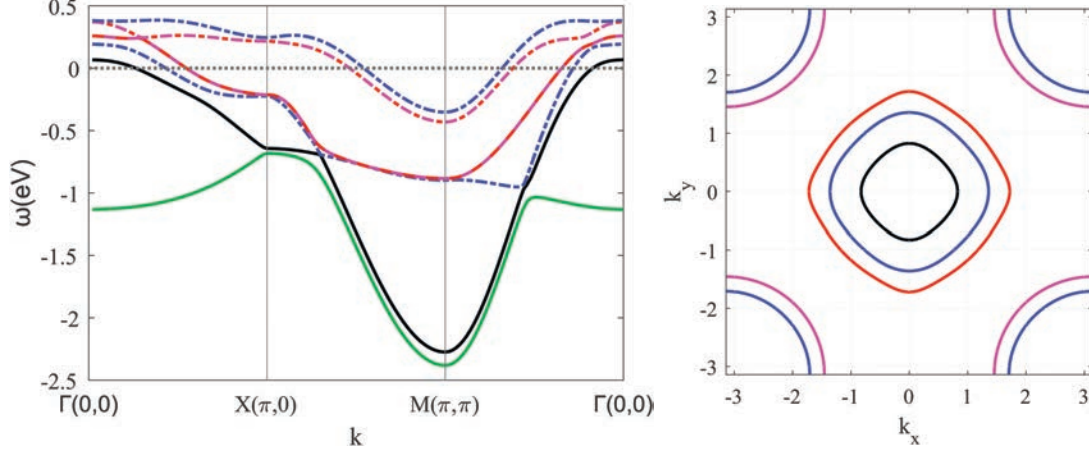


Fig. 4. The same as in Fig. 1 but for the finite intra- and inter-ion SO couplings, $\lambda = \lambda' = 100$ meV; $\mu = -0.0567$ eV

In the normal state, the change of the FS topology near the M -point should lead to the change of the dynamical spin response that is maximal at or near the wave vector connecting Γ and M -points. In the superconducting state, the obtained topology of the FS near the M -point may result in inequality between gaps on the inner and the outer FS sheets. Together with the gaps at the FS sheets near the Γ -point, it can give an explanation for the observation of the three different gaps in LiFeAs [17]. Another important consequence of the SO coupling is the possible realization of the nontrivial superconducting state. Since the spin becomes bad quantum number and the states are classified by the pseudospin, the superconducting order parameter may be composed of the spin-singlet and spin-triplet states [28].

Note that the study of LiFeAs material within the ten-orbital model [26] with the SO coupling results in qualitatively the same change of the FS topology and the band structure [27] if here we assume both finite λ and λ' . Folding procedure is absent in the ten-orbital model and the inter-ion SO interaction doesn't appear there in a direct form. Although the folding for the energies of the three-orbital model is a transparent procedure, for the wave functions it is a separate complicated task. Therefore, the introduction of the inter-ion SO interaction here in the form of Eq. (8) may be treated as an ansatz to simulate the results of the proper ten-orbital treatment that lacks the artificial folding procedure.

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Изменение топологии поверхности Ферми в трехорбитальной модели пниктидов железа со спин-орбитальным взаимодействием

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Аннотация. Изучено влияние спин-орбитального взаимодействия на зонную структуру и поверхность Ферми в трехорбитальной модели. Межорбитальная часть модели изменена для полного соответствия с симметрией решетки железа. Из-за наличия двух ионов железа в элементарной ячейке мы вводим внутри- и межзонную константы спин-орбитального взаимодействия для разделения эффектов соответствующих взаимодействий на зонную структуру и поверхность Ферми. И внутри, и межзонная части снимают вырождение некоторых зон в точке $\Gamma = (0, 0)$ и вырождение вдоль направления $(0, \pi) - (\pi, \pi)$ зоны Бриллюэна. Показано, что межзонная часть спин-орбитального взаимодействия приводит к изменению топологии поверхности Ферми, а именно к разделению листов поверхности Ферми вблизи точки $M = (\pi, \pi)$.

Ключевые слова: сверхпроводники на основе железа, спин-орбитальное взаимодействие, зонная структура, поверхность Ферми.

EDN: CYMZPJ

УДК 538.955

Study of Magnetic Anisotropy of Cobalt Films Separated by a Copper Layer Depending on the Thickness of the Layers

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Abstract. This work is devoted to the numerical study of the magnetic properties of the Co/Cu/Co system using the VASP software package. The influence of the contributions of magnetocrystalline anisotropy and shape anisotropy on the magnetic anisotropy energy was investigated. The dependence of the magnetic anisotropy, spin-orbit interaction, and dipole-dipole interaction energies on the thickness of the magnetic layer of cobalt was obtained.

Keywords: magnetic anisotropy, ultrathin magnetic films, spin-density functional theory.

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Introduction

Thin film magnetism is an active area of research with a particular impact on technological progress. Such materials are widely used in spintronics, nanoelectronics, and microelectronics devices. For numerical Monte Carlo simulation [1] and comparison of the obtained results with experiment, it is necessary to know the value of the anisotropy parameter. For structures based on ultrathin magnetic films, the mutual orientation of magnetic moments of different layers depends on the distance between them, therefore it is possible to calculate and predict at what thickness of the nonmagnetic layer the structural configuration will be more energetically favorable. Calculations of the structure at different directions of magnetic moments (in the film plane or perpendicular to it) give us knowledge about the magnitude and type of magnetic anisotropy, which is one of the most important physical properties affecting the magnetic microstructure, remagnetization processes, and other magnetic properties [2].

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1. Model and methods

In this study, calculations were performed using the density functional theory (DFT) with the projector-augmented wave (PAW) method and the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation. The Co/Cu/Co system was considered, with the following parameters set: the number of k-points (kpoints) was $24 \times 24 \times 1$, and the energy cutoff (encut) parameter was set at 600 eV.

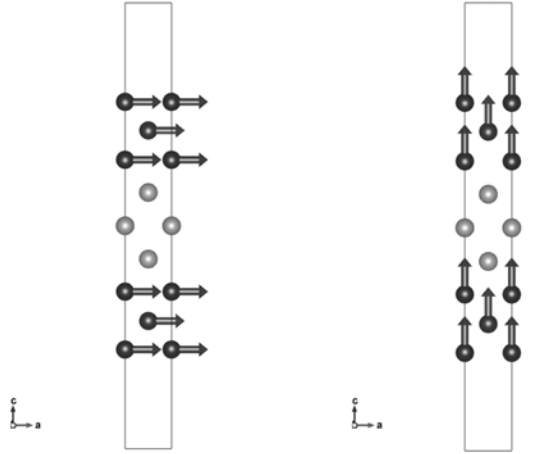


Fig. 1. The crystal structure of the Co/Cu/Co system with different orientations of the magnetic moments of the atoms

Fig. 1 shows the lattice arrangement in the Co/Cu/Co system with different orientations of the magnetic moments of the constituent atoms. The direction of the magnetic moments can influence the type of anisotropy of the system.

It is important to know the value of the magnetic anisotropy parameter as it is part of the Hamiltonian of the system with anisotropy.

$$H_{\parallel} = - \sum_{i,j} J_{i,j} \{ S_i^x S_j^x + S_i^y S_j^y + (1 - \Delta_{\parallel}) S_i^z S_j^z \}. \quad (1)$$

Let us find the dependence of the delta parameter $\Delta_{\parallel}$ - the magnetic anisotropy parameter on the number of layers of the magnetic material of cobalt.

Suppose that all spins are directed along the x-axis, then $S_i^x = S_j^x = S$; $S_i^y = S_j^y = 0$; $S_i^z = S_j^z = 0$. Let's substitute these values into (1) and get:

$$E_{\parallel} = - \sum_{i,j} J_{i,j} S^2. \quad (2)$$

Now let us direct the spins along the z-axis and, having done the same steps, we obtain:

$$E_{\perp} = - \sum_{i,j} J_{i,j} S^2 (1 - \Delta_{\parallel}). \quad (3)$$

The anisotropy parameter is therefore given by

$$E_{\parallel} < E_{\perp} \Rightarrow \Delta_{\parallel} = \frac{E_{\parallel} - E_{\perp}}{E_{\parallel}}. \quad (4)$$

The magnetic anisotropy energy $E_{MA} = E_{\perp} - E_{\parallel}$ can be defined as the difference of the total energies of the system at the direction of the magnetic moments in the film plane $E_{\parallel}$ and perpendicular to it $E_{\perp}$.

The magnetic anisotropy energy can be divided into two contributions: $E_{MA} = E_{MKA} + \Delta E_{dd}$, one of which is the magnetocrystalline anisotropy energy $E_{MKA} = \Delta E_{SOC}$ induced by spin-orbit coupling, and the other is the shape anisotropy energy ΔE_{dd} due to magnetic dipole interactions [3]. The shape anisotropy energy can be represented as the difference of the dipole-dipole interaction energies: $\Delta E_{dd} = E_{dd}(\vec{m}_{\perp}) - E_{dd}(\vec{m}_{\parallel})$, in turn the expression for finding the dipole-dipole interaction energy is as follows:

$$E_{dd} = \frac{1}{c^2} \sum_{i \neq j} \left[\frac{\vec{m}_i \cdot \vec{m}_j}{|\vec{R}_{ij}|^3} - 3 \frac{(\vec{m}_i \cdot \vec{R}_{ij})(\vec{m}_j \cdot \vec{R}_{ij})}{|\vec{R}_{ij}|^5} \right]. \quad (5)$$

2. Results

The information in Tab. 1 presents the computed magnetic moment values of atoms within cobalt films ranging from 1 to 8 monolayers, accompanied by varying copper thicknesses from 3 to 9 monolayers in comparison with experimental data.

Table 1. Values of magnetic moments of Co atoms in the film

№, ат.	N _{Co} =1			N _{Co} =2			N _{Co} =3		
	$\mu_{tot}/at, \mu_B$								
		⊥	Exp.		⊥	Exp.		⊥	Exp.
1	1.870	1.863	1.891 [4]	1.789	1.789	1.730 [5]	1.874	1.873	1.810 [6]
2				1.596	1.598		1.648	1.648	
3							1.654	1.654	

№, ат.	N _{Co} =4		N _{Co} =5		N _{Co} =6	
	$\mu_{tot}/at, \mu_B$					
		⊥		⊥		⊥
1	1.841	1.841	1.852	1.852	1.846	1.846
2	1.659	1.661	1.656	1.657	1.657	1.658
3	1.642	1.644	1.694	1.696	1.670	1.670
4	1.600	1.603	1.642	1.644	1.669	1.669
5			1.627	1.630	1.647	1.647
6					1.617	1.616

It can be concluded that the different spin configurations have a weak effect on the values of magnetic moments.

Fig. 2, we show the dependence of the energies of magnetic anisotropy, spin-orbit interaction and shape anisotropy on the thickness of the considered material. These values were calculated in the VASP [7–10] program complex using the [11] technique. The number of monolayers of cobalt magnetic material varied from 1 to 8. The thickness of copper was constant and equal to 3 monolayers.

The graph shows the curves describing the behavior of the magnetic anisotropy energy. It was calculated as $(\Delta E_{SOC} + \Delta E_{dd})$ as the sum of the energies of spin-orbit and dipole-dipole interactions and (E_{MA}) as the difference of the total energies of the system at different directions of magnetic moments. We observe concurrence in the behavior of curves.

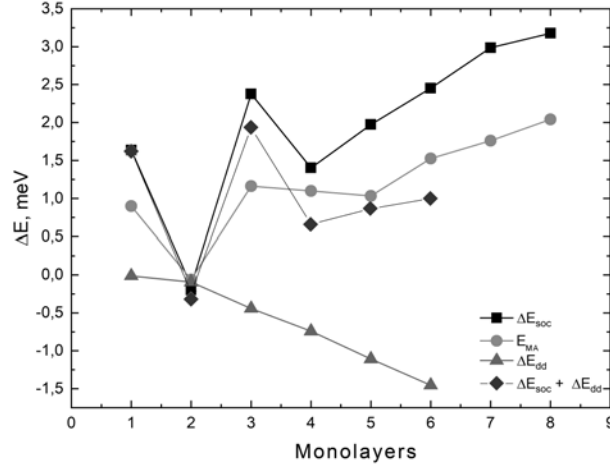


Fig. 2. Values of magnetic anisotropy, spin-orbit interaction and shape anisotropy energies as a function of the number of monolayers of magnetic cobalt material

The extended set of magnetic anisotropy, spin-orbit interaction and shape anisotropy energy values, taking into account the variable copper thickness (from 3 to 9 monolayers), is given in Tab. 2.

Table 2. Values of magnetic anisotropy, spin-orbit interaction and shape anisotropy energies as a function of the number of monolayers of copper substrate and cobalt magnetic material

N_{Co}	N_{Cu}	E_{MA}	ΔE_{SOC} , meV	E_{dd} , meV
1	3	0,898	1,636	-0,01218
	5	0,944		
	7	0,899	1,560	-0,00195
	9	0,930	1,641	-0,00139
2	3	-0,062	-0,142	-0,095
	5	0,092	0,053	-0,080
	7	0,038	0,073	-0,077
	9	-0,104	0,043	-0,076
3	3	1,161	2,377	-0,442
	5	1,508	2,257	-0,422
	7	1,273	2,353	-0,415
	9	1,152	2,311	-0,412

Weak magnetic anisotropy was found for the thickness of the magnetic material of cobalt 2 monolayers. The values of the spin-orbit interaction energy difference obtained by us are significantly larger than the energy difference of the dipole-dipole interaction, hence we can conclude that the main influence on the magnetic anisotropy energy is the contribution of magnetocrystalline anisotropy due to the spin-orbit interaction.

The ferromagnetic configuration of the system was calculated for facet (100) with a copper thickness varying from 3 to 9 monolayers and cobalt thickness from 1 to 8 monolayers.

Fig. 3 shows the calculated values of the anisotropy parameter compared with the values $\Delta = 1 - \frac{T_c(N)}{T_c(\infty)}$ obtained from the experimental critical temperatures $T_c(N)$ for Co/Cu(100) [12, 13] as a function of the number of cobalt monolayers. The data obtained are in good

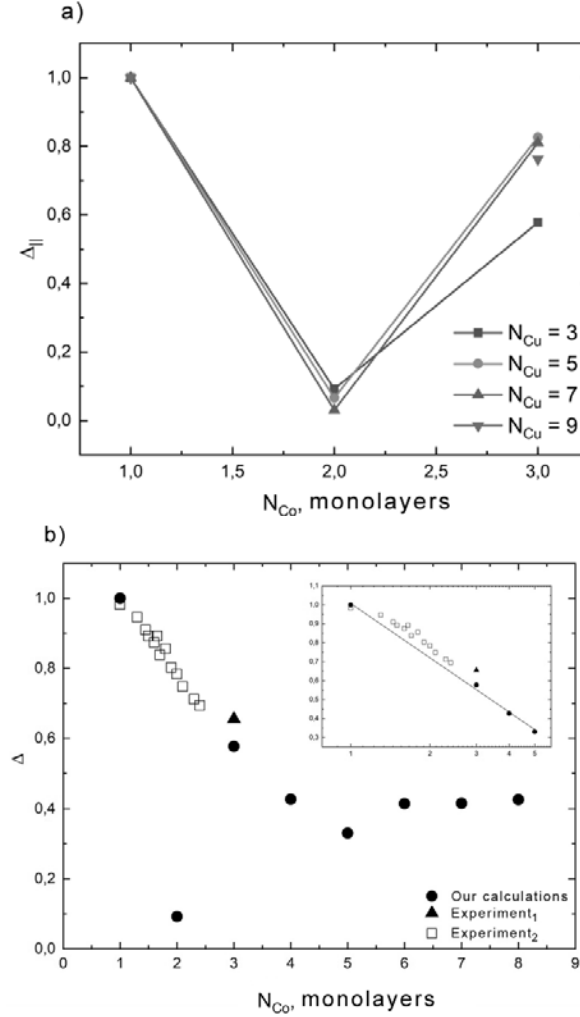


Fig. 3. (a) Calculated values of the magnetic anisotropy parameter as a function of the number of cobalt and copper monolayers (b) Calculated values of the anisotropy parameter as a function of the number of cobalt monolayers compared to values $\Delta = 1 - \frac{T_c(N)}{T_c(\infty)}$ obtained from experiments $T_c(N)$ for Co/Cu(100) (Experiment 1) [12] and (Experiment 2) [13]. The inset in the main plot shows the approximation of the values of the magnetic anisotropy parameter as a function of the number of cobalt monolayers on a logarithmic scale

agreement with experiment. The inset in the main plot shows the approximation of the values of the magnetic anisotropy parameter as a function of the number of cobalt monolayers (except $N_{Co} = 2$) on a logarithmic scale. The parameter $\lambda = 1.007$ calculated by us as the slope of the approximation line is close to the value of the scaling parameter $\lambda = 1.02$ given in the paper by [13]. For the cobalt thickness of 2 monolayers, a "dropout" of the value of the magnetic anisotropy parameter associated with the change of the magnetic anisotropy energy both in sign and value is observed, which is summarized in Tab. 2. For the cobalt thickness of 2 monolayers, a "dropout" of the magnetic anisotropy parameter value associated with the change of the magnetic anisotropy energy both in sign and value is observed, which is shown in Tab. 2. Increasing the surface cell size (from 1 atom per layer to 4) and the thickness of the non-magnetic interlayer (from 3 to 9 monolayers) did not eliminate this feature. In contrast to the experiment, where

we considered partially filled layers, i.e., the film thickness varied from 1 to 2 monolayers with a step of 0.2, we considered a thickness multiple of a monolayer. The calculation of magnetic anisotropy requires the use of non-collinear calculations, the large resource intensity of which strongly depends on the number of atoms in the supercell. In the case of fully filled monolayers, it is sufficient to use 1 atom per layer, which was used in these calculations. In the case of partially filled monolayers, it is necessary to take 4 or 9 atoms in a layer, which increases the resource intensity of calculations by an order of magnitude.

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DOI: 10.1103/PhysRevB.49.3962

Исследование магнитной анизотропии пленок кобальта, разделенных слоем меди, в зависимости от толщины слоев

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Аннотация. Данная работа посвящена численному исследованию магнитных свойств системы Co/Cu/Co с использованием программного пакета VASP. Исследовано влияние вкладов магнито-кристаллической анизотропии и анизотропии формы на энергию магнитной анизотропии. Получена зависимость энергий магнитной анизотропии, спин-орбитального взаимодействия и диполь-дипольного взаимодействия от толщины магнитного слоя кобальта.

Ключевые слова: магнитная анизотропия, сверхтонкие магнитные пленки, теория функционала спиновой плотности.

EDN: POECNP

УДК 538.945

Symmetry Analysis of Radial Profiles of Magnetic Skyrmions

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Abstract. The search for analytical profiles of chiral magnetic structures such as 2D magnetic skyrmions (MS) is important for their theoretical study. Since the Euler–Lagrange (EL) equations for such excitations are not solved exactly, the MSs are described using analytical ansatzs. In this work, we validate one of the widely used ansatzs based on a symmetry analysis of the 1D analog of the EL equations, which characterizes the radial profile of the MS. As a development of this approach, a profiles of skyrmion bags are proposed.

Keywords: magnetic skyrmions, chiral interactions, symmetry analysis of differential equations, skyrmion bags.

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Introduction

Skyrmions are topologically nontrivial field configurations that are solutions of nonlinear differential equations of physics. At the first time, such solutions has been considered by T. Skyrme in nuclear physics for the baryon field [1, 2]. Later, the similar fields, $\vec{m}(\vec{r})$, were found in magnetic systems [3] and liquid crystals, where $\vec{m} \in \mathbb{S}^2$, and $\vec{r} \in \mathbb{R}^{2,3}$. Recently, magnetic skyrmions were found experimentally [4–6], arousing scientific interest in themselves as promising objects for logic and memory devices of a new generation [7, 8]. The main practical interest to MS is due to their topological stability or, in other words, the inability to convert a nontrivial distribution $\vec{m}(\vec{r})$ with a topological index Q into the another one (corresponding to homotopy class with different Q) without overcoming a high energy barrier. The vast majority of studies considered the MSs with $Q = -1$ (in what follows we fix the chirality of the MS). However, the more exotic magnetic vortices has recently been predicted numerically [9–11] and experimentally confirmed [12, 13]. The last ones are, for example, magnetic skyrmionium ($Q = 0$) or skyrmion bags of nontrivial morphology (arbitrary Q).

The finding of new 2D magnetic structures actualized the problem of their analytical description. So, when describing the ordinary MS with $Q = -1$, the three types of analytical ansatz are widely used: i) linear, ii) exponential, and iii) 2π -domain wall ansatz. In this paper, we

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demonstrate that the type iii) ansatz correspond to the symmetry of EL equations describing the MS profile in the XoZ plane. We also generalize type iii) ansatz to the case of skyrmion bags.

1. Energy and profile of the magnetic skyrmion

Let us consider a continuum model of a 2D chiral magnet with competing interactions. We introduce the magnetization field $\vec{m}(\vec{r})$: $|\vec{m}| = 1$, $\vec{r} \in \mathbb{R}^2$. Then, the excitation energy of an arbitrary configuration, $\vec{m}(\vec{r})$, compared to the ferromagnetic case, $m_z(r) \equiv 1$, has the form:

$$E_{cont} = \int \int \left[\mathcal{J} \cdot ((\nabla m_x)^2 + (\nabla m_y)^2 + (\nabla m_z)^2) + \mathcal{D} \cdot \left(m_z \frac{\partial m_x}{\partial x} - m_x \frac{\partial m_z}{\partial x} + m_z \frac{\partial m_y}{\partial y} - m_y \frac{\partial m_z}{\partial y} \right) + \mathcal{A} \cdot (1 - m_z^2) \right] dx \wedge dy \quad (1)$$

The first two terms describe the Heisenberg and the Dzyaloshinskii–Moriya interactions of the strengths $\mathcal{J}$ and $\mathcal{D}$, respectively. These interactions are competing, making it possible to exist of non-trivial configurations, such as MS. The third term describes the "easy-axis" single-ion anisotropy with an amplitude of $\mathcal{A}$, which defines a chosen space direction along the z -axis.

The MSs are axially symmetric configurations, and are standardly described by the parametrization:

$$m_x = \sin \tilde{\Theta}(r) \cos \varphi, \quad m_y = \sin \tilde{\Theta}(r) \sin \varphi, \quad m_z = \cos \tilde{\Theta}(r). \quad (2)$$

Here φ is the azimuthal magnetization angle, which coincides with the polar angle of the vector $\vec{r}$ in XoY plane, $\tilde{\Theta}(r)$ is the «skyrmion angle», depending on the polar radius r (the origin of polar coordinates coincides with MS center). The $\tilde{\Theta}$ vs. r dependence can be found from the EL equation:

$$l_w^2 \Delta_r \tilde{\Theta} - \frac{(l_w^2 + r^2)}{2r^2} \sin 2\tilde{\Theta} + 2\mathcal{E} \frac{\sin^2 \tilde{\Theta}}{r/l_w} + \frac{\mathcal{H}}{2\mathcal{K}} \sin \tilde{\Theta} = 0. \quad (3)$$

where $l_w = \sqrt{\frac{\mathcal{J}}{\mathcal{K}}}$, $\mathcal{E} = \frac{\mathcal{D}}{2\sqrt{\mathcal{J}\mathcal{K}}}$, $\Delta_r \tilde{\Theta}(r) = \frac{1}{r} (\partial_r \tilde{\Theta} + \partial_r^2 \tilde{\Theta} r)$ – the Laplace operator in spherical coordinates. The equation (3) can be obtained by substituting (2) into the functional (1) and varying the latter against the $\tilde{\Theta}(r)$ and its derivatives. By now, no exact solution of Eq. (3) has been obtained. In order to simulate the numerical solutions the 2π -domain wall ansatz is widely used

$$\tilde{\Theta}_{-1}(r, R, w) = 2 \operatorname{arctg} \left(\frac{\cosh R/w}{\sinh r/w} \right) \quad (4)$$

with two parameters R and w , where w describes the domain wall width, whereas R is the distance from the skyrmion center to domain wall middle.

Due to complexity of Eq. (3), let us consider a simplified situation where

$$m_x = \sin \Theta(x), \quad m_y = 0, \quad m_z = \cos \Theta(x). \quad (5)$$

Such situation correspond to the flat domain wall, varying along the Ox axis. Since morphologically MS is an axially symmetric domain wall, one can assume that $\Theta(x)$ approximates the radial dependence $\tilde{\Theta}(r)$ of the skyrmion angle for large MS (see Fig. 1). This assumption does

not take into account azimuthal distortions of the MS texture. However, for this picture, it is possible to obtain a simple EL equation

$$\frac{d^2\Theta}{dx^2} = \frac{\mathcal{A}}{2\mathcal{J}} \sin 2\Theta \quad (6)$$

for the angle $\Theta(x)$. The exact solution

$$\Theta(x) = 2 \arctan(e^{x/w}), \quad (7)$$

of Eq. (6) can be obtained if $\mathcal{J} > 0$, $\mathcal{A} > 0$. It describes the π -domain wall, where $\vec{m}$ wraps by an angle π in the XoZ plane. The solution (7) is a partial one of the Eq. (6). Another solutions can be obtained by symmetry analysis of the Eq. (6).

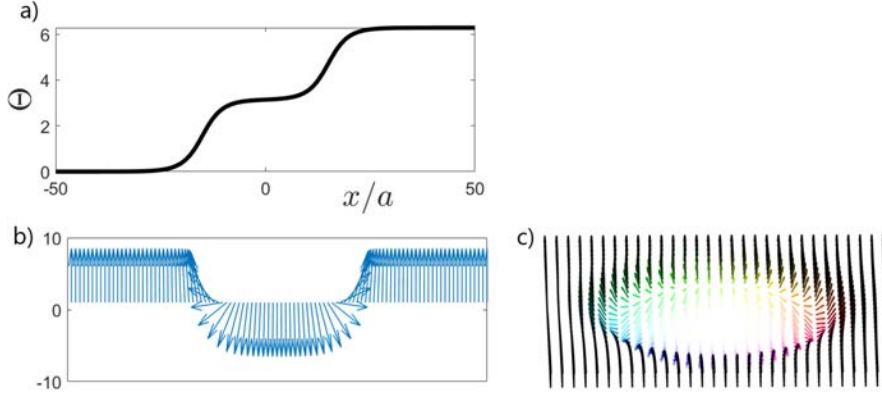


Fig. 1. a) visualization of a 2π -domain wall using a vector field (5) with an angle function $\Theta(x)$ defined by Eq. (22) with parameters $R = 15$, $w = 3$; b) the function $\Theta(x)$ profile; c) visualization of the vector field MS (see Eq. (2)) with the function $\Theta(r)$, see Eq. (4), with parameters $R = 15$, $w = 3$. The lengths are defined in the lattice parameter $a = 1$

2. Symmetry analysis of the approximate equation

Let us perform a symmetry analysis of the Eq. (6), and first consider its discrete symmetries. It can be seen that the equation retains under transformations: $x \rightarrow -x$ and $\Theta(x) \rightarrow -\Theta(x)$. Then, taking into account Eq. (7), one get a set of four exact solutions of Eq. (6):

$$\Theta(x) = \pm 2 \arctan(e^{\pm x/w}). \quad (8)$$

Now consider the continuous (Lie) symmetries of the equation. Then the replacing of variables

$$x' = F(t; x, \Theta, \dot{\Theta}), \quad \Theta' = \Phi(t; x, \Theta, \dot{\Theta}), \quad \dot{\Theta}' = \Psi(t; x, \Theta, \dot{\Theta}) \quad (9)$$

becomes continuously dependent on the parameter t , forming an infinite set. Hereafter $\dot{\Theta} = d\Theta/dx$, and Θ' denotes a new function retaining the form of Eq. (6) in new variables. In contrast to the discrete transformations (8), it can be shown that continuous ones should involve both the variable x as well as the functions Θ and $\dot{\Theta}$, see Eq. (9). The main goal of this changing is to find the mappings, that retain the form of Eq. (6). Then, knowing the discrete

set of solutions (8), as well as the infinite set of mappings (9), one can obtain new solutions that depends on t .

It is convenient to implement this idea in the framework of the so called geometric approach to differential equations [16]. Let us consider a second-order equation of the general form

$$\mathcal{F}(x, \Theta, \dot{\Theta}, \ddot{\Theta}) = \ddot{\Theta} - f(x, \Theta, \dot{\Theta}), \quad (10)$$

as a surface in the space of *independent* parameters $\mathcal{D} = \{x, \Theta, \dot{\Theta}, \ddot{\Theta}\}$. In the geometric approach, the replacement of variables (9) corresponds to continuous paths in the space $\mathcal{D}$, depended on the starting points $\{x_0, \Theta_0, \dot{\Theta}_0, \ddot{\Theta}_0\}$ (in fact being a discrete set of solutions (8)) on the surface $\mathcal{F}$. This idea is visualized in Fig. 2 in the reduced space $\{x, \Theta, \dot{\Theta}\}$ for simplicity. By this

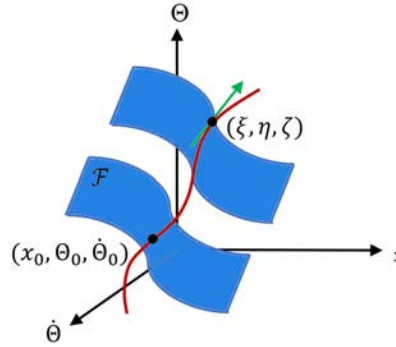


Fig. 2. Visualization of the idea of continuous paths in space depending on the initial points on the initial surface

approach, the invariance of the Eq. 10 under the mapping (9) means preserving the form of the surface $\mathcal{F}$ with a deformation where the points $\{x_0, \Theta_0, \dot{\Theta}_0, \ddot{\Theta}_0\}$ move along the paths defined by Eq. (9). At the same time, these paths determine at each point of the surface $\mathcal{F}$ the tangent vectors $\vec{l} = (\xi, \eta, \zeta)$:

$$\xi = \left. \frac{\partial x'}{\partial t} \right|_{t=0} = \left. \frac{\partial F}{\partial t} \right|_{t=0}; \quad \eta = \left. \frac{\partial \Theta}{\partial t} \right|_{t=0}; \quad \zeta = \left. \frac{\partial \dot{\Theta}}{\partial t} \right|_{t=0}.$$

Then, the problem of invariance of the surface $\mathcal{F}$ is reduced to finding its tangents with zero directional derivative:

$$\begin{aligned} \left. \frac{d}{dt} \mathcal{F} \right|_{t=0} &= \frac{\partial \mathcal{F}}{\partial x'} \frac{\partial x'}{\partial t} + \frac{\partial \mathcal{F}}{\partial \Theta'} \frac{\partial \Theta'}{\partial t} + \frac{\partial \mathcal{F}}{\partial \dot{\Theta}'} \frac{\partial \dot{\Theta}'}{\partial t} = \\ &= \left(\xi \frac{\partial \mathcal{F}}{\partial x'} + \eta \frac{\partial \mathcal{F}}{\partial \Theta'} + \zeta \frac{\partial \mathcal{F}}{\partial \dot{\Theta}'} \right) = (\xi, \eta, \zeta) \cdot \nabla \mathcal{F}|_{t=0} = 0. \end{aligned} \quad (11)$$

Hence, the generator of Lie symmetries of Eq. (10) is determined as

$$\begin{aligned} \hat{O} &= \xi \partial_x + \eta \partial_\Theta + \zeta \partial_{\dot{\Theta}}, \\ \hat{O} \mathcal{F} &= 0. \end{aligned} \quad (12)$$

This property, together with the Eq.(10), defines the necessary conditions for finding the functions ξ , η and ζ . However, these conditions are not complete yet. So, considering in the space $\mathcal{D}$ the variables x , Θ , $\dot{\Theta}$ and $\ddot{\Theta}$ as independents, we neglected in fact the bonds between them:

$$d\Theta - \dot{\Theta} dx = 0, \quad d\dot{\Theta} - \ddot{\Theta} dx = 0. \quad (13)$$

In order to account the last ones, let us continue the space $\mathcal{D} = \{x, \Theta, \dot{\Theta}, \ddot{\Theta}\}$ to the expanded space $\tilde{\mathcal{D}} = \{x, \Theta, \dot{\Theta}, \ddot{\Theta}, d\Theta, d\dot{\Theta}, dx\}$, continuing the symmetry generator as well:

$$\hat{O} \rightarrow \hat{\tilde{O}} = \xi \partial_x + \eta \partial_\Theta + \zeta \partial_{\dot{\Theta}} + \dots$$

Then, in the expanded space, besides the invariance of the equation, it is necessary to consider the invariance of differential bonds in a similar way. The latter is the subject of the theory of continuations described in [16]. This approach allows us to consider the invariance of the *differential*, and not the *algebraic* equation. Then, following the Ref. [16], the sufficient conditions for searching ξ , η and ζ are reduced to the so called system of splittings:

$$\eta_{xx} = f_x \xi + f_\Theta \eta - f \eta_\Theta + 2f \xi_x; \quad (14)$$

$$2\eta_{x\Theta} - \xi_{xx} = 3f \xi_\Theta; \quad (15)$$

$$\eta_{\Theta\Theta} - 2\xi_{x\Theta} = 0; \quad (16)$$

$$-\xi_{\Theta\Theta} = 0. \quad (17)$$

Here, the lower indices indicate the derivatives of the corresponding variables. In our case, $f = a \sin 2\Theta$, see Eq. (6) and Eq. (10).

Starting solving system from the last equation (16), we find:

$$\xi = A\Theta + B; \quad A = A(x), \quad B = B(x) \quad (18)$$

Next, from Eq.(15) we find also solution for η :

$$\eta = A'\Theta^2 + C\Theta + D; \quad C = C(x), \quad D = D(x). \quad (19)$$

Going to Eq. (14), we obtain that $3A''\Theta + 2C' = B'' + 3aA \sin 2\Theta$. Differentiating this equation by Θ , we get:

$$\frac{A''(x)}{A(x)} = f_\Theta(\Theta) \Rightarrow \frac{A''}{A} = f_\Theta = K = \text{const}. \quad (20)$$

Finally, consider Eq. (14). So that the solution is not trivial, let us put $A = 0$. Then, we get: $C_{xx}\Theta + D_{xx} = 2aC\Theta \cos 2\Theta + 2Da \cos 2\Theta - aC \sin 2\Theta + 2B_x a \sin 2\Theta$. Expanding $\cos 2\Theta$ and $\sin 2\Theta$ into a Taylor series and considering coefficients at various degrees of Θ as independent ones, we find that $D, C, B_x = 0$. If we consider again the operator $\hat{O}$, we will see that $\hat{O} = \partial_x$. Thus, the group of Lie symmetries of Eq. (6) is poor and reduces to translations $x \rightarrow x + C$. As a result, the analysis of Eq.(6) allows to generate from the exact solution (7) a set of others:

$$\Theta(x) = \pm 2 \arctan(e^{(\pm x + C)/w}). \quad (21)$$

Such solutions describe a set of domain walls, with the middle at points $x = \pm C$, where the angle $\Theta(x)$ changes from 0 to $\pm\pi$ (or from $\pm\pi$ to 0) when x changes from $x = -\infty$ to $x = \infty$. It can be shown that if two domain walls $\Theta_1(x)$ and $\Theta_2(x)$ from the class of solutions (21) overlap slightly, then the sum of $\Theta = \Theta_1 + \Theta_2$ is also an approximate solution of Eq. (6). This allows to build combinations of solutions of the type 0π - or 2π -domain walls. The latter one is:

$$2 \arctan(e^{\frac{-x-R}{w}}) + 2 \arctg(e^{\frac{x-R}{w}}) = 2 \arctan\left(\frac{\cosh \frac{x}{w}}{\sinh \frac{R}{w}}\right) \quad (22)$$

It is a 1D variant of the already mentioned ansatz for MS, see Eq. (22).

Fig. 1a shows the dependence of the function $\Theta(x)$ (22). Next, Fig. 1b shows a visualization of the vector field $\vec{m}(x)$, see Eq. (5), for such a domain wall. Finally, Fig. 1c shows a similar visualization of the MS profile, see Eq. (2), where the free parameters for the functions $\hat{\Theta}(r)$ and for $\Theta(x)$ are chosen the same. It can be seen that the MS profile can be considered with good accuracy as a 1D 2π - "rotation body" of the domain wall. This makes it possible to generalize the MS profile to more exotic magnetic structures.

3. Some generalizations of the skyrmion profile

Summarizing above, it is possible to construct other combinations that will describe axially symmetric domain walls. So, by choosing a combination of four exact solutions of Eq. (6) with replacing $x \rightarrow r$, one can obtain an expression for the so-called skyrmionium angle

$$\tilde{\Theta}_0(r) = \pi - 2 \arctan \left(\frac{\cosh(R_1/w_1)}{\sinh(r/w_1)} \right) - 2 \arctan \left(\frac{\cosh(R_2/w_2)}{\sinh(r/w_2)} \right), \quad (23)$$

generalizing the two-parameter MS ansatz (4). Magnetic excitation, which profile is described by (2) and (23) is called magnetic skyrmionium. It has a topological charge $Q = 0$, which is depicted by the lower subscript on the radial function $\Theta_0(r)$. A comparison of the radial dependencies of the skyrmion and skyrmionium magnetizations, $m_z = \cos \tilde{\Theta}$, is shown in Fig. 3. It can be seen that the latter can be considered as an axially symmetric pair of successive domain walls. At the same time, in (23), the parameters w_1 and R_1 (w_2 and R_2) characterize the width and distance to the middle of the first (second) domain wall. Recently, magnetic skyrmioniums have been discovered experimentally [12].

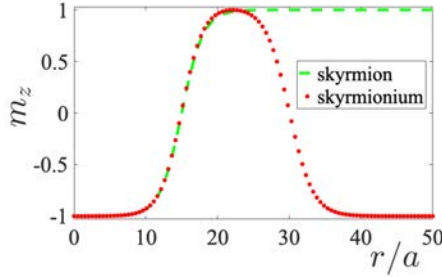


Fig. 3. Dependencies of the component $m_z = \cos \tilde{\Theta}$ for the skyrmion (4) and the skyrmionium (23) on the polar radius r . Parametrization for the skyrmionium corresponds to situations where the component $m_z = \cos \Theta_0 = -1$ at the points $r = 0$ and $r = \infty$. The parameters of ansatz (4) and (23) in units of a are: $R = R_1 = 15$, $R_2 = 30$, $w_1 = 3$, $w_2 = 3.3$

The knowledge of the skyrmionium profile allows us to construct profiles of C_g -symmetric exotic magnetic structures — the skyrmion bags. They have a topological charge $Q = g - 1$ and have recently been discovered both numerically [9–11] and experimentally [13]. Numerical experiments have shown [9] that to stabilize skyrmion bags with $Q = g - 1$, g magnetic skyrmioniums located near to each other should be chosen as the seed magnetic configuration. A similar idea can be developed in the analytical construction of skyrmion bags profiles. We will consider the skyrmionium (23) to be the generating function of the skyrmion bags. Consider the cut of the magnetic skyrmionium m_z -profile by the plane XoZ . This dependence has a two-peak structure, which can be obtained by mirroring the dependency in Fig. 3. Now, let us shift the origin from the point $(x, y) = (0, 0)$ to the peak point, $(x, y) = (-a, 0)$, and move on to the polar system centered at the latter. So, we first replace the variables in dependence $\vec{m}(r(\tilde{x}, \tilde{y}), \varphi(\tilde{x}, \tilde{y}))$ in the following way:

$$\begin{pmatrix} \tilde{x} \\ \tilde{y} \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \tilde{x} \\ \tilde{y} \end{pmatrix} + \begin{pmatrix} a \\ 0 \end{pmatrix}, \quad (24)$$

followed by the second replacement $\tilde{x} \rightarrow r'$, $\tilde{y} \rightarrow 0$. At this stage, we have introduced a new parametrization of the magnetic skyrmionium, whose field now depends on the variables r' , θ of

the new polar system: $\vec{m} = \vec{m}(r', \theta)$. Now, let us make the third substitution: $r' \rightarrow r^g$, $\theta \rightarrow g \cdot \varphi$, where $r \in [0, \infty)$, $\varphi \in [0, 2\pi)$. As a result, the g replicas of the skyrmionium profile are obtained, that are slightly deformed at small r and large g . Let us write the Eq. (24) in matrix form: $\tilde{\vec{r}} \rightarrow \vec{r} = \tilde{O}(\theta) \cdot \tilde{\vec{r}} + \tilde{\vec{a}}$ and apply the described transformations to the function $\tilde{\Theta}_0(\vec{r})$, see Eq. (23). Thus, one obtain the so-called angle of skyrmion bags $\tilde{\Theta}_g(r, \varphi)$:

$$\Theta_0(\tilde{\vec{r}}) \rightarrow \tilde{\Theta}_g(r, \varphi) = \pi - 2 \arctan \left(\frac{\cosh(R_1/w_1)}{\sinh(\sqrt{a^2 + r^{2g} + 2a \cdot r \cdot \cos g\varphi} / w_1)} \right) - 2 \arctan \left(\frac{\cosh(R_2/w_2)}{\sinh(\sqrt{a^2 + r^{2g} + 2a \cdot r \cdot \cos g\varphi} / w_2)} \right). \quad (25)$$

The profiles of the components $m_z = \cos \tilde{\Theta}_g$ are shown in Fig. 4 for the cases: a) $g = 1$ – magnetic skyrmionium; b) $g = 2$ and c) $g = 3$. It can be shown that functions b) and c) approximate with a good accuracy the numerical profiles of skyrmion bags with topological charges $Q = 1$ and $Q = 2$, respectively.

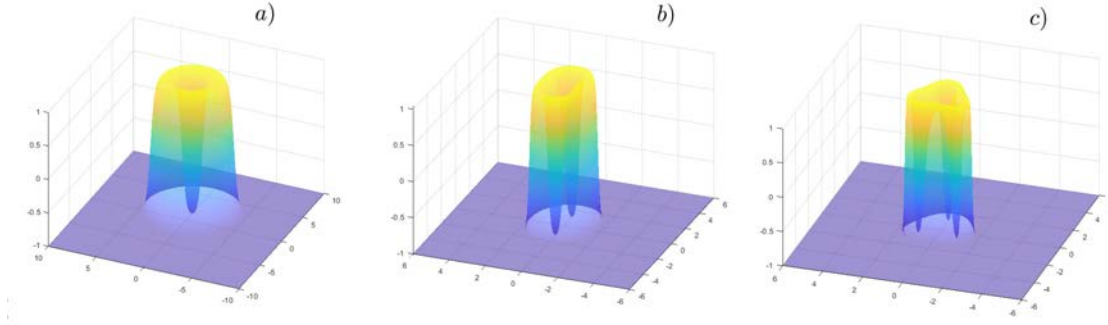


Fig. 4. Dependencies $m_z(r) = \cos \Theta_g$, where Θ_g is defined by (25). Parameters $\{R_1, R_2, w_1, w_2, g\}$ are: a) $\{7.66, 26.47, 2.80, 3.08, 1\}$, b) $\{7.86, 26.45, 2.83, 3.08, 2\}$, c) $\{7.56, 26.34, 2.90, 3.09, 3\}$

However, note that the Eq. (25) approximates the numerical profiles of the skyrmion bags with good accuracy only for small topological indices Q . For large Q , the peculiarities in the analytical profiles, that are absent in numerical one appear. A more correct construction of analytical profiles for skyrmion bags will be the subject for further studies.

Summary

The Euler–Lagrange equations are found for magnetic excitations in 2D chiral magnets with competing interactions: Heisenberg, Dzyaloshinskii–Moriya, and the "easy axis" single-ion anisotropy as well. These equations were found for two classes of magnetic excitations: i) axially symmetric magnetic vortices, such as magnetic skyrmions, and ii) flat domain walls. It is noted that in the case i) the EL equations cannot be solved analytically, and their numerical solutions can be approximated by an axially symmetric 2π -domain wall analytical ansatz. Whereas in case ii), the partial solutions as well as the Lie symmetries of corresponding EL equations can be obtained exactly, despite the approximate character of the description.

The symmetry analysis of the EL equations for the case ii) was carried out in the framework of the so-called "continuation theory". The main idea of latter is to find new exact solutions

from the previous ones with the use of the symmetries of the EL equation. It was shown that the EL equations describing the flat domain walls (case ii) have only the translation symmetry of the domain wall middle. Moreover, the sum of solutions for several domain walls displaced by a significant distance relative to each other is also an approximate solution of the equation.

The latter feature allowed us to justify the choice of a 2π -domain wall type ansatz for a magnetic skyrmion. Moreover, continuing this idea for more complex combinations of closed domain walls, including axially asymmetric ones, allowed us to propose analytical ansatzs for more exotic magnetic vortex structures — the skyrmion bags, recently discovered both numerically and experimentally. A detailed analysis and justification of the proposed parametrization will be the subject for further study.

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Анализ симметрии радиальных профилей магнитных скирмионов

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Аннотация. Поиск аналитических профилей киральных магнитных структур типа 2D магнитных скирмионов (МС) является важным при их теоретическом описании. Поскольку уравнения Эйлера–Лагранжа (ЭЛ) для таких возбуждений не решаются точно, описание МС проводят с помощью аналитических пробных функций — анзацев. В настоящей работе проводится обоснование одного из широко используемых анзацев на основе симметричного анализа 1D версии уравнений ЭЛ, определяющего радиальный профиль МС. В развитии такого подхода предлагаются профили скирмионных мешков с топологическими зарядами $Q \geq 0$.

Ключевые слова: магнитные скирмионы, киральные взаимодействия, симметричный анализ дифференциальных уравнений, скирмионные мешки.

EDN: BVFFHK

УДК 538.945

Fermion Parity of Phases Supporting Multiple Majorana Modes in a Superconducting Nanowire

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Abstract. The fermion parity of the ground state is determined in various topological phases of the semiconducting nanowire under external magnetic field with proximity-induced superconductivity and strong spin-orbit interaction. Electron hopping as well as spin-flip hopping due to spin-orbit coupling and superconducting pairings in the second coordination sphere are taken into account. The connection between the fermion parity and the parity of the BDI topological invariant is shown. The formation of topological phases with three and four pairs of Majorana modes has been demonstrated.

Keywords: Fermion parity, superconducting nanowire, Majorana modes, topological invariant, topological phase diagram.

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Majorana modes (MMs) that emerge in topological superconductors are of both fundamental and practical interest. This is due to their spatial nonlocality and the potential for implementing non-Abelian statistics through permutations. These characteristics suggest the possibility of utilizing MMs for quantum computing, providing protection against decoherence processes [1–4]. Permutations of non-Abelian anyons result in alterations to the system's ground state wave function. Unlike Abelian anyons (such as fermions and bosons), this change is not solely confined to the introduction of a phase factor.

One of the most renowned systems where the realization of MMs is feasible involves an InSb (or InAs) semiconductor nanowire brought into contact with an s-wave superconductor. For instance, this could be achieved by depositing a 3–5 nm thick layer of aluminum on the nanowire's surface and subjecting it to an external magnetic field [5, 6]. The formation of MMs in such heterostructures was initially predicted in several studies [7–9]. The joint implementation of strong spin-orbit coupling, spin-singlet s-wave superconductivity, and a magnetic field allows the realization of a Kitaev chain regime with effective triplet p-wave superconducting pairings [10]

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in the system under consideration. It's essential to note that the experiments have yet to conclusively confirm that the observed signals, such as the conductance peak at zero bias voltage, are precisely associated with MMs and not with trivial Andreev bound states [11, 12].

Typically, in superconducting nanowires, a Majorana bound state is considered, consisting of two MMs localized at opposite ends of the wire. However, the one-dimensional (1D) Hamiltonian used to describe superconducting wires belongs to the BDI symmetry class [13–15], enabling not only non-trivial topology in 1D systems but also the formation of multiple MMs on a single end of a finite-dimensional chain. In this scenario, different MMs on one edge of the chain do not hybridize with each other due to the effective symmetry similar to time-reversal symmetry [16]. Indeed, the realization of a phase with two pairs of MMs (two modes at each edge of the chain) was shown in Refs. [16, 17] when considering extended s-wave superconducting pairings. This implies the presence of pairing interaction between electrons at one site of the chain, as well as between nearest sites. The transition to such a phase can be induced by on-site Hubbard repulsion of electrons [17].

Previously, the formation of topological phases with multiple edge MMs was also shown in chains containing magnetic atoms [18] and under the influence of periodically time-varying external fields [19]. The interest in topological phases with several MMs stems from their realization with arbitrarily small Zeeman splitting of spin subbands (in zero field, they transform into Majorana-Kramers doublets [16]). Additionally, it offers the potential to permute multiple MMs simultaneously in quantum wires of T-, X-, and Y-structures, as was shown for single MMs in [2].

It is natural to assume that in the mentioned systems there also exist non-zero amplitudes of electron hopping, including spin-flip hopping due to spin-orbit coupling, and superconducting pairings between next-to-nearest neighbors in the chain. In this work, considering the processes in the second coordination sphere (CS), we demonstrate the formation of phases with three and four pairs of MMs in the model of a superconductor-semiconductor nanowire with spin-orbit coupling, placed in a magnetic field. Notably, for the classification of regions in the topological phase diagrams, a clear method is employed in addition to the BDI topological invariant [14, 16]. This method is associated with determining the fermion parity (FP) of the system ground state and was previously proposed in Refs. [20–22].

1. Model of a superconducting nanowire in the presence of processes in the second coordination sphere

The Hamiltonian of the model of a superconducting nanowire with open boundary conditions in the tight binding approximation has the form [21]:

$$\begin{aligned}
 H_W = & \sum_{\sigma, n=1}^N (-\mu - \eta_{\sigma} h) C_{n\sigma}^{\dagger} C_{n\sigma} + \sum_{\sigma, n=1}^{N-1} t (C_{n\sigma}^{\dagger} C_{n+1\sigma} + C_{n+1\sigma}^{\dagger} C_{n\sigma}) + \\
 & + \sum_{n=1}^{N-1} [(-\alpha)(C_{n\uparrow}^{\dagger} C_{n+1\downarrow} - C_{n\downarrow}^{\dagger} C_{n+1\uparrow}) + H.c.] + \sum_{n=1}^N [\Delta_0 C_{n\uparrow}^{\dagger} C_{n\downarrow}^{\dagger} + H.c.] + \\
 & + \sum_{n=1}^{N-1} [\Delta(C_{n\uparrow}^{\dagger} C_{n+1\downarrow}^{\dagger} + C_{n+1\uparrow}^{\dagger} C_{n\downarrow}^{\dagger}) + H.c.].
 \end{aligned} \tag{1}$$

The first term in (1) characterizes the electron energy measured from the chemical potential μ taking into account the Zeeman splitting h in an external magnetic field. The second term describes hoppings between nearest sites with amplitude t . The third and fourth terms define the Rashba spin-orbit interaction with amplitude α and on-site superconducting pairings with amplitude Δ_0 , respectively. The last term is due to superconducting pairings between electrons at nearby sites with the amplitude Δ . It was shown in [16,17] that the formation of phases with two pairs of MMs is possible at $2\Delta > \Delta_0$. To consider this regime in further calculations, it is convenient to set $\Delta_0 = 0$, keeping in mind that the results will be preserved at a qualitative level at $\Delta_0 \neq 0$ and $2\Delta > \Delta_0$. The parameter η_σ determines the sign and takes the following values depending on the projection of the spin momentum σ : $\eta_\sigma = +1$ (at $\sigma = \uparrow$), $\eta_\sigma = -1$ (at $\sigma = \downarrow$).

The Hamiltonian (1) describes only processes in the first CS. Therefore, to take into account long-range hopping, including spin-flip processes, as well as superconducting pairings in the second CS, we add the following terms:

$$H_2 = \sum_{\sigma, n=1}^{N-2} t_2 (C_{n\sigma}^\dagger C_{n+2\sigma} + C_{n+2\sigma}^\dagger C_{n\sigma}) + \sum_{n=1}^{N-2} \left[(-\alpha_2) (C_{n\uparrow}^\dagger C_{n+2\downarrow} - C_{n\downarrow}^\dagger C_{n+2\uparrow}) + H.c. \right] + \sum_{n=1}^{N-2} \left[\Delta_2 (C_{n\uparrow}^\dagger C_{n+2\downarrow}^\dagger + C_{n+2\uparrow}^\dagger C_{n\downarrow}^\dagger) + H.c. \right]. \quad (2)$$

The first part of (2) determines the hopping between next nearest sites with amplitude t_2 , the second and third terms are responsible for the effective spin-orbit interaction with the parameter α_2 and superconducting pairing with the parameter Δ_2 for the second CS. As a result, the Hamiltonian of the model under consideration is written as:

$$H = H_W + H_2. \quad (3)$$

2. Fermion parity and topological phase diagram

The Hamiltonian of the model in the reciprocal space under periodic boundary conditions takes the form:

$$H = \sum_{k, \sigma} \xi_{k\sigma} C_{k\sigma}^\dagger C_{k\sigma} + \sum_k \left[\Delta_k C_{k\uparrow}^\dagger C_{-k\downarrow}^\dagger + \Delta_k^* C_{-k\downarrow} C_{k\uparrow} \right] + \sum_k i\alpha_k \left[C_{k\uparrow}^\dagger C_{k\downarrow} - C_{k\downarrow}^\dagger C_{k\uparrow} \right], \quad (4)$$

where

$$\xi_{k\sigma} = t_k - \mu - \eta_\sigma h, \quad t_k = 2t \cos(k) + 2t_2 \cos(2k), \quad (5)$$

$$\alpha_k = 2\alpha \sin(k) + 2\alpha_2 \sin(2k), \quad (6)$$

$$\Delta_k = 2\Delta \cos(k) + 2\Delta_2 \cos(2k). \quad (7)$$

It was previously shown that the fermion parity (FP) of the ground state of a superconducting system is related to the structure of the Hamiltonian (1) at special points of the first Brillouin zone [20–22]. Here, the special points are considered as those values of the quasimomentum at which either the superconducting order parameter Δ_k or the spin-orbit coupling integral α_k vanishes while having non-zero amplitudes α , Δ , α_2 , Δ_2 . The identification of such special points is justified by the fact that, as it will be shown later, the Bogoliubov transformations have a different form for these points compared to other BZ points. It is worth noting that for models,

like Bardeen, Cooper, Schrieffer (BCS) model, special points are limited to the nodal points of the superconducting order parameter, where $\Delta_k = 0$. When considering the spin-orbit interaction, the energy spectrum takes a form distinct from BCS theory, although the superconducting gap in the spectrum still closes at the nodal points upon tuning model parameters. However, it can be easily shown in the considered model, that the condition $\alpha_k = 0$ also results in the closing the gap in the elementary excitation spectrum, which could be the cause of a topological transition with a possible change in the ground state FP. In this case, the gap is closed at points where $\alpha_k = 0$ but the superconducting order parameter Δ_k is non-zero. It is important to emphasize that it applies specifically to the closure of the superconducting gap, as in the absence of superconductivity the spectrum becomes gapless on the Fermi contour (lines in BZ).

From (6) and (7) it is obtained that the parameter Δ_k vanishes at the following points:

$$K_{\pm} = \pi - \arccos(\tilde{\Delta}_{\pm}), -K_{\pm}; \quad \tilde{\Delta}_{\pm} = \frac{1}{4} \frac{\Delta}{\Delta_2} \pm \frac{1}{4} \sqrt{\left(\frac{\Delta}{\Delta_2}\right)^2 + 8} \in [-1, 1], \quad (8)$$

and the spin-orbit coupling integral α_k vanishes at the points

$$Q = 0, \pi, \quad P = \pi - \arccos(\tilde{\alpha}), -P; \quad \tilde{\alpha} = \frac{\alpha}{2\alpha_2} \in [-1, 1]. \quad (9)$$

Thus, the Hamiltonian (4) is modified to the following sum of quadratic forms:

$$H = h(0) + h(\pi) + r(P) + r(K_+) + r(K_-) + \sum_{k>0, k \neq Q, P, K_{\pm}} r(k), \quad (10)$$

where $r(k) = h(k) + h(-k)$ (this definition is also valid for points $P, K_{\pm}$), and

$$h(Q) = \sum_{\sigma} \xi_{Q\sigma} C_{Q\sigma}^{\dagger} C_{Q\sigma} + \Delta_Q C_{Q\uparrow}^{\dagger} C_{Q\downarrow}^{\dagger} + \Delta_Q^* C_{Q\downarrow} C_{Q\uparrow}, \quad (11)$$

$$h(P) = \sum_{\sigma} \xi_{P\sigma} C_{P\sigma}^{\dagger} C_{P\sigma} + \Delta_P C_{P\uparrow}^{\dagger} C_{-P\downarrow}^{\dagger} + \Delta_P^* C_{-P\downarrow} C_{P\uparrow}, \quad (12)$$

$$h(K_{\pm}) = \sum_{\sigma} \xi_{K_{\pm}\sigma} C_{K_{\pm}\sigma}^{\dagger} C_{K_{\pm}\sigma} + i\alpha_{K_{\pm}} \left[C_{K_{\pm}\uparrow}^{\dagger} C_{K_{\pm}\downarrow} - C_{K_{\pm}\downarrow}^{\dagger} C_{K_{\pm}\uparrow} \right], \quad (13)$$

$$h(k) = \sum_{\sigma} \xi_{k\sigma} C_{k\sigma}^{\dagger} C_{k\sigma} + \Delta_k C_{k\uparrow}^{\dagger} C_{-k\downarrow}^{\dagger} + \Delta_k^* C_{-k\downarrow} C_{k\uparrow} + i\alpha_k \left[C_{k\uparrow}^{\dagger} C_{k\downarrow} - C_{k\downarrow}^{\dagger} C_{k\uparrow} \right]. \quad (14)$$

In [22], it was demonstrated that the eigenfunction of the term $r(k)$ (refer to (14)) can always be expressed as a superposition of states with an even number of fermions, akin to the wave function in BCS theory. Consequently, the structure and FP of the system's ground state are determined by the eigenvectors of the operators $h(Q)$, $r(P)$, and $r(K_{\pm})$.

As delineated in [22], the eigenvectors of the term of $h(Q)$ comprise two states: one with a single fermion, denoted as $|F_{Q\sigma}\rangle = C_{Q\sigma}^{\dagger}|0\rangle$, and two with an even number of fermions, represented as $|\Phi_{Q\pm}\rangle = \left(A_Q^{\pm} + B_Q^{\pm} C_{Q\uparrow}^{\dagger} C_{Q\downarrow}^{\dagger} \right) |0\rangle$. Of particular interest is the scenario where the state containing one fermion possesses the lowest energy. For $h > 0$ (which is the case always considered here), this reduces to the condition $E_{Q\downarrow}^F < E_{Q,-}^{\Phi}$. Consequently, the following inequalities emerge:

$$\begin{aligned} h &> \sqrt{\{2t_2 + 2t - \mu\}^2 + |2\Delta_2 + 2\Delta|^2}, \\ h &> \sqrt{\{2t_2 - 2t - \mu\}^2 + |2\Delta_2 - 2\Delta|^2}. \end{aligned} \quad (15)$$

The term $r(P)$ has the BCS form, and its eigenvectors also consist of an even number of fermions, leaving FP of the system's ground state wave function unaffected. In contrast, the term $h(K_{\pm})$ possesses an eigenvector that can be expressed as a superposition of states with a single fermion: $(a_{K_{\pm}}C_{K_{\pm}\sigma}^{\dagger} + b_{K_{\pm}}C_{K_{\pm}\bar{\sigma}}^{\dagger})|0\rangle$ ($\bar{\sigma}$ represents the opposite spin projection to σ). Nevertheless, it's evident that the state $(a_{-K_{\pm}}C_{-K_{\pm}\sigma}^{\dagger} + b_{-K_{\pm}}C_{-K_{\pm}\bar{\sigma}}^{\dagger})|0\rangle$, which is appropriate for the term $h(-K_{\pm})$, has the same energy. Therefore, even in cases where these states have lower energies than states with an even number of fermions, FP of the ground state wave function remains unchanged because the one-fermion states at points $K_{\pm}$ and $-K_{\pm}$ are filled simultaneously. Consequently, the conditions for achieving the ground state of a system with odd FP are dictated by the inequalities (15).

It is important to note that using the quadratic forms highlighted in (10), it is feasible to describe transitions that change the ground state wave function even when FP remains the same. By applying the Bogoliubov transformation, we can determine the excitation energies at the selected k points that may change signs with parameter variations.

$$\begin{aligned} E_{1k} &= \sqrt{(t_k - \mu)^2 + |\Delta_k^2|} - h, \quad k = Q, P, -P, \\ E_{2,3k} &= (-\mu) \pm \sqrt{\alpha_k^2 + h^2}, \quad k = K_{\pm}, -K_{\pm}. \end{aligned} \quad (16)$$

From the considerations given above, it follows that FP of the ground state for a given set of parameters coincides with the parity of the number of negative energies (16). The energies E_{1Q} at points 0 and π assume negative values under the conditions of (15), respectively. We will now outline the inequalities that lead to the remaining energies becoming negative:

$$E_{1P, -P} < 0, \quad h > \left\{ [4t_2(\tilde{\alpha}^2 - 1/2) - 2\tilde{\alpha}t - \mu]^2 + |4\Delta_2(\tilde{\alpha}^2 - 1/2) - 2\tilde{\alpha}\Delta|^2 \right\}^{1/2}, \quad (17)$$

$$E_{2K_{\pm}, -K_{\pm}} < 0, \quad h < \{\mu^2 - \mu_c^2\}^{1/2}, \quad \mu > \mu_c, \quad (18)$$

$$E_{3K_{\pm}, -K_{\pm}} < 0, \quad h > \{\mu^2 - \mu_c^2\}^{1/2}, \quad \mu < -\mu_c, \quad (19)$$

$$E_{3K_{\pm}, -K_{\pm}} < 0, \quad \mu > -\mu_c, \quad (20)$$

where $\mu_c = 2\sqrt{(1 - \tilde{\Delta}_{\pm}^2)(\alpha - 2\alpha_2\tilde{\Delta}_{\pm})}$.

Whenever the sign of any of the energies changes, it leads to a modification in the ground state wave function. It is easy to verify that the same conditions correspond to the closure of the gap in the elementary excitations spectrum at special points such as $Q, \pm P, K_{\pm}, -K_{\pm}$, while at other points the spectrum remains gapped until the amplitudes of superconducting pairings are non-zero.

In further calculations, we will assume that all parameters within the same CS are equal to each other. In this context, we will consider $\lambda_1 = \alpha = \Delta = t$ and $\lambda_2 = \alpha_2 = \Delta_2 = t_2$ and explore the case where $\lambda_2 = 1.5\lambda_1$, as it allows for the maximum number of MMs to be realized within the given model. Moreover, to establish the relationship between FP and the parity of the topological invariant, we compute the value of the topological index BDI [14, 16]:

$$N_{BDI} = \frac{-i}{\pi} \int_{k=0}^{k=\pi} \frac{dz(k)}{z(k)}, \quad z(k) = \frac{\det(Q(k))}{|\det(Q(k))|}, \quad Q(k) = \xi_k \sigma_0 - h \sigma_z - \alpha_k \sigma_y - i \Delta_k \sigma_y, \quad (21)$$

where σ_0 is the identity matrix, σ_y, σ_z are Pauli matrices.

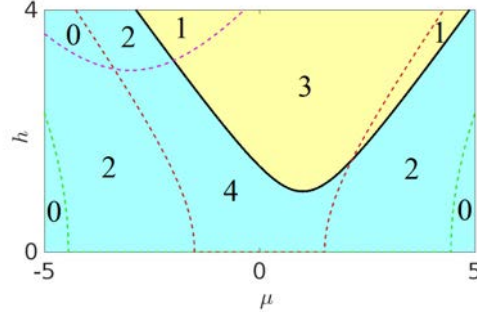


Fig. 1. Topological phase diagram in the variables chemical potential μ — Zeeman splitting h (in units of λ_1) for the amplitude $\lambda_2 = 1.5\lambda_1$. The lines are obtained from the conditions of changing the sign of the energies of one-fermion states at special points of the Brillouin zone. The value of the topological index is marked in each phase. Different colors indicate different fermion parities of the ground state wave function

Fig. 1 illustrates parameter lines $h(\mu)$ where the energy sign changes (16). As indicated in Fig. 1, these lines delineate the regions with distinct values of the topological index. Additionally, these regions are characterized by distinct numbers of negative energies (16). The solid line depicts the dependence (15). Traversing this line results in an energy sign change and a parity shifts in the count of negative energies. The other dependencies, determined from the inequalities (17)–(19), are shown by the dashed lines in Fig. 1. The dashed line obtained from (17) is highlighted in purple, the dependencies (18) and (19) are shown by green and red lines, respectively (see color version). Crossing these lines does not change FP but does modify the form of the ground state wave function. Similarly, when passing through the dashed lines, the topological index changes without affecting its parity. Consequently, the solid line serves as the boundary of the regions in the topological phase diagram with different FP of the wave function of the ground state. The regions highlighted in grey (turquoise in color version) have even FP, and the light (yellow) ones have odd FP.

It can be concluded that the parity of the topological invariant coincides with the parity of the number of one-fermion states that have negative energy in a given region. In this work, we do not specify the precise forms of the ground state wave functions. Their construction is similar to the method described in [22].

It is seen in Fig. 1, that the phases with $N_{BDI} = 3$ and $N_{BDI} = 4$ are formed in the presence of processes in the second CS. It should be noted that the $N_{BDI} = 3$ phase appear even when $\lambda_2 < \lambda_1$ (approximately for $\lambda_2 > 0.5\lambda_1$). And this phase can be realized at low values of Zeeman splitting ($\sim 0.1t$). On the other hand, the $N_{BDI} = 4$ phase is implemented for $\lambda_2 > \lambda_1$.

3. Majorana modes in phases with $N_{BDI} = 3$ and $N_{BDI} = 4$

To describe features of MMs in the new phases, we consider the Hamiltonian (3) with open boundary conditions and assume that the number of sites is $N = 150$. The transition from the original Fermi operators to Bogoliubov operators is carried out:

$$\begin{aligned} a_j &= \sum_{n=1}^N u_{jn\uparrow} C_{n\uparrow} + u_{jn\downarrow} C_{n\downarrow} + v_{jn\downarrow} C_{n\downarrow}^\dagger + v_{jn\uparrow} C_{n\uparrow}^\dagger, \\ a_j^\dagger &= \sum_{n=1}^N u_{jn\downarrow}^* C_{n\downarrow}^\dagger + u_{jn\uparrow}^* C_{n\uparrow}^\dagger + v_{jn\uparrow}^* C_{n\uparrow} + v_{jn\downarrow}^* C_{n\downarrow}, \quad j = [-2N, 2N]. \end{aligned} \quad (22)$$

The coefficients $u_{jn\sigma}$ and $v_{jn\sigma}$ in this case are components of the eigenfunctions of the Hamiltonian matrix. They can be used to obtain the distribution of quasiparticle operators over the sites of the chain. However in order to spatially separate MMs in the chain, it is convenient to use Majorana operators:

$$\begin{aligned} b' &= a_j + a_j^\dagger = \sum_{n,\sigma} \omega_{n\sigma} \gamma_{An\sigma} + z_{n\sigma} \gamma_{Bn\sigma}, \\ b'' &= i(a_j^\dagger - a_j) = \sum_{n,\sigma} \tilde{\omega}_{n\sigma} \gamma_{An\sigma} + \tilde{z}_{n\sigma} \gamma_{Bn\sigma}, \\ \omega_{n\sigma} &= 2\text{Re}(u_{jn\sigma} + v_{jn\sigma}), \quad z_{n\sigma} = -2\text{Im}(u_{jn\sigma} - v_{jn\sigma}), \\ \tilde{\omega}_{n\sigma} &= 2\text{Im}(u_{jn\sigma} + v_{jn\sigma}), \quad \tilde{z}_{n\sigma} = 2\text{Re}(u_{jn\sigma} - v_{jn\sigma}), \end{aligned} \quad (23)$$

where $\gamma_{An\sigma} = C_{n\sigma} + C_{n\sigma}^\dagger$, $\gamma_{Bn\sigma} = i(C_{n\sigma}^\dagger - C_{n\sigma})$ are Majorana self-conjugate operators.

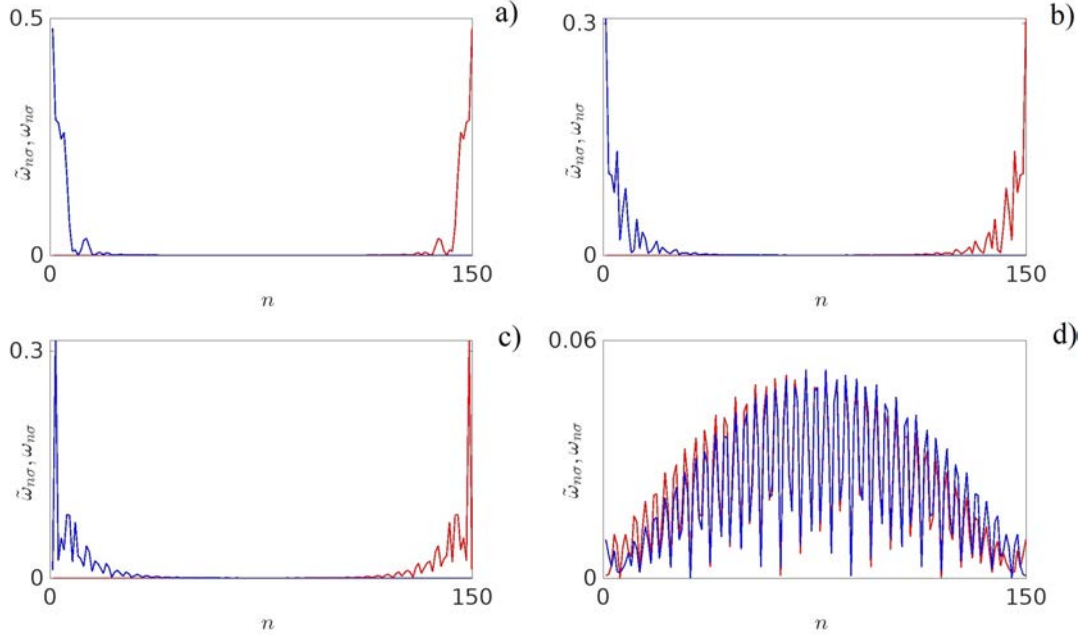


Fig. 2. Distributions of Majorana modes depending on the chain site for the first (a), second (b), third (c) and fourth (d) excitation energies in the phase with $N_{BDI} = 3$. The ratio between amplitudes in the second and first coordination spheres is $\lambda_2/\lambda_1 = 2$

In Fig. 2, the distributions of MMs on the chain sites (coefficients $\omega_{n\sigma}$ and $\tilde{\omega}_{n\sigma}$ in (23)) are presented for the phase with $N_{BDI} = 3$. The dependencies for coefficients $z_{n\sigma}$ and $\tilde{z}_{n\sigma}$ are qualitatively the same. Figures (a-c) reveal that the formation of MMs with edge localization occurs only for the first three excitation energies. These three excitations have near-zero energies (counted from the Fermi level). In the case of the fourth excitation with nonzero energy (d), the probability density of quasiparticles is minimal at the edges of the chain and increases towards its middle. The probability density of edge states in $N_{BDI} = 3$ phase exhibits an exponential decay with oscillations.

Fig. 3 shows the distributions of MMs for the phase with $N_{BDI} = 4$. As evident from figures (a-d), MMs are formed for all four excitation energies in this phase, resulting in the formation of four pairs of MMs with near-zero energies.

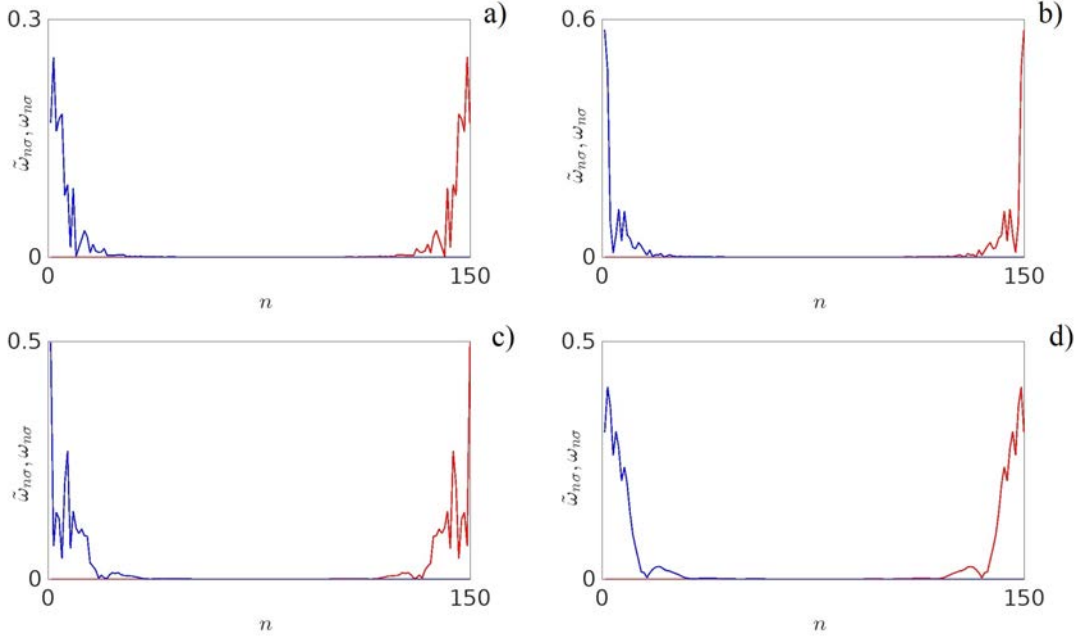


Fig. 3. Site distributions of Majorana modes for the first (a), second (b), third (c) and fourth (d) excitation energies in the phase with $N_{BDI} = 4$ ($\lambda_2/\lambda_1 = 2$)

Conclusion

The model of the superconducting nanowire under external magnetic field is considered with the periodic as well as open boundary conditions taking into account electron hoppings, including spin-flip hoppings due to Rashba spin-orbit coupling, as well as superconducting pairings between nearest and next-nearest neighbors. It is shown for the periodic boundary conditions that different phases on the topological phase diagram can be characterized by the number of occupied one-fermion states that have negative energy at special points of the Brillouin zone. As a result, the region of formation of odd fermion parity of the ground state is determined. It is obtained that the parity of the ground states coincides with the parity of the topological invariant, which in turn indicates the number of Majorana bound states, each of which contains one pair of Majorana modes localized at opposite edges of the chain. We show that taking into account the processes between next-nearest neighbors in a chain with open boundary conditions leads to the formation of phases with three and four pairs of Majorana modes.

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Фермионная четность в фазах с множеством майорановских мод в сверхпроводящей нанопроволоке

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Аннотация. Для полупроводниковой нанопроволоки, помещенной во внешнее магнитное поле, с наведенной сверхпроводимостью и сильным спин-орбитальным взаимодействием определена фермионная четность основного состояния в различных топологических фазах при учете перескоков электронов, включая перескоки с переворотом спина за счет спин-орбитальной связи, и сверхпроводящих спариваний во второй координационной сфере. Показана связь фермионной четности и четности BDI топологического инварианта. Продемонстрировано формирование топологических фаз с тремя и четырьмя парами майорановских мод.

Ключевые слова: фермионная четность, сверхпроводящая нанопроволока, майорановские моды, топологический инвариант, топологическая фазовая диаграмма.

EDN: SAFZUJ

УДК 538.9

Scaling Behavior of Complex Low-dimensional Spin Systems

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Abstract. We studied the non-equilibrium properties of spin-valve structures like a main elements of spintronics. We observed the aging effects in non-equilibrium critical behavior like a two-time dependence of measurement quantities: waiting time and observation time. The other essential property of ageing systems is dynamic scaling which we detected in our systems.

Keywords: dynamic scaling, spin-valve structures, thin films, aging effects, phase transitions and critical phenomena.

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In the early 1990s, there was a significant rise in the physics of magnetic phenomena due to the discovery of the giant magnetoresistance effect in the Fe/Cr structure [1]. This is how spintronics appeared — a new area of research. The basic elements of spintronics include spin valve nanostructures and multilayer.

Spin valves consist of magnetic metals (Fe, Co, Ni) separated by a nonmagnetic interlayer. The magnetization direction of one of the ferromagnetic films is fixed due to the strong bond with the antiferromagnetic metal layer (IrMn, FeMn, FeO) unlike multilayer structures. The structure of the spin valve is mainly represented as NiFe/NM/FM/FeMn, where one of the ferromagnetic films is the NiFe alloy — permalloy. The advantage of spin valves is their high sensitivity to an external magnetic field. For comparison, saturation fields at room temperature

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in multilayer structures are 100–2000 Oe, while in spin valves 5–50 Oe. This makes them the most advantageous in practical applications [2].

One of the fundamental properties of spin valves is magnetic anisotropy. In its absence, these elements do not have the characteristic features of changing magnetic and magnetoresistive properties. Magnetic anisotropy is one of the dominant factors influencing the behavior and properties of magnetic materials [3]. Symmetry breaking at the boundaries is always present during crystal growth due to the influence of various external and internal factors. This why the properties of a ferromagnetic system dependence on the direction of the magnetization vector. It is possible to create nanostructures from several layers using various materials. The magnetic moments of these layers will be directed along or across the plane of the film surface. It is convenient to describe the properties of such structures using the anisotropic Heisenberg model, which belongs to statistical spin models with slow dynamics.

Critical slowing down, aging effects, memory effects [4] are one of the most phenomena which we can see in statistical systems with slow dynamics. That is why these systems have theoretical and experimental interest.

The critical slowing down is the increasing of the autocorrelation time τ_{cor} near the critical temperature T_c . In the spin system this critical slowing down characterized by the universal dynamical critical index z

$$\tau_{\text{cor}} \sim |T - T_c|^{-\nu z} \quad (1)$$

with the correlation length ξ critical exponent ν .

1. Model and Methods

We simulated the systems with size $N_s = L \times L \times N$. Here N is number of layers and $L \times L$ is number of atom in one layer. We updated spin configurations with help of Metropolis algorithm. In this paper, we investigated time dependencies next quantities. For ferromagnetic films we defined the magnetization

$$m = \left\langle \frac{1}{N_s} \left[\left(\sum_i^{N_s} S_i^x \right)^2 + \left(\sum_i^{N_s} S_i^y \right)^2 + \left(\sum_i^{N_s} S_i^z \right)^2 \right]^{1/2} \right\rangle, \quad (2)$$

where angle brackets denote the statistical averaging.

In this paper, we calculated time dependencies of two-time autocorrelation function in order to inspected ageing phenomena

$$C(t, t_w) = \left\langle \frac{1}{N_s} \sum_i \vec{S}_i(t) \vec{S}_i(t_w) \right\rangle - m(t) \cdot m(t_w), \quad (3)$$

where t is the observation time is time from the sample preparation; t_w is the waiting time is the time characterizing the time got by from the moment of sample preparation to the measurement of its characteristics.

In this paper we calculated the correlation time. But Monte Carlo simulation have some troubles in obtaining the relaxation time. One of them is slow dynamics of our systems. The second problem is the method of determining the relaxation time. This method requires accurate evaluation of a long-time tail of the autocorrelation function. In order to fixed this problems, in

paper [5] had suggested a dimensionless dynamic correlation function

$$R(t, t_w) = \frac{C(t, t_w)}{\sqrt{\left\langle \left(\frac{1}{N_s} \sum_i \vec{S}_i(t) \vec{S}_i(t_w) \right)^2 \right\rangle}} \sim e^{-\delta t / \tau_{cor}}. \quad (4)$$

Equation (4) is the ratio of an odd to an even moment of the autocorrelation function.

2. Dynamic scaling in thin films

We researched the relaxation of process in thin Heisenberg films. In this paper, we studied how initial state influences these processes. In order to simulate relaxation processes in our systems we used Heisenberg hamiltonian with easy axis anisotropy

$$H = -J \sum_{\langle i, j \rangle} \left[(1 - \Delta)(S_i^x S_j^x + S_i^y S_j^y) + S_i^z S_j^z \right] - h \sum_i S_i^z. \quad (5)$$

We examined the thickness dependence of the Curie temperature T_C for thin films of $Ni(111)/W(110)$ [6] to determine the effective anisotropy constant $\Delta(N)$ [7, 8].

We simulated our systems at critical temperature $T_c = 1.15$ for $N = 3$ monolayer (ML), $T_c = 1.31$ for $N = 5$ ML, $T_c = 1.39$ for $N = 7$ ML [7–9] from height-temperature initial state $m_0 = 0.0001$. Simulation was carry out for different values of the waiting time $t_w = 200, 100, 70, 50, 20, 0$ Monte-Carlo steps per spin (MCS/s).

Autocorrelation function $C(t, t_w)$, which we calculated, depends on two time t and t_w . For different values of waiting time t_w the data are characterized the different curves. This fall on different curves means that time-translation invariance is broken. There are the effects of aging.

Therefore, the aging dynamics of systems may depend on the entire background history of our sample. In the aging regime for $t - t_w \sim t_w \gg 1$, the two-time dependence of the autocorrelation function is characterized by a scaling form:

$$C(t, t_w) \sim t_w^{-c} F_c(t/t_w), \quad (6)$$

where the index c is expressed in terms of the critical indices $c = 2\beta/z\nu$ at the quenching temperature $T_s = T_c$. When we plotted the same data for $C(t, t_w)$ depending on t/t_w , they fall onto a one curve. We observed this effect only the time t_w is large enough. In Fig. 1 the dynamic scaling for autocorrelation function presented for $N = 3$ ML and $N = 7$ ML. The collapse in Fig. 1 means that $C(t, t_w) = f(L(t)/L(t_w))$. Presence of dynamic scaling is the one of the properties of ageing systems.

We calculated the dimensionless dynamic correlation function $R(t, t_w)$ [10] to estimate the correlation time of our systems. For sufficiently long times, $R(t, t_w)$ decreases exponentially:

$$R(t, t_w) \sim \exp(-t/\tau_{cor}). \quad (7)$$

The values of the correlation time are presented in table:

N	$t_w = 50$	$t_w = 70$	$t_w = 100$	$t_w = 200$
3	2255(47)	2704(5)	2835(6)	3039(8)
5	1858(3)	2326(3)	2398(4)	2626(4)
7	1498(2)	1987(3)	2074(3)	2237(4)

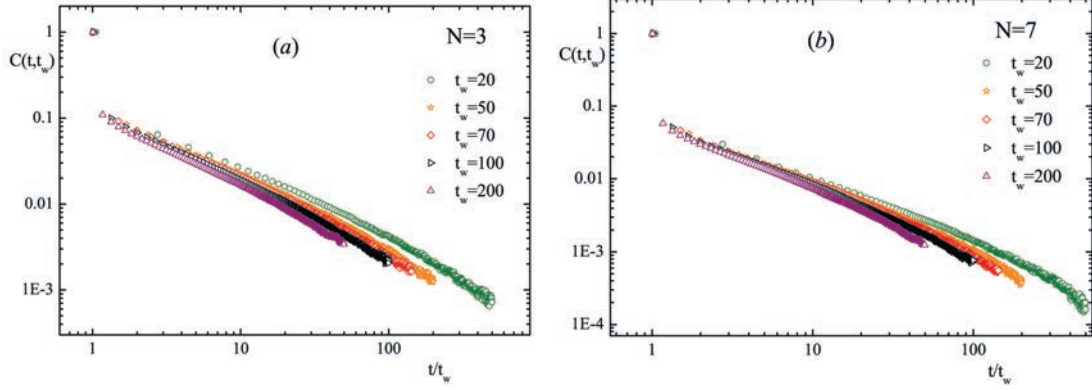


Fig. 1. Dynamical scaling of the two-time autocorrelation function $C(t, t_w)$ for several values of the waiting time $t_w = 200; 100; 70; 50; 20$ MCS/s with $N = 3$ ML (a) and $N = 7$ ML (b) for initial state $m_0 \ll 1$

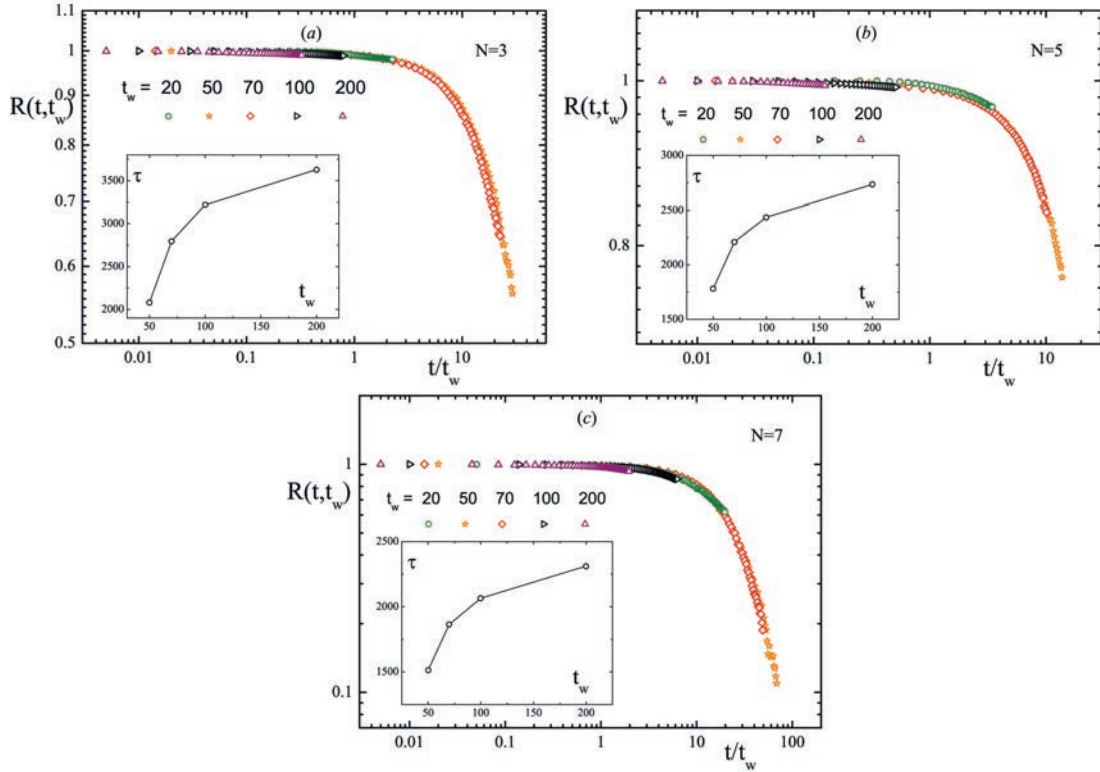


Fig. 2. Dynamical scaling of the two-time dimensionless dynamic correlation function $R(t, t_w)$ for several values of the waiting time $t_w = 200; 100; 70; 50; 20$ MCS/s with $N = 3$ ML (a), $N = 5$ ML (b) and $N = 7$ ML (c) for initial state $m_0 \ll 1$

The dynamic scaling for $R(t, t_w)$ is presented on Fig. 2 for thin film with different thickness. $R(t, t_w)$ is dimensionless correlation function. That is why their dynamical scaling form is given by [5]

$$R(t, t_w) \sim F_R(t/t_w) \quad (8)$$

without a factor involving the power of t_w , as for $C(t, t_w)$ in Eq. (6) or magnetization $m(t, t_w)$ in Eq. (9).

3. Dynamic relaxation of the complicated spin valve

Thin films from magnetic metals are the main elements of magnetic spin valve structures. This spin valve structures are used in devices with giant magnetoresistance (GMR) effects [1,11].

In order to simulate dynamic processes in spin valve structures we used Heisenberg hamiltonian with easy axis anisotropy (5). We considered system with size $L \times L \times N$. Our structures presented on Fig. 3. There are $N_1 = 3$ ML, $N_2 = 3$ ML, $N_3 = 3$ ML. The neighboring spins interacted with exchange integral J_1 . This exchange integral is assumed to be $J_1/kT = 1$. Other exchange integrals between the neighboring spins and the interlayer interaction are $J_{01} = 0.4J_1$, $J_{02} = 0.75J_1$, $J_2 = 0.01J_1$, $J_3 = -3.0J_1$, $J_4 = J_5 = -2.0J_1$ [2].

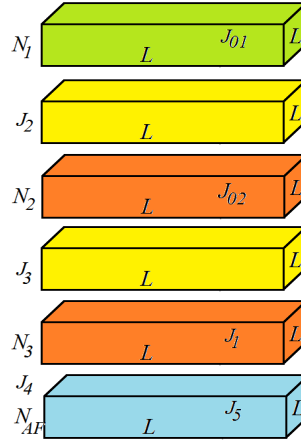


Fig. 3. Model of complicated spin valve. L , N_1 , N_2 , N_3 , N_{AF} – linear sizes; $J_{01} > 0$, $J_1 > 0$, $J_2 > 0$, $J_3 < 0$, $J_4 < 0$, $J_5 < 0$ – exchange interaction constants

In order to investigate the dynamic properties of spin valve structure we calculated time dependencies of magnetization $m_1^z(t)$. The simulation was carry out for waiting times $t_w = 10, 20, 30, 40, 60, 80, 100$ MCS/s. The following form of scaling dependence of magnetization is predicted from the theory of non-equilibrium processes:

$$m(t, t_w) \sim t_w^{-b} F_m(t/t_w), \quad (9)$$

where parameter $b = \beta/\nu z$ at temperature $T_s = T_c$. The time dependence of magnetization is construct in double logarithmic scale. It is necessary for finding the exponents, which characterize power-law character behavior of magnetization. In Fig. 4, there is a collapse of data for various t_w on a universal curve corresponding to the scaling form from the formula (9). We determined the value of the parameter $b = 0.0146(1)$ for system in critical point $T = T_c$ and $b = 0.0071(1)$ for system at $T = T_c/2$. We used next procedure to determine value of the parameter b : we chose the time interval according to the minimum of mean-square approximation error for the magnetization for each waiting time t_w ; then, we picked up the parameter b at the selected intervals so that the data collapsed on universal curve.

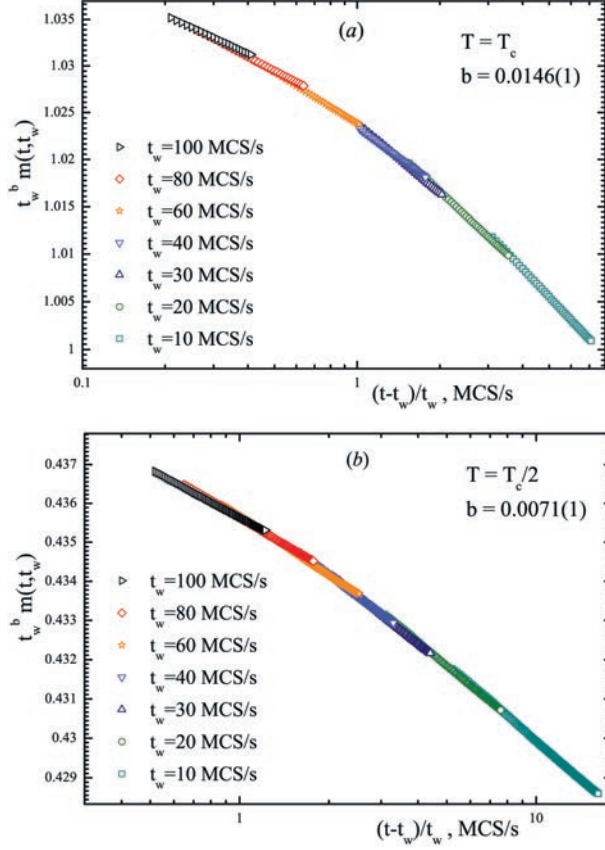


Fig. 4. Scaling dependence of magnetization $t_w^b \cdot m(t, t_w)$ from ratio $(t - t_w)/t_w$ at temperature $T = T_c$ (a) and $T = T_c/2$ (b)

Conclusion

In this paper, we studied the non-equilibrium behavior of a complicated spin-valve structure. The dependence of the reduced magnetization for free ferromagnetic film $m_1^z/m_0^z(t, t_w)$ on the observation time was shown. It characterized the behavior of magnetization relaxation. In the systems we study, the effects of critical slowdown of relaxation processes are observed. Our slow-dynamic systems exhibit the effects of aging.

We considered the structure with thicknesses $N_1 = N_2 = N_3 = 3$ ML. In this structure, the study of the manifestation of aging effects becomes clearer with an increase in the external magnetic field h . For $h = 2.0 J_1$, the system relaxes over a longer period of time with an increase in the waiting time t_w . This confirms that aging effects are observed in this dependence in the complicated spin-valve structure. In this paper, the presence of aging effects is confirmed by the data collapse for dynamic scaling dependencies for magnetization $m(t, t_w)$, autocorrelation function $C(t, t_w)$ and dimensionless autocorrelation function $R(t, t_w)$.

An important feature of non-equilibrium behavior in systems with slow dynamics is the occurrence of aging effects, which are observed in the present study. For different values of t_w the data clearly fall on different curves which means that time-translation invariance is broken. When we plotted the same data for $C(t, t_w)$ depending on t/t_w , they collapse onto a single

curve. This effect we observed only the time t_w is large enough. The observed effects of aging in magnetic structures such as thin films are in good agreement with the experimental results presented in [12].

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Скейлинговое поведение сложных низкоразмерных спиновых систем

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Аннотация. Изучены неравновесные свойства спин-вентильных структур как основных элементов спинтроники. Эффекты старения были обнаружены в неравновесном критическом поведении в виде двувременной зависимости измеряемых величин: времени ожидания и времени наблюдения. Другим важным свойством стареющих систем является динамическое масштабирование, которое было обнаружено в рассматриваемых системах.

Ключевые слова: динамический скейлинг, спин-вентильные структуры, тонкие пленки, эффекты старения, фазовые переходы и критические явления.

EDN: SHZXUG

УДК 538.945.91

Magnetization and Flux Capture in a Superconducting Ring Made of High- T_c 2G Tape

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Abstract. This article presents a theoretical examination of magnetization processes and magnetic flux capturing in a superconducting ring. Additionally, experimental investigations of magnetic flux trapping in a ring made of second-generation high-temperature superconducting tape were performed. The obtained experimental results are in excellent agreement with the theoretical model. Furthermore, the critical current in zero magnetic field was measured using a well-established non-contact method based on the capture of magnetic flux within the ring, and these data align with manufacturer-provided values. The method for determining the critical current was further modified by applying a local magnetic field to a small region of the ring. This modification allowed for the study of the critical current dependence on the magnitude and orientation of the magnetic field, providing a straightforward and reliable means of determining the critical current and its anisotropy.

Keywords: superconductivity, high-temperature superconducting tape, superconducting ring, flux capture, critical current, anisotropy.

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Second-generation high-temperature superconducting (2G HTSC) tapes, characterized by high critical current at liquid nitrogen temperature, find applications as cables, transformers, current limiters, and others [1]. One of the key parameters of 2G HTSC tapes is the critical current, denoted as I_c , and its dependence on the value of magnetic field H and field orientation relative to the tape plane (α). To determine the critical current, the resistive method is widely used, providing highly accurate values of $I_c(H, \alpha)$. The use of the resistive method for 2G HTSC tapes with high critical currents requires creating low-resistance current contacts with minimal heat dissipation to the thin superconducting layer (typically 1–2 μm) covered with normal conducting layers. Therefore, non-contact methods for measuring $I_c(H)$ are more favorable. A well-known non-contact method is the measurement of superconductor magnetic field dependence of magnetization $M(H)$ and calculation of critical current from magnetization hysteresis. However, the use of this method relies on the applicability of the critical state model and leads to noticeable errors at low fields [2]. Another non-contact method to determine I_c , which is the focus of this study, uses the capture of magnetic flux in a superconducting ring, allowing the precise determination of the maximum dissipationless current in the tape. This method does not rely on any specific models but requires the creation of a continuous superconducting ring. At a temperature below the critical temperature (T_c) of the superconducting material, a magnetic

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flux is introduced into the ring's cavity, inducing a current within the HTSC tape that generates a magnetic flux opposing the applied flux. By measuring the magnetic field of this current the value of current can be calculated. This non-contact method for measuring critical currents is well-established and has been employed in several studies [3–5].

Rings made of 2G HTSC tapes are used for the design of magnetic systems for various purposes [6–8]. In these studies, significant attention was paid to determining the critical current of the tape and its dependence on the magnetic field. More information about the process of penetration and capture of magnetic flux is contained in the complete hysteresis curve of the magnetic field dependence of the ring magnetization. Previously, the hysteresis dependence of the magnetization of a bulk 2G HTSC ring on the flux of an internal solenoid was discussed in [5].

In the present work we analyze the magnetization curve of a superconducting ring and perform experimental studies of the penetration and capture of magnetic flux within a ring made from 2G HTSC tape produced by SuperOx. This study allowed us to determine the tape critical current dependences on the magnitude and orientation of the magnetic field with respect to the tape plane. In the designed setup an elongate solenoid is placed inside the ring, and at $T < T_c$, a current is introduced into the solenoid to generate magnetic flux within the ring. Subsequently the solenoid turns off, some magnetic flux is captured in the ring and the field of the ring current is recorded by Hall sensor situated in the ring center.

1. Penetration and capture of magnetic flux in a 2G HTSC tape ring

The process of flux penetration and capture in the ring is based on the ability of a superconducting ring to preserve the enclosed magnetic flux until the critical current is reached. This process is schematically illustrated in Figure 1a, where the dependences of the magnetic flux created by the ring's current (Φ_{ring} , blue) and the total magnetic flux $\Phi_{\Sigma} = \Phi_{ring} + \Phi_{sol}$ (red) on the flux created by the solenoid (Φ_{sol}) are depicted.

As the magnetic flux generated by the solenoid (Φ_{sol}) increases, a current I arises within the superconducting ring, creating a magnetic flux $\Phi_{ring} = I * L_{ring}$, where L_{ring} is the inductance of the ring. In the initial part of the dependence, the total magnetic flux $\Phi_{\Sigma} = \Phi_{ring} + \Phi_{sol} = 0$, that is complete compensation of the applied magnetic flux. This complete compensation persists until the magnetic flux reaches $\Phi_{sol} = \Phi_c$, where $\Phi_c = I_c * L_{ring}$, and I_c represents the tape critical current at the given temperature. If, at this point, the solenoid current is reduced to zero, the ring's current also decreases to zero, and no flux capture occurs in the ring. However, if the solenoid flux is increased beyond Φ_c , the ring's flux Φ_{ring} remains constant, as the ring current reached its maximum value I_c , and the total flux Φ_{Σ} starts to increase. The captured flux in the ring (Φ_{sol} switched off) also increases, reaching its maximum at $\Phi_{sol} = 2\Phi_c$. This corresponds to the change in the ring's current from $-I_c$ to $+I_c$. If the solenoid flux decreases from even higher values, the captured flux in the ring remains constant at Φ_c .

A similar reasoning of magnetization and flux capture in the superconducting ring with negative Φ_{sol} values allow to construct the full hysteresis dependence of the ring's magnetization on the magnetic flux generated by the solenoid. The analysis of hysteresis magnetization allows us to obtain the relation between the captured magnetic flux in the ring and the magnitude of the flux initially created by the solenoid and subsequently reduced to zero (Fig. 1b).

2. Experiment

The rings were created from the 12 mm wide 2G HTSC tape (SuperOx). The tape critical current, as specified by the manufacturer, is $I_c = 514$ A at the liquid nitrogen temperature ($T = 77$ K). Using a 0.1 mm thick diamond disc, a longitudinal cut with a length of 10 cm was

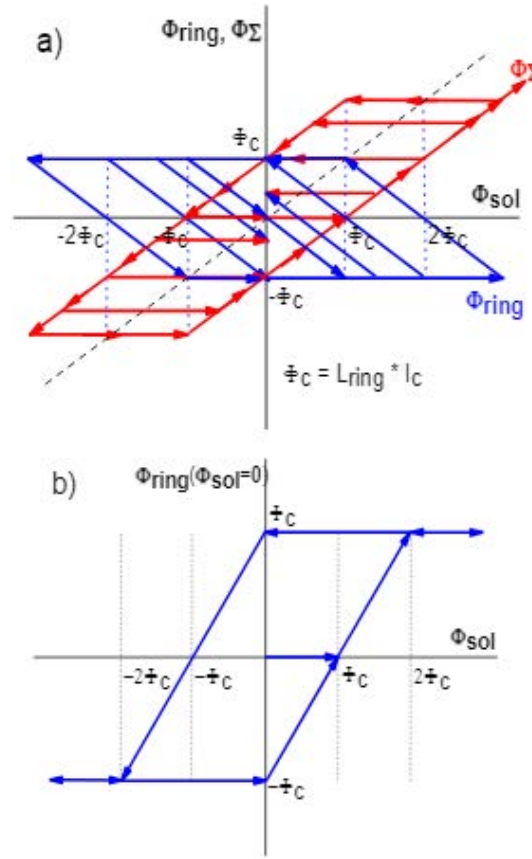


Fig. 1. Process of magnetic flux penetration and capture in the superconducting ring. a) — dependences of the magnetic flux created by the ring's current (Φ_{ring} , blue) and the total magnetic flux Φ_{Σ} (red) on the flux generated by the solenoid (Φ_{sol}); b) — dependence of the captured magnetic flux in the ring $\Phi_{ring}(\Phi_{sol}=0)$ on the magnitude of the flux Φ_{sol} initially created by the solenoid and subsequently reduced to zero

made along the middle of the tape width, not reaching the edges. Fig. 2 presents the components of the experimental setup. A solenoid (1) with a 3 cm diameter was inserted into the slit of the tape (2). The Hall sensor was placed at the center of the solenoid. This component of the setup alone allows to measure $I_c(H=0)$ by measuring the magnetic field H of ring's current I using the Hall sensor and calculating rings current by the well-known relation: $H = I/2R$, where R is the ring radius.

The setup was additionally equipped with a mechanism (3) to move NdFeB magnets (4) closer and to increase the magnetic field acting on a local region of the tape. Mechanism (5) allows changing the tape orientation (angle α) relative to the local magnetic field with a 5-degree rotation step. By varying the magnitude of the local magnetic field using device (3) and the angle of its orientation using device (5), it is possible to obtain I_c dependences on the magnitude and orientation of the magnetic field.

All measurements were performed at the liquid nitrogen temperature $T = 77$ K. Initially, the ring is considered to be completely demagnetized, i.e. $I = 0$. The setup, along with the HTSC tape ring, is cooled down, and a current is applied to the solenoid to generate magnetic flux within the ring. Then, the solenoid switched off and the Hall sensor records the magnetic field of the ring's current in the center of the ring.

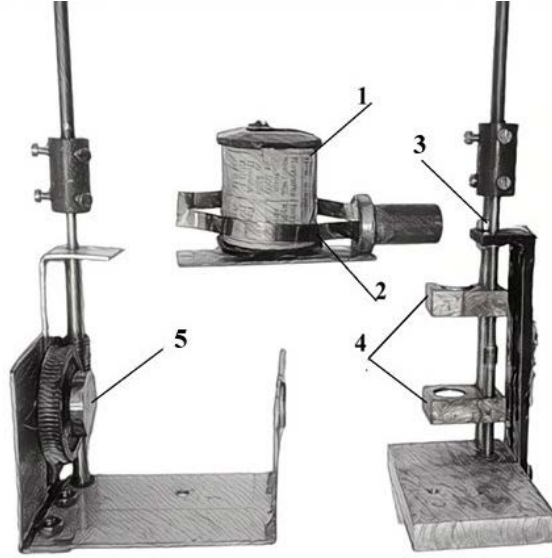


Fig. 2. Setup for non-contact measurements of the $I_c(H, \alpha)$ dependences. Three main setup components are depicted. In the center is the solenoid (1) with the 2G HTSC tape (2) and the Hall sensor at the center of the solenoid. Additional components include: the mechanism for tape orientation changing (3) relative to the field of permanent magnets (4) located on the local field changing mechanism (5)

3. Results

Using the devised setup, a dependence of the magnetic field at the center of the ring H_{center} measured by the Hall sensor and proportional to the ring flux $\Phi_{ring} = L \cdot I$, was obtained as a function of the magnitude of the previously applied (and subsequently reduced to zero) magnetic field of the solenoid H_{sol} (Fig. 3). The field at the center H_{center} is created by the ring current and is therefore proportional to the ring flux $\Phi_{ring} = L \cdot I$, and the field H_{sol} is proportional to the flux Φ_{sol} . Thus, the H_{center}/H_{sol} dependence (Fig. 3) should correspond to the Φ_{ring}/Φ_{sol} dependence (Fig. 1b).

It is evident that there is good agreement between the experiment and the simple model based on the preservation of magnetic flux in the superconducting ring up to its critical current. It is worth noting that the field at which the captured flux starts to rise sharply (250 Oe) is half of the field at which the captured flux saturates (500 Oe), in line with the model analysis. The saturation of the captured flux corresponds to the ring's current reaching its critical value and allows for its estimation using the relation $H_{max} = I_c/2R$. H_{sol} higher than 500 Oe does not lead to an increase in the magnetic flux captured by the ring. Reducing H_{sol} has almost no impact on the value of the captured flux in the ring until the magnetic field changes its sign, and the flux capture process repeats with the opposite sign.

4. Dependence of the critical current on the magnitude and orientation of the magnetic field

The critical current of the HTSC tape decreases with an increase in the magnetic field, and the application of a magnetic field locally on a small region of the tape results in a reduction in the overall critical current of the ring. Using permanent magnets to create magnetic field acting

locally on a small region of the tape, the dependence of the critical current on the external magnetic field was measured. The magnitude of the local magnetic field was varied by changing the distance between the permanent magnets.

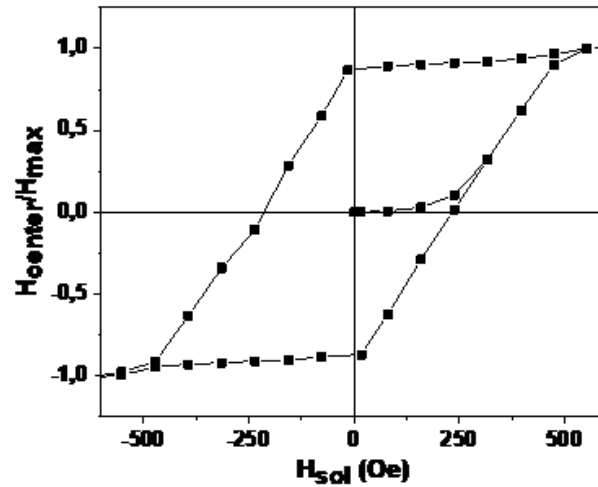


Fig. 3. Dependence of captured flux, created by the currents in the HTSC ring, on H after reducing the magnetic field of the solenoid from H_{sol} to zero

Fig. 4 shows the dependence of the critical current on the external magnetic field oriented perpendicular to the tape plane at $T = 77$ K. Each point of the dependence was obtained after ring cooling to 77 K in zero magnetic field.

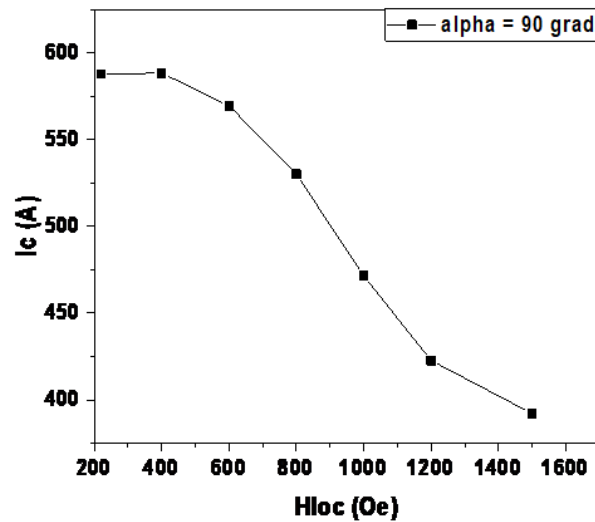


Fig. 4. Dependence of the critical current of the 12mm wide SuperOx tape on the magnitude of the local magnetic field, oriented perpendicular to the tape plane, at $T=77$ K

It's worth noting the high value of the critical current of the tape (580 A) at low magnetic fields (200 Oe), which corresponds to the value provided by the tape manufacturer. When the magnetic field increases to 1.5 kOe, the critical current decreasing is 30%.

By changing the orientation of the tape relative to the local magnetic field orientation, the dependence of the critical current on the angle between the tape plane and the direction of the magnetic field was obtained.

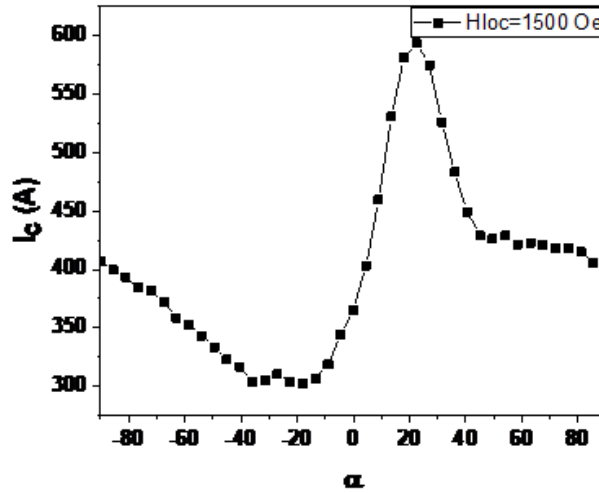


Fig. 5. Dependence of the critical current of the 12mm wide SuperOx tape at $T=77K$ on the angle between the tape plane and magnetic field orientation. Local magnetic field $H_{loc} = 1500$ Oe

Conclusion

Experimentally obtained complete hysteresis loop of the captured magnetic flux in the ring resembled closely the calculated one. The field at which the captured flux starts to rise sharply (250 Oe) is half of the field at which the captured flux saturates (500 Oe). Measuring the magnetic field at the center of the ring corresponding to the maximum trapped flux reliably allows us to determine the critical current of the tape.

For the 12 mm wide SuperOx HTSC tape the value of the critical current (580 A) at low magnetic fields (200 Oe) corresponds to the value provided by the manufacturer. When the magnetic field increases to 1.5 kOe, the critical current decreasing is 30%. The HTSC tape exhibits a significant and asymmetric dependence of the critical current on the orientation of the applied magnetic field.

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Намагниченность и захват потока в кольце из высокотемпературной сверхпроводящей ленты второго поколения

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Аннотация. В данной статье проведено теоретическое рассмотрение процессов намагниченности и захвата магнитного потока в сверхпроводящем кольце, а также проведено экспериментальное исследование захвата магнитного потока в кольце, изготовленном из высокотемпературной сверхпроводящей ленты второго поколения. Полученные экспериментальные результаты хорошо согласуются с моделью. Дополнительно в работе был измерен критический ток в нулевом поле, используя широко известный бесконтактный метод измерения критического тока на основе захваченного в кольце магнитного потока. Эти данные согласуются с значениями, предоставленными производителем. Данный метод был модифицирован путем применения локального магнитного поля к участку кольца. Это позволило исследовать зависимость критического тока от величины и ориентации магнитного поля, представляя собой простой и надежный способ определения критического тока и его анизотропии.

Ключевые слова: сверхпроводимость, высокотемпературная сверхпроводящая лента, сверхпроводящее кольцо, захват магнитного потока, критический ток, анизотропия.